SHEEP FARMING

FIELD AWARENESS AND VETERINARY MANAGEMENT GUIDE - VOLUME 2

Dr M.S.Saravanan

B.V.Sc.,M.A.,PGDCM.,LME.,

Feedback

The e-book titled *"Sheep Farming – Field Awareness and Veterinary Management Guide"*, authored by Dr. M. S. Saravanan, Deputy Director of the Animal Husbandry Department, Virudhunagar District, Tamil Nadu, represents a well-conceived and comprehensive contribution to the field of small ruminant health management and livestock extension. It is an exceptionally well-structured field guide, thoughtfully designed to meet the needs of veterinarians, students, and livestock farmers alike.

The document systematically integrates the fundamental principles of sheep production with contemporary veterinary practices, providing a balanced and practical synthesis of disease prevention, flock health management, and sustainable production strategies. It covers a wide range of essential topics, including disease management, vaccination schedules, orthopaedic conditions, parasitic infestations, antimicrobial and anthelmintic resistance, and sustainable farming practices in a clear, concise, and highly practical manner.

A notable strength of this work lies in its structured and methodical presentation. The author has effectively organized diverse subject areas into clearly defined sections, supported by appropriate pictorial representations, which significantly enhance readability and facilitate ease of understanding. The emphasis on evidence-based, field-oriented approaches ensures that the content remains both scientifically robust and operationally relevant.

The division of the e-book into Part I and Part II further enhances its pedagogical value, allowing for systematic learning among farmers, students, and practitioners. The content is presented in a simple yet highly organized format, incorporating clear headings, key messages, and actionable insights. This approach makes the guide particularly effective for field-level application, especially in rural and resource limited settings.

The e-book is especially commendable for addressing emerging global concerns such as antimicrobial resistance and sustainable livestock systems. The discussion on integrated parasite management and responsible drug use reflects a progressive and forward-looking perspective aligned with current international veterinary and public health priorities. The incorporation of the One Health approach further strengthens the work by emphasizing the interconnectedness of animal, human, and environmental health.

Additionally, the use of structured formats such as bullet points, clinical signs, treatment protocols, and preventive strategies enhances its utility as a quick reference guide for busy field veterinarians and livestock handlers. The emphasis on preventive care, hygiene, early diagnosis, and stakeholder collaboration underscores a modern and responsible approach to livestock management.

From an educational standpoint, this e-book serves as a valuable reference for veterinary practitioners, students, and extension personnel. Its clarity of expression, logical organization, and practical orientation make it an effective tool for capacity building and knowledge dissemination at the grassroots level.

Overall, this work constitutes a meaningful and impactful resource that contributes significantly to improving sheep health management practices and promoting sustainable livestock production. It holds considerable relevance for both academic dissemination and practical implementation, not only within India but also in similar agro-climatic regions globally. Overall, it stands as an excellent extension and training resource with strong potential for field-level impact in veterinary science and animal husbandry.

I extend my hearty congratulations to the author for his commendable work for the benefit of sheep farming community.

With regards,

Dr. R. Uma Rani, Ph.D., PGDECM.,
Professor,
Tamil Nadu Veterinary and Animal Sciences University,
Veterinary University Training and Diagnostic Centre,

Madurai

Tamil Nadu

Table of Contents

The Art of Disease Management in a Sheep Farm: A Preface

The stewardship of a sheep farm is a delicate balance between ancient pastoral wisdom and the rigorous precision of modern veterinary science. It is not merely a task of clinical intervention but an intricate art form where the shepherd's observation meets the clinician's diagnostic acumen. Success in this domain is measured not by the diseases we cure, but by the outbreaks we prevent.

The Philosophy of Prevention

Disease management is built upon the foundation of the Five Freedoms of Animal Welfare. When we ensure freedom from hunger, thirst, discomfort, pain, and fear, we are doing more than practicing ethics; we are fortifying the biological defences of the flock. A sheep that is nutritionally secure and environmentally comfortable is a sheep that possesses the innate resilience to withstand pathogens.

The One Health Perspective

In the modern era, the sheep farm does not exist in isolation. We must operate under the One Health paradigm, recognizing that the health of our livestock is inextricably linked to the health of the environment and the human communities that depend on them. Managing zoonotic risks and ensuring the judicial use of antimicrobials are responsibilities that elevate the veterinarian from a farm consultant to a guardian of public health.

Precision and Tradition

The future of small ruminant health lies in the integration of Precision Farming. By utilizing data-driven insights—from genetic selection of hardy indigenous breeds to the strategic monitoring of grazing patterns—we can move away from "blanket treatments" toward targeted, effective interventions. This is particularly vital in the conservation of our native germplasm, where maintaining genetic diversity is our best insurance against emerging disease threats.

The Call to Action

This work is dedicated to the stakeholders of the veterinary profession: the farmers, the field clinicians, and the students. Sharing this knowledge is the highest form of professional service. It is my hope that these pages serve as a practical compass for those navigating the complexities of flock health, ensuring that our livestock thrive as productive, healthy, and respected members of our agricultural ecosystem.


How to Assess the Health Status of a Sheep Flock
FieldvetDrMSS
1 General Behaviour
Healthy sheep is alert and active
Moves with the flock
Grazes normally
Warning signs: Isolation flom flock
Depression or weakness
Abnormal posture
1 2 3 4 4 5
Very Thin 3 Ideal 5 Obese
Ideal flock BCS: 2.5-3.5
2 Body Condition Score (BCS)
Assess fat and muscle around the loin area.
Scale 1~3 5.5
1 1 2 3 3 4 4 5
Very Thin 2 Thin 3 Ideal 4 Obese
3 Appetite and Feeding Behaviour
Healthy sheep eat actively during grazing
Ruminate regularly
Warning signs:
Reduced feed intake
Refusal to graze
Slow rumination
4 FAMACHA Eye Test
Check the lower eyelid colour to detect anaemia by Hareemonchus worms.
Ieal flock BCS: 2.5-3.5
5 Mobility and Gait
Observe walking pattern.
Healthy sheep
Walk normally
No lameness
Probiems include:
Foot rot
Hoof overgrowth
Joint infections
6 Coat and Skin Condition
Healthy sheep have smooth and shiny wool/hair
Clean skin without lesions
Warning signs:
Reduced feed intake
Refusal to graze
Slow rumination
7 Respiration and Nasal Signs
Normal respiration: 12 ~ 20 breaths per minute
Warning signs:
Coughing
Nasal discharge
Rapid breathing
Preventive Health Measures
Regular vaccination
Strategic deworming
Good nutrition
Clean water supply
Veterinary supervision
8 Lamb Health Indicators
Healthy flocks typically show:
Low mortality (<5%)
Low disease incidence
Low disease incidence
Warning signs:
Weak lambs.
scours
High lamb mortality
Regular Observation is the Best Tool to Maintain a Healthy Sheep Flock.
FieldvetDrMSS
#FieldVet

Section 1: Health Management Basics

1. How to Assess the Health Status of a Sheep Flock

🐑 Importance of Flock Health Assessment

Regular health monitoring helps to:

- Detect diseases early
- Reduce mortality and economic losses
- Improve growth and reproduction
- Maintain overall flock productivity

Daily observation is the first step in flock health management.

🔍 Key Indicators to Assess Sheep Flock Health

1️⃣ General Behaviour

Healthy sheep:

- Are alert and active
- Move with the flock
- Graze normally

Warning signs:

- Isolation from flock
- Depression or weakness
- Abnormal posture

2️⃣ Body Condition Score (BCS)

Assess fat and muscle around the loin area.

- Scale 1–5

Score & Condition

- 1–Very thin
- 2–Thin
- 3—Ideal
- 4—-Fat
- 5—-Obese

Ideal flock BCS: 2.5–3.5

3️⃣ Appetite and Feeding Behaviour

Healthy sheep:

- Eat actively during grazing

- Ruminate regularly

Warning signs:

- Reduced feed intake
- Refusal to graze
- Slow rumination

4️⃣ FAMACHA Eye Test:

Used to detect anaemia caused by Haemonchus worms.

Check colour of lower eyelid:

Score - Colour-Condition

- 1 Red Healthy
- 2 Red-Pink Good
- 3 Pink Moderate
- 4 Pale Anaemia
- 5 White Severe infection

Animals with score 4–5 need treatment.

5️⃣ Coat and Skin Condition

Healthy sheep have:

- Smooth and shiny wool/hair
- Clean skin without lesions

Warning signs:

- Rough coat
- Wool loss
- Skin lesions or ticks

6️⃣ Mobility and Gait

- Observe walking pattern.

Healthy sheep:

- Walk normally
- No lameness

Problems include:

- Foot rot
- Hoof overgrowth
- Joint infections

7️⃣ Respiration and Nasal Signs

Normal respiration:

- 12–20 breaths per minute

Warning signs:

- Coughing
- Nasal discharge
- Rapid breathing

May indicate pneumonia or respiratory infection.

8️⃣ Faecal Condition

Normal faeces:

- Firm pellets

Abnormal signs:

- Diarrhoea
- Worm segments
- Blood or mucus

Often linked to parasites or digestive diseases.

9️⃣ Lamb Health Indicators

Monitor lambs for:

- Good suckling
- Active movement
- Normal growth

Warning signs:

- Weak lambs
- Scours
- High lamb mortality

🔟 Mortality and Morbidity Records

Healthy flocks typically show:

- Low mortality (<5%)
- Low disease incidence

Keep records for:

- Deaths
- Treatments

- Vaccinations

🛡️ **Preventive Health Measures**

- Regular vaccination
- Strategic deworming
- Good nutrition
- Clean water supply
- Proper housing and hygiene
- Veterinary supervision

📢 **Key Message**

"Regular Observation is the Best Tool to Maintain a Healthy Sheep Flock."


Simple Diagnostic Equipment
for Common Ailments of Sheep
(Field-Level & Farm Diagnostic Tools)
FieldvetDrMSS
1 Clinical Examination Tools
Stethoscope
• Used for:
• Heart rate & rhythm
• Lung sounds (pneumonia, bronchitis)
• Ruminal movements
Diagnoses: Pneumonia, cardiac disorders, indigestion, rumen stasis
2 Basic Observation & Handling Tools
Torch / Headlamp
• Used for:
• Oral cavity examination
• Eye inspection
• Wound examination
Diagnoses: Orf, FMD, eye infections, injuries
Diagnoses: Foot rot, foot abscess, laminitis
2 Basic Observation & Handling Tools
Torch / Headlamp
• Used for:
• Oral cavity examination
• Eye inspection
• Wound examination
Diagnoses: Orf, FMD, eye infections, injuries
4 Hematological & Blood Testing Tools
Syringes, Vacutainers & Needles
• Blood sample collection
Diagnoses: Anemia, metabolic disorders, infectious diseases
Diagnoses: Anemia, parasitc infestiatations
3 Parasitic Disease Diagnostic Tools
Microscope
• Used for: nenration
• Fecal examination
• Blood smear examination
Diagnoses: Gastrointestinal parasites., Coccidiosis, (Blood parasites (Anaplasmosis)
5 Digestive System Diagnostic Tools
Rumen Stomach Tube
• Used for:
• Rumen fluid collection
• Relief of bloat
Diagnoses: Rumen acidosis, Frothy bloat, indigestion
4 Reproductive & Urinary System Tools
Disposable Gloves
• Vaginal & rectal examination
Diagnoses: Pregnancy status, dystocia, uterine infections
8 Basic Supportive Diagnostic Tools
Weighing Tape
• Body weight estimation
Purpose: Drug dose calculation, growth nontoring
7 Skin & Extemal Parasite Tools
Magnitying Lens
• Detection of lice, ticks, mites
Diagnoses: Ectoparasitic infestations, mange
8 Basic Supportive Diagnostic Tools
Weighing Tape
• Body weight ellimation
Record Book / Mobile App
Purpose: Drug dose calculation, growth monitoring
Key Message
Early diagnosis using simple tools reduces mortality, treatment cost, and production losses in sheep.
#Fieldvet

2. Simple Diagnostic Equipment for Common Ailments of Sheep

(Field-Level & Farm Diagnostic Tools)

1. Clinical Examination Tools:

□ Stethoscope

Used for:

- Heart rate & rhythm
- Lung sounds (pneumonia, bronchitis)
- Ruminal movements

Diagnoses: Pneumonia, cardiac disorders, indigestion, rumen stasis

🌡 Digital Thermometer:

Used for:

- Measuring rectal temperature

Normal temperature: 38.5 – 40.0 °C

Diagnoses: Fever, septicemia, viral & bacterial infections

2. Basic Observation & Handling Tools:

👁 Torch / Headlamp

Used for:

- Oral cavity examination
- Eye inspection
- Wound examination

Diagnoses: Orf, FMD, eye infections, injuries

🐑 Hoof Tester / Hoof Knife:

Used for:

- Hoof examination
- Detecting pain points

Diagnoses: Foot rot, foot abscess, laminitis

3. Parasitic Disease Diagnostic Tools:

🔬 Microscope

Used for:

- Fecal examination
- Blood smear examination

Diagnoses:

- Gastrointestinal parasites
- Coccidiosis
- Blood parasites (Anaplasmosis)

Fecal Examination Kit:

(Slides, coverslips, flotation solution)

Used for:

- Detection of worm eggs & oocysts

Diagnoses: Helminthiasis, coccidiosis

4. Hematological & Blood Testing Tools:

Syringes, Vacutainers & Needles

Used for:

- Blood sample collection

Diagnoses: Anemia, metabolic disorders, infectious diseases

Hemoglobin Meter / PCV Tubes:

Used for:

- Estimation of hemoglobin / packed cell volume

Diagnoses: Anemia, parasitic infestations

5. Digestive System Diagnostic Tools:

Rumen Stomach Tube

Used for:

- Rumen fluid collection
- Relief of bloat

Diagnoses:

- Rumen acidosis
- Frothy bloat
- Indigestion

pH Paper:

Used for: Measuring rumen fluid pH

Normal rumen pH: 6.0 – 7.0

Diagnoses: Acidosis, alkalosis

6. Reproductive & Urinary System Tools:

🧤 **Disposable Gloves**

Used for:

- Vaginal & rectal examination

Diagnoses: Pregnancy status, dystocia, uterine infections

▯ **Urine Collection Container:**

Used for:

- Urine examination

Diagnoses: Urolithiasis, urinary infections

7. Skin & External Parasite Tools:

🔍 **Magnifying Lens**

Used for:

- Detection of lice, ticks, mites

Diagnoses: Ectoparasitic infestations, mange

8. Basic Supportive Diagnostic Tools:

⚖ **Weighing Tape**

Used for:

- Body weight estimation

Purpose: Drug dose calculation, growth monitoring

🗒 Record Book / Mobile App:

Used for:

- Recording temperature, treatments, vaccination, deworming

Importance: Disease monitoring & prevention planning

Key Message:

▯ **Early diagnosis using simple tools reduces mortality, treatment cost, and production losses in sheep.**


Essential Vaccines
for Sheep Health
Protect Your Sheep from Deadly Diseases!
FieldvetDrMSS
1 Enterotoxaemia (ET) Vaccine
Pulpy Kidney Disease
Age: 3 months onward
Booster: Annually (before monsoon)
2 Sheep Pox Vaccine
Sheeppox Virus
Age: 3 months onward
Booster: Once yearly
Important in endemic areas
FMD VACCINE
3 PPR Vaccine
Goat Plague (PPR)
Age: 3-4 months onward
Important in endemic areas
PPR
4 Foot and Mouth Disease (FMD) Vaccine
Foot & Mouth Disease
Age: 4 months onward
Booster: Every 6 months
Govt. supported vaccination in many states
FMD
5 Hemorrhagic Septicemia (HS) Vaccine
Pasteurella multocida
More common in cattle but used in sheep in endemic areas
HS VACCINE
6 Black Quarter (BQ) Vaccine
Clostridium chauvoei
Used where outbreaks occur
Booster: Annual
7 Anthrax Vaccine
Bacillus anthracis
Used only in endemic areas
Booster: Annual
8 Bluetongue Vaccine
Bluetongue Virus
Recommended in high-risk southern & western states
Booster: Annual (before vector season)
BT
Suggested Annual Vaccination Calendar
(India - Semi-arid Region)
Jan-Feb | PPR | PPR (if due)
Feb-Mar | ET (Booster) | ET (Booster)
May-June | HS + BQ | HS + BQ
June-July | FMD | FMD
Before Monsoon | Sheep Pox | Sheep Pox
Vector season (regice specifis) | Bluetongue | Bluetongue
Important Field Guidelines:
Vaccinate Only Healthy Animals
Avoid Vaccinating during High Fever or Stress
Maintain Cold Chain (2-8 ′C)
Deworm 2-3 weeks Before Vaccination (if required):
#Fieldvet

Section 2: Vaccination & Preventive Care

3. Essential Vaccines for Sheep Health

Here is a practical field-oriented list of vaccines commonly used for sheep flocks in India, useful for flock health planning and extension education.

1️⃣ Enterotoxaemia (ET) Vaccine

- Protects against: Clostridium perfringens (Type D)
- Also called: Pulpy Kidney disease
- Age: 3 months onward
- Booster: Annually (before monsoon)

2️⃣ Sheep Pox Vaccine

- Protects against: Sheep pox virus
- Age: 3 months onward
- Booster: Once yearly
- Important in endemic areas

3️⃣ PPR Vaccine

- Protects against: Peste des petits ruminants virus
- Disease: PPR (Goat Plague)
- Age: 3–4 months onward
- Booster: Usually once in 3 years (as per state program)

4️⃣ Foot and Mouth Disease (FMD) Vaccine

- Protects against: Foot-and-mouth disease virus
- Age: 4 months onward
- Booster: Every 6 months
- Government-supported vaccination in many states

5️⃣ Hemorrhagic Septicemia (HS) Vaccine

- Protects against: Pasteurella multocida
- More common in cattle but used in sheep in endemic areas
- Annual vaccination before monsoon

6️⃣ Black Quarter (BQ) Vaccine

- Protects against: Clostridium chauvoei
- Used where outbreaks occur
- Annual vaccination

7 Anthrax Vaccine

- Protects against: Bacillus anthracis
- Used only in endemic areas
- Annual vaccination

8 Bluetongue Vaccine

- Protects against: Bluetongue virus
- Recommended in high-risk southern and western states
- Annual (before vector season)

Suggested Annual Vaccination Calendar (India – Semi-Arid Region):

Month & Vaccine

- Jan–Feb - PPR (if due)
- Feb–Mar - ET (Booster)
- May–June - HS + BQ
- June–July - FMD

Before Monsoon

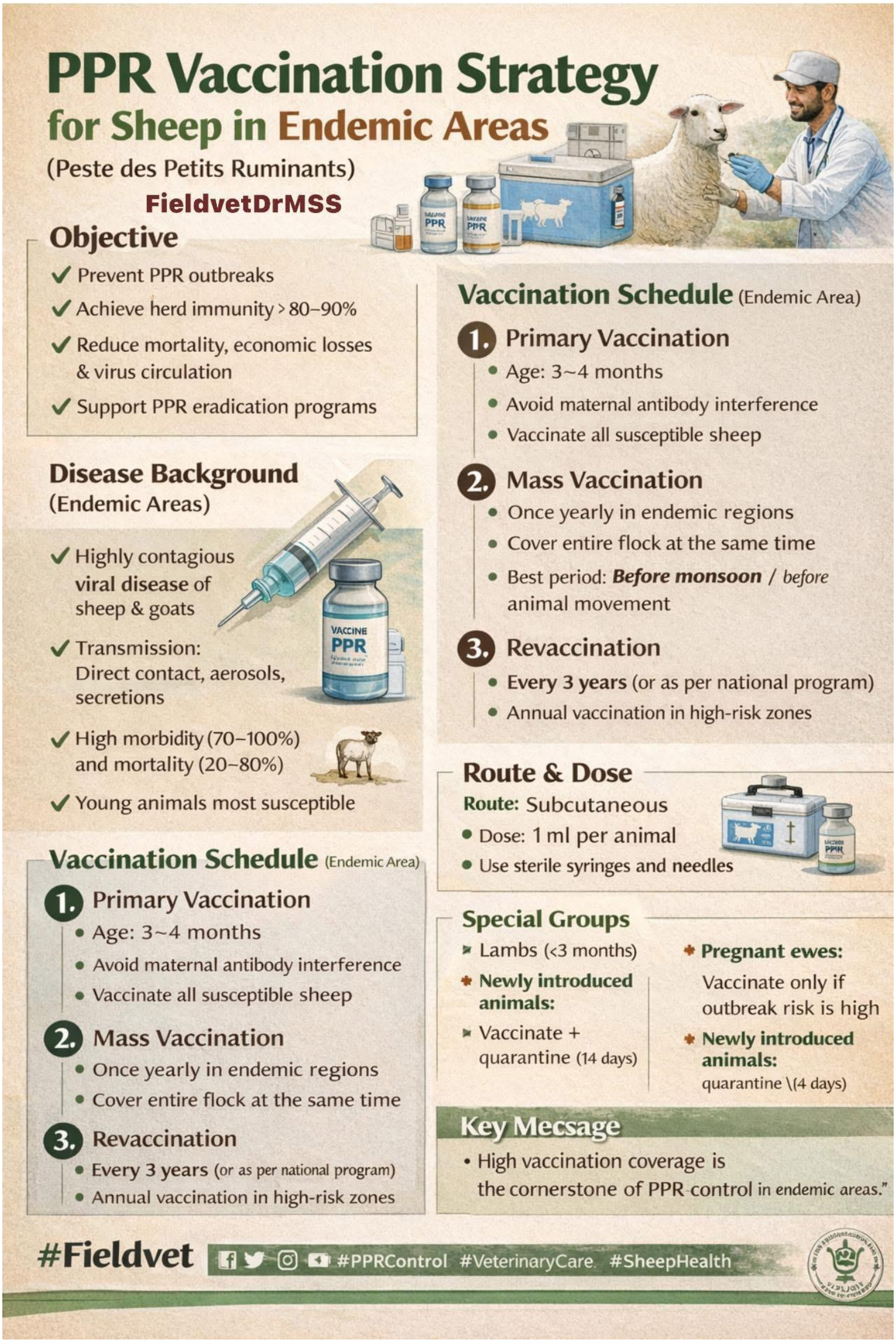
- Sheep Pox
- Vector season (region-specific)
- Bluetongue.

Vaccination Schedule for Lambs:

- 3 months: ET + PPR
- 4 months: FMD
- 6 months: Sheep Pox
- Annual boosters as per protocol

⚠ Important Field Guidelines:

✓ Vaccinate only healthy animals
✓ Avoid vaccinating during high fever or stress
✓ Maintain cold chain (2–8°C)
✓ Deworm 2–3 weeks before vaccination (if required)
✓ Use sterile syringes.


PPR Vaccination Strategy
for Sheep in Endemic Areas
(Peste des Petits Ruminants)
FieldvetDrMSS
PPR
PPR
Objective
✔ Prevent PPR outbreaks
✔ Achieve herd immunity > 80–90%
✔ Reduce mortality, economic losses & virus circulation
✔ Support PPR eradication programs
Disease Background
(Endemic Areas)
✔ Highly contagious viral disease of sheep & goats
✔ Transmission: Direct contact, aerosols, secretions
VACCINE
PPR
✔ High morbidity (70–100%) and mortality (20–80%)
✔ Young animals most susceptible
Vaccination Schedule (Endemic Area)
1. Primary Vaccination
• Age: 3~4 months
• Avoid maternal antibody interference
• Vaccinate all susceptible sheep
2. Mass Vaccination
• Once yearly in endemic regions
• Cover entire flock at the same time
3. Revaccination
• Every 3 years (or as per national program)
• Annual vaccination in high-risk zones
Vaccination Schedule (Endemic Area)
1. Primary Vaccination
• Age: 3~4 months
• Avoid maternal antibody interference
• Vaccinate all susceptible sheep
2. Mass Vaccination
• Once yearly in endemic regions
• Cover entire flock at the same time
• Best period: Before monsoon / before animal movement
3. Revaccination
• Every 3 years (or as per national program)
• Annual vaccination in high-risk zones
Route & Dose
Route: Subcutaneous
• Dose: 1 ml per animal
• Use sterile syringes and needles
PPR
Special Groups
Lambs (<3 months)
Newly introduced animals:
Vaccinate + quarantine (14 days)
Pregnant ewes: Vaccinate only if outbreak risk is high
Newly introduced animals: quarantine \(4 days)
Key Mecsage
• High vaccination coverage is the cornerstone of PPR control in endemic areas."
#Fieldvet
#PPRControl #VeterinaryCare #SheepHealth

4. PPR Vaccination Strategy for Sheep in Endemic Areas

(Peste des Petits Ruminants)

🎯 **Objective:**

- Prevent PPR outbreaks
- Achieve herd immunity ≥ 80–90%
- Reduce mortality, economic losses & virus circulation
- Support PPR eradication programs

🦠 **Disease Background (Endemic Areas):**

- Highly contagious viral disease of sheep & goats
- Transmission: Direct contact, aerosols, secretions
- High morbidity (70–100%) and mortality (20–80%)
- Young animals most susceptible

💉 **Vaccine Used:**

- Live attenuated PPR vaccine
- Thermostable (preferred for field use)
- Provides long-term immunity (3–4 years)
- Safe in sheep & goats

📅 **Vaccination Schedule (Endemic Area):**

1️⃣Primary Vaccination

- Age: 3–4 months
- Avoid maternal antibody interference
- Vaccinate all susceptible sheep

2️⃣Mass Vaccination

- Once yearly in endemic regions
- Cover entire flock at the same time
- Best period: Before monsoon / before animal movement

3️⃣Revaccination

- Every 3 years (or as per national program)
- Annual vaccination in high-risk zones

🐑 **Special Groups:**

- Lambs (<3 months): Delay vaccination
- Pregnant ewes: Vaccinate only if outbreak risk is high
- Newly introduced animals: Vaccinate + quarantine (14 days)

▯ Cold Chain & Handling:

- Maintain 2–8 °C
- Do not freeze vaccine
- Use reconstituted vaccine within 2 hours
- Protect from direct sunlight

💉 Route & Dose:

- Route: Subcutaneous
- Dose: 1 ml per animal
- Use sterile syringes and needles

🚫 Do Not Vaccinate:

- Sick or febrile animals
- Severely stressed sheep
- During active outbreak (unless ring vaccination is advised)

🌍 Integrated Control Measures:

- Restrict animal movement
- Quarantine new stock
- Improve nutrition & biosecurity
- Early disease reporting
- Proper carcass disposal

📊 Monitoring & Records:

- Maintain vaccination records
- Identify vaccinated animals (ear tag/marking)
- Monitor post-vaccination immunity
- Report adverse reactions (rare)

🏁 Key Message:

"High vaccination coverage is the cornerstone of PPR control in endemic areas."


Vaccination Side Effects in SHEEP
FieldvetDrMSS
Common Reactions
Swelling at injection site
Mild fever
Reduced feed intake
Moderate Reactions
Large swelling
Lameness
Facial swelling
Severe Reactions
Anaphylaxis (Severe Allergy)
Sudden collapse
Breathing distress
Frothing
Immediate Treatment:
Adrenaline Injection
Antihistamines
Supportive Fluids
Abortion Risk
Avoid in early pregnancy
Precautionary Measures
Vaccinate only healthy sheep
Keep vaccines cold (2-8°C)
Use sterile needles
Keep emergency medicins ready
Monitoring After Vaccination
Observe animals for 30 minutes
Recheck flock after 24 hours
Record adverse reactions
FieldvetDrMSS

5. Vaccination side effects in Sheep flock

Vaccination in sheep flocks is essential for preventing major diseases, but mild to severe adverse reactions can occasionally occur. As a field veterinarian, understanding these reactions helps in planning safer flock-level immunization programs.

1️⃣ Common (Mild & Expected) Side Effects:

These are usually self-limiting and resolve within 24–72 hours:

◈ Local Reactions

- Swelling at injection site
- Warmth and mild pain
- Small granuloma or sterile abscess (especially with oily adjuvant vaccines)

◈ Systemic Reactions

- Mild fever (0.5–1.5°C rise)
- Lethargy
- Reduced feed intake for 1–2 days
- Temporary drop in milk yield (in lactating ewes)

2️⃣ Moderate Reactions:

- Large injection-site swelling (>5 cm)
- Persistent abscess formation
- Stiffness or lameness (if injected incorrectly)
- Mild hypersensitivity (urticaria, facial swelling)

These may require NSAIDs or antihistamines.

3️⃣ Severe (Rare but Critical) Reactions:

⚠ Anaphylaxis

- Occurs within minutes

Signs:

- Sudden collapse
- Dyspnea
- Frothing
- Pale mucous membranes
- Convulsions

Immediate Treatment:

- Adrenaline (Epinephrine) IM or IV
- Antihistamines

- Corticosteroids
- Supportive fluids

4️⃣ Abortion Risk:

Some vaccines (especially live vaccines) may cause abortion if:

- Given during early pregnancy
- Cold chain was compromised
- Animals were immunocompromised

Always verify gestational stage before vaccination.

5️⃣ Vaccine-Specific Reactions (Common in Sheep):

- Enterotoxaemia vaccine – Local swelling common
- PPR vaccine – Mild fever in some animals
- Sheep pox vaccine – Local pox lesion at injection site
- Brucella Rev-1 – May cause abortion if given to pregnant ewes

6️⃣ Risk Factors in Flocks

- Poor cold chain maintenance
- Vaccinating stressed animals (transport, heat stress, parasitism)
- Incorrect injection site (IM vs SC)
- Reusing needles
- Overdosing
- Mixing vaccines improperly

7️⃣ Preventive Measures for Flock Safety

✓ Vaccinate only healthy animals
✓ Avoid extreme heat hours (important in Indian field conditions)
✓ Maintain cold chain (2–8°C)
✓ Use sterile needles and change frequently
✓ Follow correct route (usually SC in sheep)
✓ Keep adrenaline ready during mass vaccination
✓ Avoid vaccinating heavily pregnant ewes unless recommended

8️⃣ Field Monitoring Protocol (Recommended)

For flock vaccination programs:

- Observe animals for 30 minutes post-vaccination
- Recheck flock after 24 hours
- Record batch number and manufacturer
- Maintain adverse event log


VACCINATION SCHEDULE FOR SHEEP –
ASIA REGION
(General guideline – Adapt to local disease prevalence & govt programmes)
FieldvetDrMSS
Lamb Vaccination Schedule
Age
Vaccine
Disease Protected
Birth – 7 days
Tetanus Antitoxin
Tetanus (if dam not vaccinated)
4–6 weeks
Enterotoxemia + Tetanus (1st Dose)
Pulpy Kidney, Tetanus
8–10 weeks
Enterotoxemia + Tetanus (Booster)
Clostridial Diseases
3 months
PPR Vaccine
Peste des Petits Ruminants
3-4 months
FMD Vaccine
Foot and Mouth Disease
3-4 months
Sheep Pox Vaccine
Sheep pox
Risk Areas
Orf Vaccine
Contagious Ecthyma
Adult Sheep Vaccination Schedule
• Enterotoxemia + Tetanus Once Yearly
• FMD Every 6 Months
• PPR Once Yearly*
• Sheep Pox Endemic Areas
• Anthrax Before Monsoon (Risk Zones)
• Orf As per Outbreak Risk
Pregnant Ewes (Pre-Lambing)
• Clostridial Booster 4–6 weeks before lambing
• Ensures Strong Colostral Immunity
Core Vaccines (Asia Region)
• PPR
• FMD
• Enterotoxemia
• Tetanus
• Sheep Pox (Endemic Areas)
Risk-Based Vaccines
• Anthrax
• Orf
• Bluetongue
• Enzootic Abortion
Important Guidelines
Vaccinate Healthy Animals
Keep Vaccines Cold (2–8°C)
No Deworming on Vaccination Day
Record Date & Batch No.
Asia-Specific Note
• Mass PPR & FMD Campaigns
• Consult Local Veterinarians
#FieldVet • Prevent Disease • Improve Productivity • Protect Livelihoods

6. Vaccination Schedule - Asia

(Field-Level Preventive Health Calendar)

Vaccination is a key component of flock health management in India. A proper schedule protects sheep from major infectious diseases that cause high mortality and economic loss.

Recommended Vaccination Schedule

Age / Time	**Disease**	**Vaccination Schedule**
1–2 months	Enterotoxaemia (Pulpy Kidney)	First vaccination
1–2 months	PPR (Peste des Petits Ruminants)	Single dose
2–3 months	Sheep Pox	First vaccination
3 months	Foot and Mouth Disease (FMD)	First dose
3–4 months	Anthrax	Vaccinate in endemic areas
3–4 months	Black Quarter (BQ)	Vaccinate in endemic areas
Pregnant ewes (1 month before lambing)	Enterotoxaemia	Booster to protect lambs
Every 6 months	FMD	Booster
Every year	Enterotoxaemia	Annual booster
Every year	Sheep Pox	Annual vaccination
Every 3 years	PPR	Revaccination
Every year	Anthrax	In endemic areas

Major Diseases Covered

1️⃣ Peste des Petits Ruminants (PPR)

- Highly contagious viral disease
- Causes high mortality in sheep and goats
- National control programme implemented in India

2️⃣ Enterotoxaemia (Pulpy Kidney)

- Caused by Clostridium perfringens
- Sudden death in well-fed lambs

3️⃣ Sheep Pox

- Viral disease-causing skin lesions and fever
- Leads to severe production loss

4️⃣ Foot and Mouth Disease (FMD)

- Highly contagious disease affecting cloven-hoofed animals
- Causes lameness, mouth lesions, and production loss

5️⃣ Anthrax

- Zoonotic disease affecting livestock and humans
- Vaccination recommended in endemic regions

6️⃣ Black Quarter (Blackleg)

- Bacterial disease affecting muscles
- Occurs mainly in young animals grazing contaminated soil

⚠ Important Vaccination Guidelines

✓ Vaccinate healthy animals only
✓ Maintain vaccine cold chain (2–8°C)
✓ Avoid vaccination during severe stress or extreme heat
✓ Use sterile needles and proper dosage
✓ Maintain vaccination records

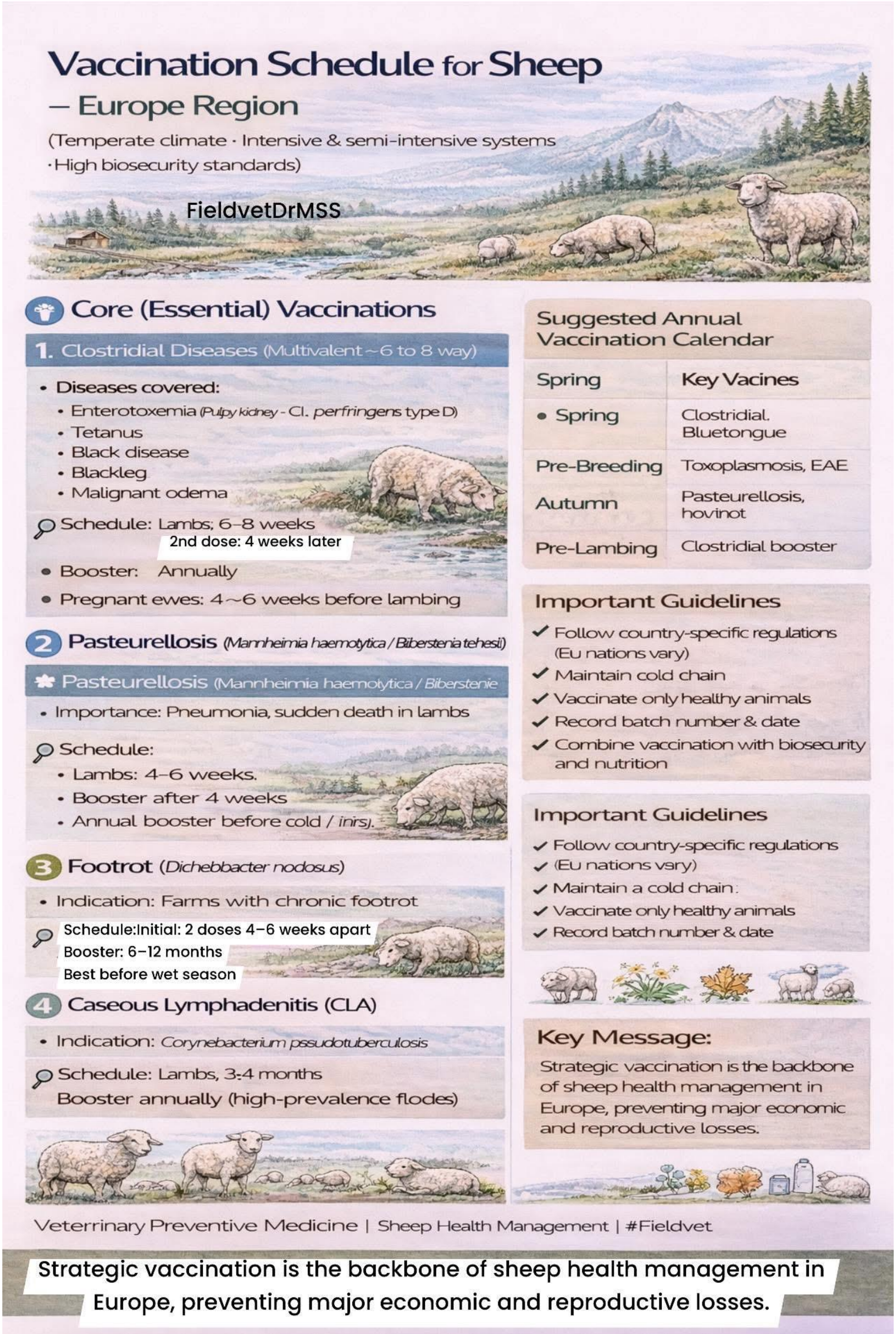
🌱 Significance for Sheep Farmers

A proper vaccination program helps to:

- Reduce mortality in sheep flocks
- Improve meat production and flock productivity
- Prevent major disease outbreaks
- Increase farm profitability

✅ Key Message

"Prevention through vaccination is the most economical and effective strategy in sheep flock health management."


Vaccination Schedule for Sheep
– Europe Region
(Temperate climate · Intensive & semi-intensive systems
· High biosecurity standards)
FieldvetDrMSS
Core (Essential) Vaccinations
1. Clostridial Diseases (Multivalent ~6 to 8 way)
• Diseases covered:
• Enterotoxemia (Pulpy kidney - Cl. perfringens type D)
• Tetanus
• Black disease
• Blackleg
• Malignant odema
Schedule: Lambs; 6–8 weeks
2nd dose: 4 weeks later
• Booster: Annually
• Pregnant ewes: 4~6 weeks before lambing
2 Pasteurellosis (Mannheimia haemotytica / Bibersteria tehesii)
Pasteurellosis (Mannheimia haemolytica / Biberstenie
• Importance: Pneumonia, sudden death in lambs
Schedule:
• Lambs: 4–6 weeks.
• Booster after 4 weeks
• Annual booster before cold / inrs).
3 Footrot (Dichebbacter nodosus)
• Indication: Farms with chronic footrot
Schedule:Initial: 2 doses 4–6 weeks apart
Booster: 6–12 months
Best before wet season
4 Caseous Lymphadenitis (CLA)
• Indication: Corynebacterium pssudotuberculosis
Schedule: Lambs, 3-4 months
Booster annually (high-prevalence flodes)
Suggested Annual Vaccination Calendar
Spring | Key Vacines
• Spring | Clostridial. Bluetongue
Pre-Breeding | Toxoplasmosis, EAE
Autumn | Pasteurellosis, hovinot
Pre-Lambing | Clostridial booster
Important Guidelines
✓ Follow country-specific regulations (Eu nations vary)
✓ Maintain cold chain
✓ Vaccinate only healthy animals
✓ Record batch number & date
✓ Combine vaccination with biosecurity and nutrition
Important Guidelines
✓ Follow country-specific regulations
✓ (Eu nations vary)
✓ Maintain a cold chain:
✓ Vaccinate only healthy animals
✓ Record batch number & date
Key Message:
Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.
Veterrinary Preventive Medicine | Sheep Health Management | #Fieldvet
Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.

7. Vaccination Schedule – Europe Region

(Temperate climate • Intensive & semi-intensive systems • High biosecurity standards)

🐑 **Core (Essential) Vaccinations:**

(Recommended for all sheep flocks across Europe)

1. Clostridial Diseases (Multivalent – 6 to 8 way)

Diseases covered:

- Enterotoxemia (Pulpy kidney – Cl. perfringens type D)
- Tetanus
- Black disease
- Blackleg
- Malignant oedema

Schedule:

- Lambs:
- 1st dose: 6–8 weeks
- 2nd dose: 4 weeks later
- Booster: Annually
- Pregnant ewes: 4–6 weeks before lambing

🐑 **2. Pasteurellosis (Mannheimia haemolytica / Bibersteinia trehalosi):**

Importance: Pneumonia, sudden death in lambs

Schedule:

- Lambs:
- 1st dose: 4–6 weeks
- Booster after 4 weeks
- Annual booster before cold / housing season

🦠 Region-Specific / Risk-Based Vaccinations:

3. Bluetongue (Serotype specific – EU regulated)

Vector: Culicoides midges

Status: Mandatory in endemic or outbreak zones

Schedule:

- As per national veterinary authority
- Usually annual before vector season (spring)

4. Footrot (Dichelobacter nodosus):

Indication: Farms with chronic footrot

Schedule:

- Initial: 2 doses 4–6 weeks apart
- Booster: 6–12 months
- Best before wet season

5. Caseous Lymphadenitis (CLA):

Cause: Corynebacterium pseudotuberculosis

Schedule:

- Lambs: 3–4 months
- Booster annually (high-prevalence flocks)

6. Enzootic Abortion of Ewes (EAE):

Cause: Chlamydia abortus

Schedule:

- Replacement ewe lambs: Before first mating
- Single dose (long-term immunity)

7. Toxoplasmosis:

Risk: Abortion storms

Schedule:

- Ewe lambs: At least 3 weeks before mating
- Single lifetime vaccination.

Optional / Special Situations:

- Orf (Contagious ecthyma)
- Only in infected flocks
- Q Fever
- Zoonotic risk farms
- Louping ill
- Tick-endemic upland regions.

Suggested Annual Vaccination Calendar:

Season & Key Vaccines

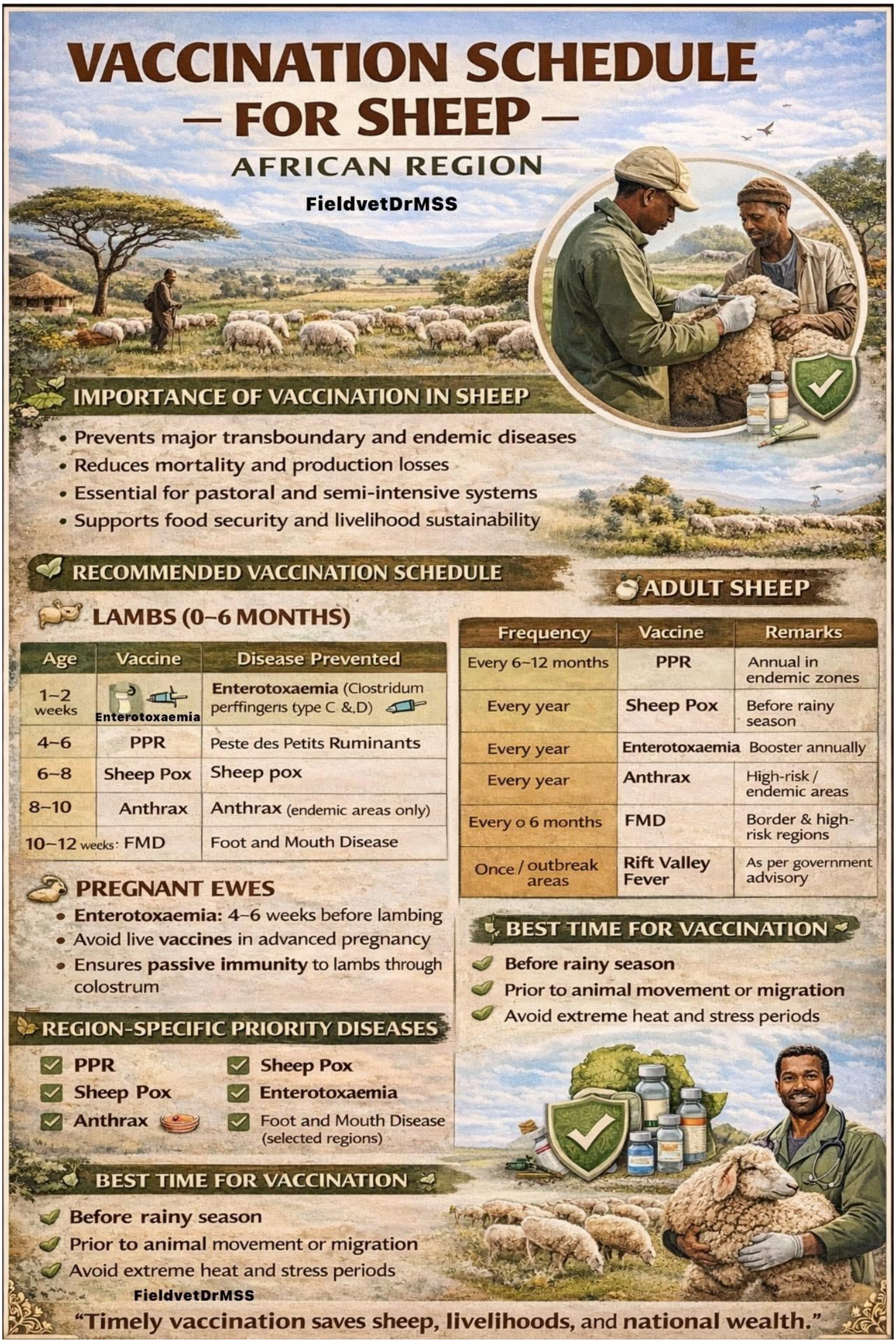
- Spring —Clostridial, Bluetongue
- Pre-Breeding —Toxoplasmosis, EAE
- Autumn —Pasteurellosis, Footrot
- Pre-Lambing—Clostridial booster.

Important Guidelines

- Follow country-specific regulations (EU nations vary)
- Maintain cold chain
- Vaccinate only healthy animals
- Record batch number & date
- Combine vaccination with biosecurity and nutrition

Key Message:

Strategic vaccination is the backbone of sheep health management in Europe, preventing major economic and reproductive losses.


VACCINATION SCHEDULE
– FOR SHEEP –
AFRICAN REGION
FieldvetDrMSS
IMPORTANCE OF VACCINATION IN SHEEP
• Prevents major transboundary and endemic diseases
• Reduces mortality and production losses
• Essential for pastoral and semi-intensive systems
• Supports food security and livelihood sustainability
RECOMMENDED VACCINATION SCHEDULE
LAMBS (0–6 MONTHS)
Age | Vaccine | Disease Prevented
1–2 weeks | Enterotoxaemia | Enterotoxaemia (Clostridium perffingens type C &,D)
4–6 | PPR | Peste des Petits Ruminants
6–8 | Sheep Pox | Sheep pox
8–10 | Anthrax | Anthrax (endemic areas only)
10–12 weeks: FMD | Foot and Mouth Disease
ADULT SHEEP
Frequency | Vaccine | Remarks
Every 6–12 months | PPR | Annual in endemic zones
Every year | Sheep Pox | Before rainy season
Every year | Enterotoxaemia | Booster annually
Every year | Anthrax | High-risk / endemic areas
Every o 6 months | FMD | Border & high-risk regions
Once / outbreak areas | Rift Valley Fever | As per government advisory
PREGNANT EWES
• Enterotoxaemia: 4–6 weeks before lambing
• Avoid live vaccines in advanced pregnancy
• Ensures passive immunity to lambs through colostrum
BEST TIME FOR VACCINATION
Before rainy season
Prior to animal movement or migration
Avoid extreme heat and stress periods
REGION-SPECIFIC PRIORITY DISEASES
PPR
Sheep Pox
Anthrax
Sheep Pox
Enterotoxaemia
Foot and Mouth Disease (selected regions)
BEST TIME FOR VACCINATION
Before rainy season
Prior to animal movement or migration
Avoid extreme heat and stress periods
FieldvetDrMSS
"Timely vaccination saves sheep, livelihoods, and national wealth."

8. Vaccination Schedule – African Region

(General Field Guideline for Flock Health Protection)

Vaccination is essential in African sheep production systems to prevent highly contagious and economically important diseases. The schedule may vary slightly by country depending on disease prevalence and veterinary regulations

Recommended Vaccination Schedule

Age / Time	**Disease**	**Vaccine Schedule**
At 1–2 weeks of age	Enterotoxaemia (Pulpy Kidney)	First vaccination
4–6 weeks	Enterotoxaemia booster	Booster dose
2–3 months	Sheep Pox	Single dose
3 months	Peste des Petits Ruminants (PPR)	Single vaccination
3–4 months	Anthrax	Annual vaccination in endemic areas
3–4 months	Black Quarter (Blackleg)	First dose
6 months	Brucellosis (female lambs only)	Single dose
Every year	PPR	Annual vaccination
Every year	Enterotoxaemia	Annual booster
Every year	Sheep Pox	Annual vaccination in risk areas
Every year	Anthrax	Annual vaccination in endemic zones

Important Diseases Covered

1 Peste des Petits Ruminants (PPR)

- Highly contagious viral disease
- Causes high mortality in sheep and goats
- Vaccination is very important across Africa

2 Enterotoxaemia (Pulpy Kidney Disease)

- Caused by Clostridium perfringens
- Often occurs in well-fed or fast-growing lambs

3 Sheep Pox

- Viral disease-causing skin lesions and severe production losses

4 Anthrax

- Zoonotic bacterial disease
- Occurs in endemic grazing areas

5 Black Quarter (Blackleg)

- Caused by Clostridium chauvoei
- Usually affects young animals

⚠ **Vaccination Management Tips**

✓ Vaccinate only healthy animals
✓ Maintain cold chain (2–8°C) for vaccines
✓ Avoid vaccination during severe stress or transport
✓ Follow booster schedules strictly
✓ Maintain proper vaccination records

Significance for Sheep Farmers

Proper vaccination helps to:

- Reduce mortality in flocks
- Improve meat and wool production
- Prevent disease outbreaks
- Protect farmer livelihood

✅ **Key Message**

A well-planned vaccination program is the backbone of profitable sheep farming and flock health management.


Essential Vaccines
for African Sheep Farming
Protect Your Sheep Flock from Deadly Diseases!
FieldvetDrMSS
1 Clostridial Diseases
Polyvalent vaccines (5-in-1, 7-in-1, 8-in-1)
Protect against:
• Enterotoxemia (Pulpy kidney)
• Tetanus
• Blackleg
• Black Disease
• Malignant Edema
2 Peste des Petits Ruminants (PPR)
Peste des petits ruminants virus
Highly endemic across Africa
Common in East & North Africa
3 Sheep Pox
Sheeppox virus
Common in East & North Africa
4 Rift Valley Fever (RVF)
Rift Valley fever virus
Periodic outbreaks (heawy raintall years)
Important zoonotic impact
5 Brucellosis
Brucella melitensis
Vaccine: Rev.1 strain
Used in control programs
6 Pasteurellosis / Mannheimiosis
Mannheimia haemolytica
Pasteurella multocida
Important in intensive & feedlot systems
7 Contagious Caprine Pleuropneumonia
Mycoplasmo capricolum
subspecies capriprieumomae.
Primarily Goat disease, but relevant in mixed flocks
8 Anthrax
Bacillus anthracis
Endemic zones
Annual vaccination
high-risk districts
Regional Vaccine Focus
East Africa PPR, Sheep pox, RVF, Clostridial
West Africa PPR, Clostridial, Brucellosis
Southern Africa Clostridial, Anthrax Pasteurellosis,
North Africa PPR, Sheep pox, Brucellosis
FIELD GUIDELINES
✓ Vaccinate Lambs at 6–8 Weeks: Clostridlal Vacine
✓ Annual Booster for All Adults
✓ PPR Vaccine to Control Mortality in Endemic Areas
✓ High-Risk Areas: RVE Vaccine Before Rainy Season
✓ Follow National Disease Control Programs
#Fieldvet

9. Core Vaccines Used in African Sheep Production

1️⃣ Clostridial Diseases (Core Vaccine – Widely Used):

Polyvalent vaccines (5-in-1, 7-in-1, 8-in-1)

Protect against:

- Clostridium perfringens (Enterotoxemia / Pulpy kidney)
- Clostridium tetani (Tetanus)
- Clostridium chauvoei (Blackleg)
- Clostridium novyi (Black disease)
- Clostridium septicum (Malignant edema)

📌 **Most essential vaccine in African small ruminant systems**

2️⃣ Peste des Petits Ruminants (PPR):

Disease: Peste des Petits Ruminants

Agent: Peste des petits ruminants virus

- Highly endemic across Sub-Saharan Africa
- Mass vaccination campaigns supported by AU-IBAR & FAO
- Critical for mortality control

3️⃣ Sheep Pox:

Disease: Sheep pox

Agent: Sheeppox virus

- Common in East and North Africa
- Used in endemic regions

4️⃣ Rift Valley Fever (RVF):

Disease: Rift Valley fever

Agent: Rift Valley fever virus

- Periodic outbreaks (heavy rainfall years)
- Vaccination in high-risk zones
- Important zoonotic significance

5️⃣ Brucellosis:

Disease: Brucellosis

Agent: Brucella melitensis

- Vaccine: Rev.1 strain
- Used in organized control programs

- Major public health relevance

6️⃣ Pasteurellosis / Mannheimiosis:

Agents:

- Mannheimia haemolytica
- Pasteurella multocida
- Important in intensive & feedlot systems
- Often seasonal (stress-associated)

7️⃣ Contagious Caprine Pleuropneumonia (where mixed flocks exist):

Agent: Mycoplasma capricolum

Primarily goat disease, but relevant in mixed sheep-goat systems.

8️⃣ Anthrax (Endemic Zones):

Disease: Anthrax

Agent: Bacillus anthracis

- Used in endemic pastoral regions
- Annual vaccination in high-risk districts

9️⃣ Foot-and-Mouth Disease (Selective Use):

Disease: Foot-and-mouth disease

Agent: Foot-and-mouth disease virus

- Sheep are minor hosts
- Used mainly in national control programs

Region & Priority Vaccines:

- East Africa - PPR, Sheep pox, RVF, Clostridial
- West Africa - PPR, Clostridial, Brucellosis
- Southern Africa - Clostridial, Pasteurellosis, Anthrax
- North Africa - PPR, Sheep pox, Brucellosis.

Typical Vaccination Schedule (General Field Guideline)

- Lambs: Clostridial at 6–8 weeks + booster
- Pre-breeding ewes: PPR, Brucellosis (as per policy)
- Annual: Clostridial booster
- High-risk seasons: RVF (before rainy season)


SHEEP VACCINATION
SCHEDULE - NZ & AUSTRALIA
FieldvetDrMSS
NEW ZEALAND
Major products commonly used include:
✓ Multine' 5-in-1 ✓ Covexin 10
✓ Nilvax Selenised ✓ Phenax Classic
1 Ewes (Breeding Females)
Pre-Lambing (4–6 weeks before lambing)
✓ 5-in-1 or 10-in-1 booster
Ensures strong colostral antibody transfer
Typical protocol:
• If previously vaccinated: Single annual booster
• If unvaccinated:
2 doses, 4–6 weeks apart (before first lambing)
2 Rams
Annual 5-in-1 or 10-in-1 booster
✓ Ideally 4 weeks before joining
3 Lambs
At Docking / Marking (4–8 weeks old)
✓ First 5-in-1 dose
✓ 4–6 Weeks Later
Second 5-in-1 dose
(critical for long-term immunity
✓ Annual Booster
Optional (Risk-Based)
✓ Orf (Scabby mouth) –
Use Phenax Classic if disease present
✓ Selenium-containing vaccines in deficient regions
AUSTRALIA
Major vaccines include:
✓ Ultravac' 5-in-1 ✓ Glanvac 6
✓ Websters 6-in-1 ✓ Gudair
1 Ewes
4–6 Weeks Before Lambing
✓ Booster 5-in-1 or 6-in-1 annually
Ensures:
✓ Protection against pulpy kidney
✓ Tetanus
✓ Black disease
✓ Malignant oedema
✓ Blackleg
2 Rams
Annual booster before breeding season
Lambs
✓ Marking (6–8 - weeks)
✓ First 5-in-1 / 6-in-1
✓ 4 Weeks Later
✓ Second dose (mandatory)
✓ Annual Booster
Additional Vaccines
NZ | NZ | Australia
5-in-1 | 5-in-1 | 5-in-1 / 6-in-1
2 doses | 2 doses | Annual, pre-lamb
Clostridial vaccines | Common | Common in endemic areas
Orf Based | Risk-based | Risk based
Healthy Flocks Start with Smart Vaccination.
FieldvetDrMSS

10. Vaccines in New Zealand and Australia

🐑 **Sheep Vaccines – New Zealand**

Clostridial disease vaccines

These protect against the major clostridial bacteria that cause tetanus, pulpy kidney (enterotoxaemia), blackleg, malignant oedema and black disease.

- Multine® 5-in-1 – Clostridial vaccine protecting against the five common clostridial diseases.
- Nilvax® Selenised – Specialist prelamb clostridial vaccine with selenium and immune stimulant.
- Covexin® 10 – Expanded 10-in-1 clostridial vaccine for broader protection (commonly used on farms with high risk).
- Lamb Vaccine (selenised or plain) – Short-term tetanus and some pulpy kidney protection for lambs, often used at docking/tailing.

Other sheep vaccines

- Phenax® Classic – Vaccine against scabby mouth (contagious ecthyma / orf) in sheep and goats.

(Note: Leptospirosis vaccines like Leptavoid-2® and Leptoshield® are registered for sheep in NZ, but these are generally cattle formulations also used off–label under vet guidance.)

🐑 **Sheep Vaccines – Australia:**

Clostridial disease vaccines

Australian sheep flocks use similar clostridial protective vaccines, often with 5- or 6-in-1 formulations:

Common products include:

- Ultravac® 5-in-1 – Standard clostridial disease vaccine.
- Glanvac® 6 B12 / Glanvac® 6S – Six-in-one vaccines with added protection and vitamin B12 variants.
- Websters® LV 5-in-1 / LV 6-in-1 – Multivalent clostridial vaccines available in different formulations.
- Guardian 6-in-1 – Clostridial vaccine offering protection against multiple clostridial agents.
- Tasvax 5-in-1 / 6-in-1 – Clostridial protection options from Cooper’s Animal Health.

Combined vaccines

• Cydectin® Eweguard products – Combined vaccination and worm control vaccines (sheep vaccine with integrated parasite protection).

Other important sheep vaccines

• Gudair® (OJD vaccine) – Controls Ovine Johne's Disease (OJD). Single dose typically for life (or long-term immune response).

• Scabby mouth / contagious ecthyma vaccine – Commonly used where scabby mouth is present or a local risk (available through veterinary suppliers).

💉 **General Notes on Use:**

Vaccination Strategy

• Clostridial vaccines are typically given in two doses 4–6 weeks apart initially, with annual boosters.

• Vaccination should be timed relative to lambing, weaning, and stress periods for best maternal antibody transfer and lamb protection.

• OJD vaccination (e.g., Gudair) has specific protocols and handling precautions due to risk of human self-injection injury; consult a veterinarian.

Veterinary Advice:

This list is illustrative; actual vaccine choices, timing, and protocols should be decided with a local veterinarian or sheep health advisor, as registration and usage guidance vary between states/territories in Australia and regions in New Zealand.

New Zealand:

Major products commonly used include:

- Multine 5-in-1
- Covexin 10
- Nilvax Selenised
- Phenax Classic

🐑 **Ewes (Breeding Females)**

Pre-Lambing (4–6 weeks before lambing)

✓ 5-in-1 or 10-in-1 booster

→ Ensures strong colostral antibody transfer

Typical protocol:

- If previously vaccinated: Single annual booster
- If unvaccinated: 2 doses, 4–6 weeks apart (before first lambing)

🐏 2️⃣Rams

- ✓ Annual 5-in-1 or 10-in-1 booster
- ✓ Ideally 4 weeks before joining

🐏 3️⃣Lambs

At Docking / Marking (4–8 weeks old)

- ✓ First 5-in-1 dose

4–6 Weeks Later

- ✓ Second 5-in-1 dose (critical for long-term immunity)

Annual Booster

- ✓ At 12 months of age

🦠 Optional (Risk-Based)

- ✓ Orf (Scabby mouth) – Use Phenax Classic if disease present
- ✓ Selenium-containing vaccines in deficient regions

AU Sheep Vaccination Schedule – Australia

Common vaccines include:

- Ultravac 5-in-1
- Glanvac 6
- Websters 6-in-1
- Gudair

🐏 1️⃣Ewes

4–6 Weeks Before Lambing

- ✓ Booster 5-in-1 or 6-in-1 annually

Ensures:

- Protection against pulpy kidney
- Tetanus
- Black disease
- Malignant oedema
- Blackleg

🐏 2️⃣Rams

- ✓ Annual booster before breeding season

🐏 3️⃣Lambs

Marking (6–8 weeks)

- ✓ First 5-in-1 / 6-in-1

4 Weeks Later

- ✓ Second dose (mandatory)

Annual Booster

- ✓ At 12 months

💊 **Additional Australian Vaccines:**

Ovine Johne's Disease (OJD)

- ✓ Gudair – Single lifetime dose (usually at marking age)

⚠ **Strict administration protocol due to operator safety concerns.**

Scabby Mouth (Orf)

- ✓ Live vaccine where disease risk exists

🧠 **Key Veterinary Considerations**

- Always complete the primary two-dose course
- Maintain cold chain (2–8°C)
- Use sterile technique
- Observe meat & wool withholding periods
- Align vaccination with parasite control & reproductive management

Anthelmintic Resistance in Sheep Flocks in Asia
FieldvetDrMSS
Anthelmintic Resistance (AR)
Anthelmintic Resistance is the ability of parasitic worms to survive doses of deworming drugs that were previously effective.
It mainly affects gastro-intestinal nematodes (GINs) in sheep.
Rapidly developing resistance to many Drugs
Situation in Asia
India
China
India
Srilanka
Southeast Asia
Malaysia & Southeast Asia
Major Parasites Showing Resistance –
Common resistant gastrointestinal nematodes in sheep:
Haemonchus contortus
China
India
Anthelmintic Drugs Classes Affected
1 Benzimidazoles
• Albendazole
• Fenbendazole
2 Imidazothiazoles
3 Macrocyclic lactones
• Ivermectin
• Ivermectin
• Moxidectin
4 Salicylanilides
Closantel, (still effective in some areas)
Several Asian countries have reported resistance to one or more major deworming Classes
Major Parasites Showing Resistance –
Haemonchus contortus
Trichostrongylus spp.
Oesophagostomum spp.
Teladorsagia spp.
These parasites, cause: Severe anemia, diarrhoea, Weight loss, Reduced wool and meat production High lamb mortality.
Causes of Anthelmintic Resistance –
✓ Frequent deworming without diagnosis.
✓ Using the dewormer repeatedly
✓ Using the same dewormer repeatedly
✓ Lack of pasture flock management
✓ Treating the entire flock frequently
Detection Methods –
✓ Faecal Egg Count Reduction Test (FECRT)
✓ Egg Hatch Assay (EHA)
✓ Larval Development Test
✓ Molecular detection of resistance genes.
Strategies to Control Anthelmintic –
1 Targeted Selective Treatment (TST)
Treat only heavily infected animals.
2 Rotational use of dewormers.
3 Pasture management
• Rotational grazing
• Mixed grating
Sustainable Parasite Control –
"Responsible Deworming Today Prevents Drug Failure Tomorrow!"
#FieldVet

Section 3: Parasitology & Resistance

11. Anthelmintic Resistance in Asia

🐑 **What is Anthelmintic Resistance (AR)?**

Anthelmintic Resistance is the ability of parasitic worms to survive doses of deworming drugs that were previously effective. It mainly affects gastro-intestinal nematodes (GINs) in sheep. The most important parasite involved is Haemonchus contortus (Barber's pole worm), which rapidly develops resistance to many drugs.

🌏 **Situation in Asia:**

Anthelmintic resistance is increasingly reported across many Asian countries.

India: Benzimidazole resistance in sheep nematodes reported in many flocks; resistance prevalence may exceed 70% in some regions.

• **China**: Sheep parasites show resistance to albendazole and ivermectin, with up to 85.7% of H. contortus populations resistant to ivermectin in some surveys.

• **Bangladesh**: Multiple drug resistance detected against albendazole, levamisole and ivermectin in sheep farms.

• **Malaysia & Southeast Asia**: High resistance to benzimidazoles, affecting about 50% of sheep farms.

Overall, several Asian countries have reported resistance to one or more major dewormer classes.

🦠 **Major Parasites Showing Resistance**:

Common resistant gastrointestinal nematodes in sheep:

- Haemonchus contortus
- Trichostrongylus spp.
- Oesophagostomum spp.
- Teladorsagia spp.

These parasites cause:

- Severe anemia
- Diarrhoea
- Weight loss
- Reduced wool and meat production
- High lamb mortality

💊 **Anthelmintic Drug Classes Affected:**

Resistance has been reported against major drug groups:

1. Benzimidazoles

- Albendazole
- Fenbendazole

2. Imidazothiazoles

- Levamisole

3. Macrocyclic lactones

- Ivermectin
- Moxidectin

4. Salicylanilides

- Closantel (still effective in some areas)

Parasites in Asia have developed resistance to multiple drug families simultaneously in some regions.

⚠ **Causes of Anthelmintic Resistance:**

- Frequent deworming without diagnosis
- Underdosing of drugs
- Using the same dewormer repeatedly
- Treating the entire flock frequently
- Lack of pasture management
- Poor veterinary supervision

Repeated exposure to drugs selects resistant parasites, allowing them to dominate the population.

🔬 **Detection Methods:**

Veterinarians use several methods to detect resistance:

- Faecal Egg Count Reduction Test (FECRT)
- Egg Hatch Assay (EHA)
- Larval Development Test
- Molecular detection of resistance genes

🛑 **Strategies to Control Anthelmintic Resistance:**

1️⃣ Targeted Selective Treatment (TST)

Treat only heavily infected animals.

2️⃣ Rotational use of dewormers

Use different drug classes strategically.

3️⃣ Pasture management

- Rotational grazing
- Mixed grazing with cattle

4️⃣ FAMACHA system

Identify sheep with anemia caused by Haemonchus.

5️⃣ Integrated parasite management

- Nutritional supplementation
- Biological control (fungi)
- Breeding resistant sheep lines

🌱 **Sustainable Parasite Control:**

Future strategies include:

- Vaccine development
- Genetic resistance breeding
- Precision parasite monitoring
- Reduced dependence on chemical anthelmintics

📢 **Key Message:**

“Responsible Deworming Today Prevents Drug Failure Tomorrow.”


Anthelmintic Resistance in Sheep Flocks in Africa
What is Anthelmintic Resistance (AR)?
Anthelmintic Resistance is the ability of parasitic worms to survive deworming drugs that were previously effective.
It mainly affects gastro-intestinal nematodes (GINs) in sheep and leads to treatment failure and heavy parasite burdens.
Major Parasites Showing Resistance
Important resistant gastrointestinal nematodes include:
Haemonchus contortus (Barber's pole worm)
Trichostrongylus spp.
Oesophagostomum spp.
Teladorsagia spp.
* Haemonchus contortus is tre mst dominant and highly resistant parasite in African sheep flocks.
Among Worms Haemonchus contortus is the most dominant and highly resistant parasite in African sheep flocks.
Causes of Anthelmintic Resistance
Major factors contributing to resistance:
1 Frequent deworming without diagnosis
2 Underdosing due to lack of weighing equipment
2 Repeated use of the same drug
5 Treating the whole flock frequently
6 Poor pasture management.
4 Lack of veterinary supervision
Underdosing and high treatment frequency are major drivers of resistance in African sheep farms.
Strategic Deworming Protects Sheep Health and Preserves Drug Effectiveness."
Situation in Africa
Sudan
Ethiopia
Kenya
Kenya
South Africa
Africa
Zimbabwe
Africa
Anthelmintic Resistance has been reported in South Africa, Zimbabwe, Ethiopia, Uganda, Kenya, Algeria, Mozambique and Sudan.
Surveys show that many sheep farms harbor parasites rresistant to at least one anthelmintic class.
Anthelmintic Drug Classes Affected
1 Benzimidazoles
Albendazole
Fenbendazole
2 Imidazothiazoles
Levamisole
Tetramisole
3 Macrocyclic lactones
Ivermectin
Moxidectin
4 Salicylanilides
Closantel
Rafoxanide
Resistance to benzimidazoles is the most widely reported in Africa.
Detection Methods
✓ Faecal Egg Count Reduction Test (FECRT)
✓ Egg Hatch Assay (EHA)
✓ Larval Development Test
✓ Molecular detection of resistance genes
Sustainable Parasite Control
1 Targeted Selective Treatment (TST)
Treat only animals with heavy parasite burdens.
2 Rotational Deworming
Use different drug classes strategically.
3 FAMACHA System
Identify sheep with anemia caused by Haemonchus infection.
#FieldVet
FieldvetDrMSS

12. Anthelmintic Resistance in Africa

What is Anthelmintic Resistance (AR)?

Anthelmintic Resistance is the ability of parasitic worms to survive deworming drugs that were previously effective. It mainly affects gastro-intestinal nematodes (GINs) in sheep and leads to treatment failure and heavy parasite burdens.

Situation in Africa

Anthelmintic resistance has become a major constraint to sheep production in Africa, especially in tropical and subtropical regions.

- Resistance has been reported in South Africa, Zimbabwe, Ethiopia, Uganda, Kenya, Algeria, Mozambique and Sudan.
- Surveys show that many sheep farms harbor parasites resistant to at least one anthelmintic class.
- In South Africa, some studies reported Haemonchus contortus resistant on nearly all surveyed farms, indicating a severe situation.
- On communal farms in Limpopo Province, about 90% of sheep carried nematodes resistant to one or more drug classes.

Major Parasites Showing Resistance

Important resistant gastrointestinal nematodes include:

- Haemonchus contortus (Barber’s pole worm)
- Trichostrongylus spp.
- Oesophagostomum spp.
- Teladorsagia spp.
- Cooperia spp.

Among them, Haemonchus contortus is the most dominant and highly resistant parasite in African sheep flocks.

Anthelmintic Drug Classes Affected

Resistance has been reported against several drug groups:

1 Benzimidazoles

- Albendazole
- Fenbendazole
- Oxfenbendazole

2 Imidazothiazoles

- Levamisole
- Tetramisole

3 Macrocyclic Lactones

- Ivermectin
- Moxidectin
- Doramectin

4 Salicylanilides

- Closantel
- Rafoxanide

Resistance to benzimidazoles is the most widely reported in Africa.

⚠ Causes of Anthelmintic Resistance

Major factors contributing to resistance:

- Frequent deworming without diagnosis
- Underdosing due to lack of weighing equipment
- Repeated use of the same drug
- Treating the whole flock frequently
- Poor pasture management
- Lack of veterinary supervision

Underdosing and high treatment frequency are major drivers of resistance in African sheep farms.

🔬 Detection Methods

Veterinarians detect resistance using:

- Faecal Egg Count Reduction Test (FECRT)
- Egg Hatch Assay (EHA)
- Larval Development Test
- Molecular detection of resistance genes

These methods help determine drug efficacy and resistance levels.

🛑 Strategies to Control Anthelmintic Resistance

1️⃣ Targeted Selective Treatment (TST)

Treat only animals with heavy parasite burdens.

2️⃣ Rotational Deworming

Use different drug classes strategically.

3️⃣ FAMACHA System

Identify sheep with anemia caused by Haemonchus infection.

4️⃣ Pasture Management

- Rotational grazing
- Mixed grazing with cattle

5️⃣ Nutritional Support

Better nutrition improves resistance to parasites.

🌱 Sustainable Parasite Control

Future strategies include:

- Genetic selection of parasite-resistant sheep breeds
- Biological control (nematophagous fungi)
- Improved grazing systems
- Reduced reliance on chemical dewormers

📢 Key Message

"Strategic Deworming Protects Sheep Health and Preserves Drug Effectiveness."

STANDARD PROTOCOL FOR ANTHELMINTIC USE IN SHEEP
(Intensive / Semi-Intensive System)
FieldvetDrMSS
WHY DEWORM STRATEGICALLY IN INTENSIVE SYSTEMS?
• High stocking density – rapid parasite build-up
• Increased risk of anthelmintic resistance
• Subclinical infections reduce growth & fertility
• Efficient control improves feed conversion
Routine Blind Deworming Accelerates Resistance
WHEN TO DEWORM (INTENSIVE SYSTEM)
RISK-BASED DEWORMING SCHEDULE
Production Stage | Recommended Timing
Lambs | First dose at 6–8 weeks
Growing lambs | Every 6–8 weeks (based on risk)
Breeding stock | 2–3 weeks before mating
Late pregnancy | Only if FEC / clinical signs indicate
Lactation | 2–3 weeks post-lambing, if required
Contraindicated in early pregnancy
Avoid mid-day (heat stress)
Avoid immediately after heavy feeding
Along with vaccination (maintain 7–10 day gap)
Prefer Faecal Egg Count (FEC)-based decisions
Use Targeted Selective Treatment (TST)
TIME OF DAY FOR ADMINISTRATION
Early morning
Before feeding
Avoid mid-day (heat stress)
Avoid immediately after heavy feeding
TIME / CONDITIONS TO AVOID DEWORMING
BENZIMIDAZOLES
Albendazole, Fenbendazole
Contraindicated in early pregnancy
Avoid repeated use without rotation
Resistance common in intensive farms
IMIDAZOTHIAZOLES
Levamisole
Avoid in Weak or stressed animals
Avoid connurrent organophosphates
Rapid adult nemotode action
MACROCYCLIC LACTONES
IVermectin, Dotamectin, Moxidectil
Avoid frequent repeat dosing
Resistance develops rapidly in – intensive actions:
Limited effect on arrested larvae
Overuse veduces long term efficacy
SALICYLANILIDES
Closantel, Oxyclozanide
Avoid overdose –risk of Blindness
Avoid routine use in young lambs
Effective against: Haemonchus Liver flukes Blood-feeding parasites
PREFERRED CONTROL STRATEGY
TARGETED SELECTIVE TREATNERT (TST)
Use: Faecal Egg Count (FEC)
Body Condition Score (BCS)
FAMACHA scoring
Treat only high-risk animals
RIGHT DRUG • RIGHT DOSE • RIGHT TIME • RIGHT ANIMAL

13. Standard Protocol for Anthelmintic Use

(Extensive Grazing System)

Why Deworm Strategically?

- Reduce production losses
- Prevent pasture contamination
- Delay anthelmintic resistance
- Improve flock health & fertility

☞ Avoid routine blind deworming

When To Deworm (Extensive System):

Strategic Seasonal Timing (India)

Season	**Recommended Time**	**Purpose**
Pre-monsoon	April–May	Reduce pasture contamination
Mid-monsoon	July–August	Peak parasite challenge
Post-monsoon	Sept–Oct	Reduce worm carryover
Winter	Only if FEC / signs present	Avoid unnecessary dosing.

Physiological Timing:

- Lambs: 6–8 weeks of age
- Growing lambs: Every 8–12 weeks (risk-based)
- Pre-breeding: 2–3 weeks before mating
- Lactation: After 2–3 weeks post-lambing (if required)

Time Of Day for Administration:

✓ Early morning or late evening
✓ Before grazing for better absorption

✗ Avoid mid-day (heat stress)

Time / Conditions to Avoid Deworming:

✗ First trimester of pregnancy

✗ Immediately after lambing (7–10 days)

✗ Severely weak, dehydrated, anaemic animals

✗ During disease outbreaks

✗ Along with vaccination (maintain 7–10 days gap)

Drug-Wise Anthelmintic Contraindications:

1. Benzimidazoles

(Albendazole, Fenbendazole, Oxfendazole)

❌ Contraindicated in first trimester of pregnancy

❌ Avoid high doses in breeding ewes

⚠ Overuse → rapid resistance in grazing systems

✓ Effective against:

- GI nematodes
- Liver flukes (Albendazole – adult stage)

2. Imidazothiazoles

(Levamisole)

❌ Avoid in:

- Weak, stressed, severely anaemic sheep
- Concurrent use with organophosphates

⚠ Narrow safety margin – accurate dosing essential

✓ Rapid action against adult nematodes

3. Macrocyclic Lactones

(Ivermectin, Doramectin, Moxidectin)

❌ Avoid unnecessary repeated use

❌ Resistance common in extensive grazing flocks

⚠ Poor efficacy against arrested larvae in some regions

⚠ Environmental concern – affects dung fauna

✓ Effective against:

- GI worms
- Lungworms
- Ectoparasites

4. Salicylanilides

(Closantel, Oxyclozanide)

❌ Do NOT use in young lambs unless indicated

❌ Overdose may cause blindness

✓ Specific for:

- Liver flukes
- Blood-sucking parasites (Haemonchus)

Good Practices – Extensive System:

- ✓ Dose based on heaviest animal in the flock
- ✓ Use calibrated drenching gun
- ✓ Keep treated sheep off pasture for 12–24 hrs (if feasible)
- ✓ Rotate drug classes – not brands
- ✓ Maintain refugia (do not treat entire flock every time)

Preferred Approach:

Targeted Selective Treatment (TST)

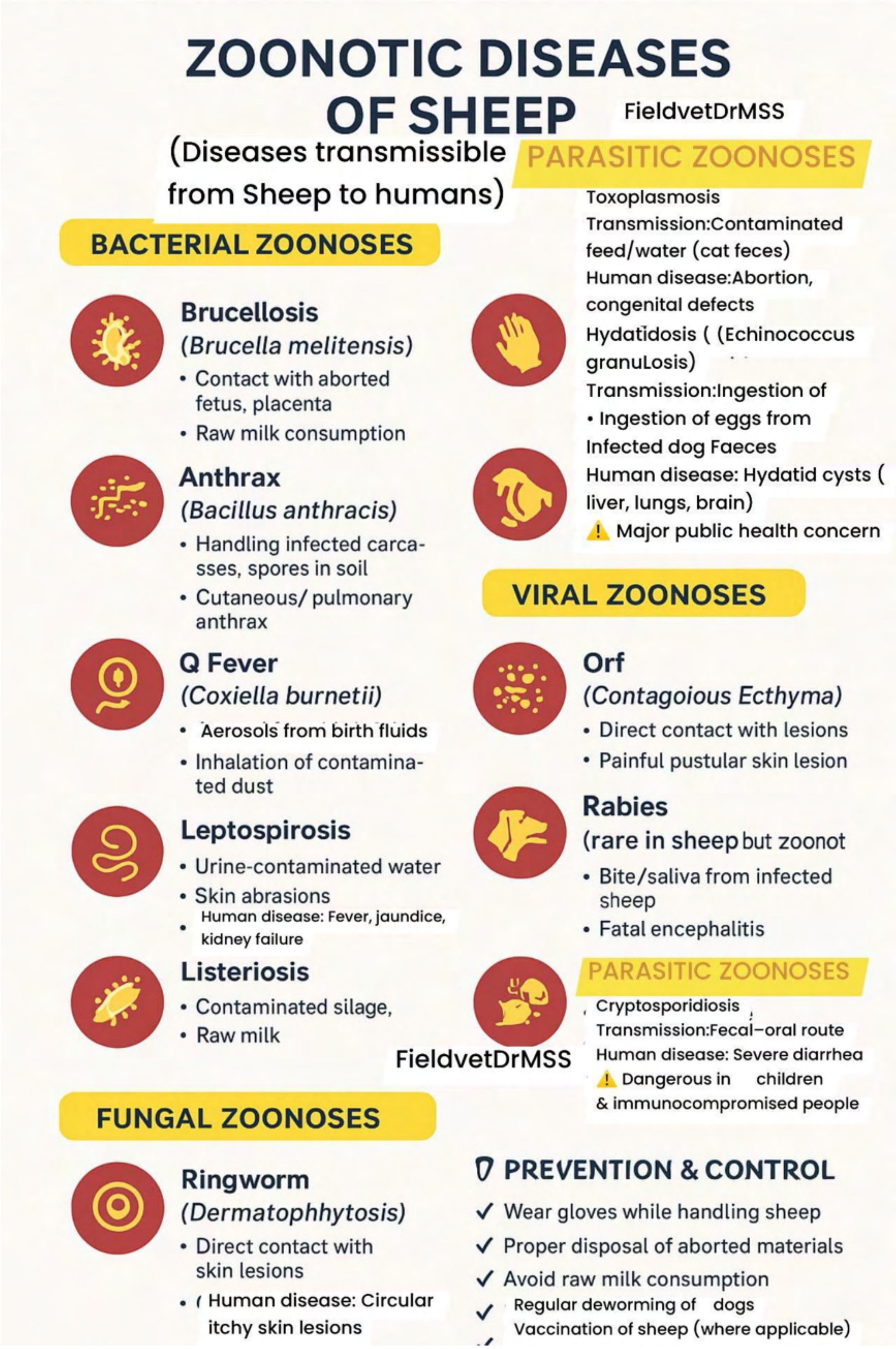
Use:

- FAMACHA scoring
- Body condition scoring
- Faecal Egg Count (where available)

☞ Treat only animals that need treatment

Key Message:

"Right drug • Right time • Right animal • Right dose"


ZOONOTIC DISEASES OF SHEEP
FieldvetDrMSS
(Diseases transmissible from Sheep to humans)
BACTERIAL ZOONOSES
Brucellosis
(Brucella melitensis)
• Contact with aborted fetus, placenta
• Raw milk consumption
Anthrax
(Bacillus anthracis)
• Handling infected carca-sses, spores in soil
• Cutaneous/ pulmonary anthrax
Q Fever
(Coxiella burnetii)
• Aerosols from birth fluids
• Inhalation of contamina-ted dust
Leptospirosis
• Urine-contaminated water
• Skin abrasions
• Human disease: Fever, jaundice, kidney failure
Listeriosis
• Contaminated silage,
• Raw milk
FieldvetDrMSS
FUNGAL ZOONOSES
Ringworm
(Dermatophhytosis)
• Direct contact with skin lesions
• (Human disease: Circular itchy skin lesions
PARASITIC ZOONOSES
Toxoplasmosis
Transmission:Contaminated feed/water (cat feces)
Human disease:Abortion, congenital defects
Hydatidosis ((Echinococcus granuLosis)
Transmission:Ingestion of
• Ingestion of eggs from Infected dog Faeces
Human disease: Hydatid cysts (liver, lungs, brain)
⚠ Major public health concern
VIRAL ZOONOSES
Orf
(Contagoious Ecthyma)
• Direct contact with lesions
• Painful pustular skin lesion
Rabies
(rare in sheep but zoonot
• Bite/saliva from infected sheep
• Fatal encephalitis
PARASITIC ZOONOSES
Cryptosporidiosis
Transmission:Fecal–oral route
Human disease: Severe diarrhea
⚠ Dangerous in children & immunocompromised people
PREVENTION & CONTROL
✓ Wear gloves while handling sheep
✓ Proper disposal of aborted materials
✓ Avoid raw milk consumption
✓ Regular deworming of dogs
Vaccination of sheep (where applicable)

Section 4: Disease Transmission & Zoonosis

14. Zoonotic Diseases of Sheep

(Diseases transmissible from sheep to humans)

🦠 **Bacterial Zoonoses:**

1️⃣ Brucellosis (Brucella melitensis)

Transmission:

- Contact with aborted fetus, placenta
- Raw milk consumption

Human disease: Undulant fever

Risk group: Veterinarians, shepherds

2️⃣ Anthrax (Bacillus anthracis):

Transmission:

- Handling infected carcasses
- Spores in soil

Human disease: Cutaneous / pulmonary anthrax

⚠ **Highly fatal if untreated**

3️⃣ Q Fever (Coxiella burnetii):

Transmission:

- Aerosols from birth fluids
- Inhalation of contaminated dust

Human disease: Fever, pneumonia, hepatitis

4️⃣ Leptospirosis:

Transmission:

- Urine-contaminated water
- Skin abrasions

Human disease: Fever, jaundice, kidney failure

5️⃣ Listeriosis:

Transmission:

- Contaminated silage
- Raw milk

Human disease: Meningitis, abortion (pregnant women)

🦠 **Viral Zoonoses**

6️⃣ Orf (Contagious Ecthyma):

Transmission:

- Direct contact with lesions

Human disease: Painful pustular skin lesions

⚠️ **Common in shepherds & veterinarians**

7️⃣ Rabies (rare in sheep but zoonotic):

Transmission:

- Bite/saliva from infected sheep

Human disease: Fatal encephalitis

🪱 Parasitic Zoonoses:

8️⃣ Hydatidosis (Echinococcus granulosus)

Transmission:

- Ingestion of eggs from dog feces

Human disease: Hydatid cysts (liver, lungs, brain)

⚠️ **Major public health concern**

9️⃣ Toxoplasmosis:

Transmission:

- Contaminated feed/water (cat feces)

Human disease: Abortion, congenital defects

🔟 **Cryptosporidiosis:**

Transmission:

- Fecal–oral route

Human disease: Severe diarrhea

⚠️ **Dangerous in children & immunocompromised people**

🧪 **Fungal Zoonoses:**

1️⃣1️⃣ Ringworm (Dermatophytosis)

Transmission:

- Direct contact with skin lesions

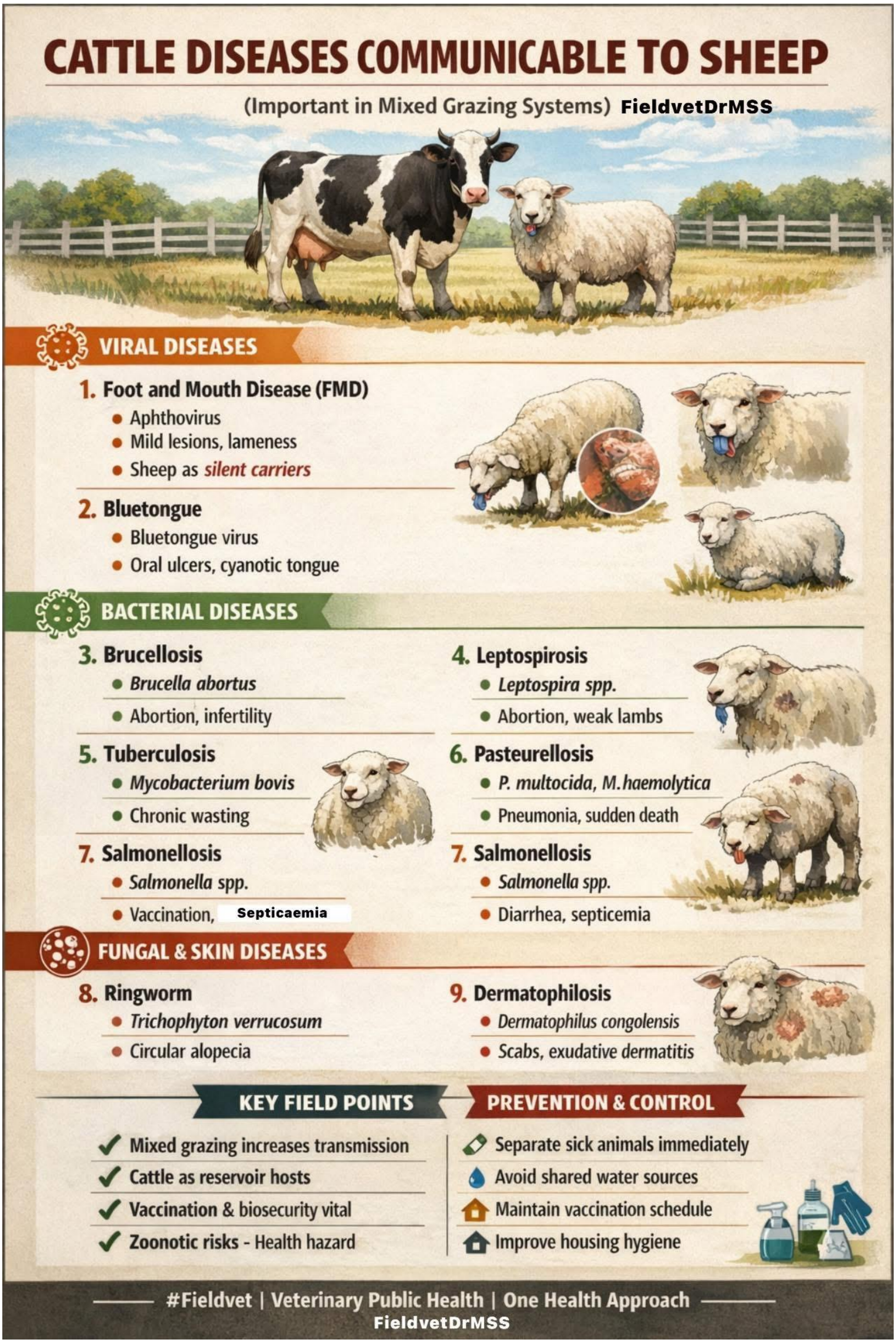
Human disease: Circular itchy skin lesions

🛡 **Prevention & Control:**

- ✓ Wear gloves while handling sheep
- ✓ Proper disposal of aborted materials
- ✓ Avoid raw milk consumption
- ✓ Regular deworming of dogs
- ✓ Vaccination of sheep (where applicable)
- ✓ Maintain farm biosecurity

📌 **One-Line Summary:**

"Sheep act as reservoirs for several zoonotic diseases including brucellosis, anthrax, Q fever, hydatidosis, orf, and toxoplasmosis, posing serious public health risks."


CATTLE DISEASES COMMUNICABLE TO SHEEP
(Important in Mixed Grazing Systems) FieldvetDrMSS
VIRAL DISEASES
1. Foot and Mouth Disease (FMD)
Aphthovirus
Mild lesions, lameness
Sheep as silent carriers
2. Bluetongue
Bluetongue virus
Oral ulcers, cyanotic tongue
BACTERIAL DISEASES
3. Brucellosis
Brucella abortus
Abortion, infertility
4. Leptospirosis
Leptospira spp.
Abortion, weak lambs
5. Tuberculosis
Mycobacterium bovis
Chronic wasting
6. Pasteurellosis
P. multocida, M.haemolytica
Pneumonia, sudden death
7. Salmonellosis
Salmonella spp.
Vaccination, Septicaemia
7. Salmonellosis
Salmonella spp.
Diarrhea, septicemia
FUNGAL & SKIN DISEASES
8. Ringworm
Trichophyton verrucosum
Circular alopecia
9. Dermatophilosis
Dermatophilus congolensis
Scabs, exudative dermatitis
KEY FIELD POINTS
Mixed grazing increases transmission
Cattle as reservoir hosts
Vaccination & biosecurity vital
Zoonotic risks - Health hazard
PREVENTION & CONTROL
Separate sick animals immediately
Avoid shared water sources
Maintain vaccination schedule
Improve housing hygiene
#Fieldvet | Veterinary Public Health | One Health Approach
FieldvetDrMSS

15. Goat Diseases Communicable to Sheep

(Important in Mixed Small-Ruminant Farming Systems)

🦠 **Viral Diseases:**

1. Peste des Petits Ruminants (PPR)

- Agent: Morbillivirus
- Transmission: Direct contact, aerosols
- Sheep signs: Fever, oral erosions, diarrhea, pneumonia
- ⚠ Highly contagious, high mortality

2. Foot and Mouth Disease (FMD)

- Agent: Aphthovirus
- Transmission: Direct contact, aerosols, fomites
- Sheep signs: Mild lesions, lameness
- Sheep often act as silent carriers

3. Orf (Contagious Ecthyma)

- Agent: Parapoxvirus
- Transmission: Direct contact, fomites
- Sheep signs: Scabby lesions on lips, gums, teats
- Zoonotic: Yes

🧫 **Bacterial Diseases:**

4. Brucellosis

- Agent: Brucella melitensis
- Transmission: Aborted materials, uterine discharge
- Sheep signs: Abortion, infertility
- Zoonotic: Yes

5. Pasteurellosis

- Agent: Mannheimia haemolytica, Pasteurella multocida
- Transmission: Respiratory secretions
- Sheep signs: Pneumonia, sudden death

6. Caseous Lymphadenitis (CLA)

- Agent: Corynebacterium pseudotuberculosis
- Transmission: Wound contamination
- Sheep signs: Abscesses in lymph nodes
- Chronic economic losses

□ Parasitic Diseases:

7. Haemonchosis

- Agent: Haemonchus contortus
- Transmission: Grazing contaminated pasture
- Sheep signs: Anemia, bottle jaw, death

8. Fasciolosis

- Agent: Fasciola spp.
- Transmission: Snail-borne, wet grazing areas
- Sheep signs: Anemia, weight loss, sudden death.

🦠 Fungal & Skin Diseases:

9. Ringworm

- Agent: Trichophyton spp.
- Transmission: Direct contact
- Sheep signs: Circular alopecia
- Zoonotic: Yes

10. Dermatophilosis

- Agent: Dermatophilus congolensis
- Transmission: Wet conditions, skin trauma
- Sheep signs: Scabby dermatitis

⚠ Key Field Points:

- ✓ Goats and sheep share most pathogens
- ✓ Mixed housing increases respiratory & skin diseases
- ✓ Parasite burden rises in shared grazing
- ✓ Several diseases are zoonotic
- ✓ PPR is the most economically devastating disease

□ Prevention & Control:

- Vaccinate against PPR, FMD, Enterotoxaemia
- Separate sick animals immediately
- Avoid overcrowding
- Strategic deworming
- Improve ventilation & hygiene
- Quarantine newly purchased goats

🧬 One Health Importance: Brucellosis & Orf pose human risk, Small-ruminant disease control supports food security


DOG DISEASES COMMUNICABLE TO SHEEP
(Important in Farm, Pasture & Shepherd Dog Systems)
FieldvetDrMSS
VIRAL DISEASES
1. Rabies
Lyssavirus
Bite wounds, saliva
Fatal & zoonotic
BACTERIAL DISEASES
2. Leptospirosis
Leptospira spp.
Urine-contaminated water/pasture
Abortion, fever, weak lambs Yes
Zoonotic: Yes
3. Brucellosis (rare)
Brucella canis
Contaminated environment
Reproductive failure (rare)
PARASITIC DISEASES (MAJOR IMPORTANCE)
4. Hydatidosis (Echinococcosis)
Echinococcus granulosus
Definitive host.
Sheep role: Cysts in liver/lungs (usually subclinical)
Intermediate host
Zoonotic:
Severe public health importance
5. Coenurosis (Gid / Sturdy)
Dog fecal contamination
Circling, head pressing, blindness
Economic loss: High
5. Toxocariasis
Toxocara canis
Eggs from dog feces
6. Toxocariasis
Toxocara canis
Ill-thrift (mainly lambs)
FUNGAL & ECTOPARASITIC DISEASES
7. Ringworm
Microsporum / Trichophyton spp.
Direct contact
Circular alopecia
PREVENTION & CONTROL
Regular deworming of dogs (Peaziquantel based)
Prevent dogs from eating sheep offal
Dispove slaughter waste properiy
Vaccinate dogs against rabies
ONE HEALTH IMPORTANCE
✓ Hydatidosis & rabies pose serious human risk
✓ Control of dog dlseases - protection of livestock & humans
#Fieldvet | Veterinary Public Health | One Health Approach

16. Dog Diseases Communicable to Sheep

(Important in Farm, Pasture & Shepherd Dog Systems)

🦠 **Viral Diseases:**

1. Rabies

- Agent: Lyssavirus
- Transmission: Bite wounds, saliva
- Sheep signs: Aggression, paralysis, sudden death

⚠ **Fatal & zoonotic**

🧫 **Bacterial Diseases:**

2. Leptospirosis

- Agent: Leptospira spp.
- Transmission: Urine-contaminated water/pasture
- Sheep signs: Abortion, fever, weak lambs
- Zoonotic: Yes

3. Brucellosis (rare)

- Agent: Brucella canis
- Transmission: Contaminated environment
- Sheep signs: Reproductive failure (rare)
- Zoonotic: Yes

□ **Parasitic Diseases (Major Importance):**

4. Hydatidosis (Echinococcosis)

- Agent: Echinococcus granulosus
- Dog role: Definitive host
- Sheep role: Intermediate host
- Sheep signs: Cysts in liver/lungs (usually subclinical)
- Zoonotic: Severe public health importance

5. Coenurosis (Gid / Sturdy)

- Agent: Taenia multiceps
- Transmission: Dog fecal contamination
- Sheep signs: Circling, head pressing, blindness
- Economic loss: High

6. Toxocariasis

- Agent: Toxocara canis
- Transmission: Eggs from dog feces

- Sheep signs: Ill-thrift (mainly lambs)

🦠 **Fungal & Ectoparasitic Diseases:**

7. Ringworm

- Agent: Microsporum / Trichophyton spp.
- Transmission: Direct contact
- Sheep signs: Circular alopecia
- Zoonotic: Yes

8. Mange (Sarcoptic / Demodectic)

- Transmission: Close contact
- Sheep signs: Pruritus, skin thickening (rare)

⚠ **Key Field Points:**

- Dogs act as definitive hosts for major sheep parasites
- Free-roaming dogs increase parasitic disease burden
- Sheep ingest infective stages from contaminated pasture
- Several diseases are zoonotic
- Shepherd & stray dogs are major risk factors

Prevention & Control:

- Regular deworming of dogs (Praziquantel-based)
- Prevent dogs from eating sheep offal
- Dispose slaughter waste properly
- Restrict stray dog access to farms
- Vaccinate dogs against rabies
- Maintain pasture hygiene

🧬 **One Health Importance:**

Hydatidosis & rabies pose serious human risk

Control of dog diseases = protection of livestock & humans


Diseases Transmitted from Ducks to Sheep
Risks in Integrated Duck-Sheep Farms
FieldvetDrMSS
1 Salmonellosis
Salmonella enterica
Transmission:
• Fecal contamination of pasture or water
• Shared ponds
• Contaminated feed areas
Clinical Signs in Sheep:
• Diarrhea
• Septicemia (especially lambs)
• Abortion in ewes
2 Leptospirosis (Environmental Link)
Leptospira interorgans
Ducks may act as mechanical carriersthrough contaminated water bodies
Transmission via:
• Urine-contaminated water
• Shared ponds or marshy grazing land
3 Campylobacteriosis
Campylobacter jejuni
Ducks are common reservors
In Sheep:
• Late-term abortion
• Weak lambs
• Stillbirth
4 Avian Influenza
(Rare Cross-Species Event)
Sheep are not typical hosts, but experimental infections show possible seroconversion
5 Pasteurellosis
(Indirect Environment Exposure)
Shared environments with bacteria-laden water may increase exposure risk.
6 Avian Influenza
(Rare Cross-Species Event)
Sheep are not typical hosts but experimental infections show possible seroconversion
6 Cryptosporidiosis
(Shared Water Source)
In Lambs:
• Profuse watery diarrhea
• Dehydration
• Reduced growth
Prevention Tips for Mixed Duck-Sheep Farms
✓ Separate water sources
✓ Avoid sheep grazing near duck ponds
✓ Regular manure management
✓ Control rodent populations
✓ Protect lambs from stagnant water
✓ Vaccinate sheep against endemic diseases
Direct duck so sheep disease transmission is uncommon,
Shared contaminated water sources pose the greatest risk,
Proper biosecurity measures and environmental hygiene are crifical.
Field based Veterinary Education
#Fieldvet
FieldvetDrMSS

17. Diseases from Ducks to Sheep

1️⃣ Salmonellosis:

Causative agent: Salmonella enterica

Transmission

- Fecal contamination of pasture or water
- Shared ponds
- Contaminated feed storage areas

Clinical Signs in Sheep

- Diarrhea
- Septicemia (especially lambs)
- Abortion in ewes

Risk Factors

- Mixed duck–sheep systems
- Poor biosecurity
- Contaminated stagnant water

2️⃣ Leptospirosis (Environmental Link):

Causative agent: Leptospira interrogans

Ducks may act as mechanical carriers through contaminated water bodies.

Transmission

- Urine-contaminated water
- Shared ponds or marshy grazing land

Clinical Signs in Sheep

- Abortion storms
- Jaundice
- Fever
- Reduced fertility

3️⃣ Campylobacteriosis:

Causative agent: Campylobacter jejuni

Ducks are common reservoirs.

In Sheep

- Late-term abortion
- Weak lambs
- Stillbirth

Transmission occurs via:

- Contaminated water
- Fecal contamination of pasture

4️⃣ Avian Influenza (Rare Cross-Species Event):

Causative agent: Influenza A virus

Sheep are not typical hosts, but experimental infections have shown possible seroconversion.

⚠ **Natural transmission to sheep is rare and not a major field concern.**

5️⃣ Pasteurellosis (Indirect Environmental Exposure):

Causative agent: Pasteurella multocida

Though not directly transmitted from ducks to sheep, shared environments with high bacterial load may increase exposure risk.

6️⃣ Cryptosporidiosis (Shared Water Source):

Causative agent: Cryptosporidium parvum

Ducks can contaminate water with oocysts.

In Lambs:

- Profuse watery diarrhea
- Dehydration
- Reduced growth

🧪 **Epidemiological Considerations**

In integrated farming systems:

- Shared water sources are the major risk factor.
- Wet grazing areas increase pathogen survival.
- Ducks often serve as reservoir hosts, not primary disease amplifiers for sheep.

🚫 **Diseases NOT Transmitted from Ducks to Sheep:**

The following are species-specific and not transmitted:

- Duck viral hepatitis virus

- Newcastle disease virus
- Duck plague virus

🛡 **Prevention in Mixed Duck–Sheep Farms**

✓ Separate water sources

✓ Avoid sheep grazing near duck ponds

✓ Regular manure management

✓ Control rodent populations

✓ Vaccinate sheep against endemic diseases

✓ Improve drainage in low-lying grazing areas

📌 **Field Conclusion:**

Direct transmission of duck-specific diseases to sheep is uncommon. However, shared environmental contamination (waterborne pathogens) poses the main risk. Proper biosecurity and environmental hygiene are critical in integrated livestock–poultry systems.

WATER-BORNE DISEASES OF SHEEP

FieldvetDrMSS

1. Leptospirosis

- Cause: Leptospira spp.
- Source: Stagnant water, ponds, urine-contaminated water
- Key signs: Fever, jaundice, abortions, hemoglobinuria
- Zoonotic: ✅ Yes

2. Colibacillosis (E. coli Diarrhoeaa)

- **Cause:** *Escherichia coli*
- Source: Contaminated drinking water
- Key signs: Profuse watery diarrhoea, dehydration, sudden death (lambs)
- Risk group: Neonatal lambs

3. Salmonellosis

Cause: *Campylobacter fetus*

- **Source:** Contaminated water
- Key signs: Fever, diarrhoea, abortions, septicemia
- Zoonotic: ✅ Yes

4. Cryptosporidiosis

- Cause: Cryptosporidium parvum
  Source: Dirty water, overcrowding
  Key signs: Watery diarrhoea, dehydration
  Zoonotic: ✅ Yes

6. Coccidiosis

- Cause: Eimeria spp.
- Source: Contaminated water & wet pens,infected adults
- Key signs: Bloody diarrhoea, weight loss, weakness
- Risk group: Lambs

7. Giardiasis

- Cause: Giardia spp.
- Source: Untreated surface water
- Key signs: Chronic diarrhoea, poor growth
- Zoonotic: ✅ Yes

8. Listeriosis

- **Cause:** *Listeria monocytogenes*
- **Source:** Water contaminated with decaying organic matter
- Key signs: Circling, paralysis, abortions
  Zoonotic: ✅ Yes

10. Schistosomiasis

- Cause: Schistosoma spp.
- Source: Snail-infested water
  Key signs: Diarrhoea, anaemia, emaciation

Prevention:Provide clean, chlorinated or filtered drinking water

Avoid stagnant ponds and marshy grazing.

Avoid stagnant ponds and marshy grazing.

Section 5: Disease Classification by Environment

18. Water-Borne Diseases

Contaminated water sources such as ponds, stagnant water, canals, and poorly maintained troughs can transmit several infectious diseases to sheep. These diseases cause diarrhoea, weakness, reduced productivity, and sometimes death, leading to economic losses for farmers

🌊 Major Water-Borne Diseases in Sheep

1️⃣Leptospirosis

Cause: Leptospira bacteria

Source of infection: Water contaminated with urine of infected animals (especially rodents)

Clinical signs

- Fever
- Jaundice
- Abortion
- Weakness
- Significance
- Zoonotic disease (can infect humans)

2️⃣Salmonellosis

Cause: Salmonella bacteria

Source: Contaminated water and feed

Clinical signs

- Severe diarrhea
- Fever
- Dehydration
- Abortion in pregnant ewes

Significance: High mortality in lambs

3️⃣Coccidiosis

Cause: Eimeria species (protozoa)

Source: Water contaminated with feces

Clinical signs

- Bloody diarrhoea
- Weight loss
- Weakness

Significance: Affects young lambs severely

4️⃣ Cryptosporidiosis

Cause: Cryptosporidium protozoa

Source: Contaminated drinking water

Clinical signs

- Watery diarrhoea
- Dehydration
- Poor growth

Significance: Important cause of lamb diarrhoea

5️⃣ Fasciolosis (Liver Fluke Disease)

Cause: Liver fluke (Fasciola species)

Source: Grazing near waterlogged areas and snail-infested water bodies

- Clinical sign
- Anemia
- Bottle jaw
- Weight loss

Significance: Major economic loss in grazing sheep

6️⃣ Blue-Green Algae Toxicity

Cause: Toxins produced by cyanobacteria in stagnant water

Clinical signs

- Sudden illness
- Nervous signs
- Death in severe cases

Risk Factors

- Stagnant water sources
- Dirty water troughs
- Grazing near swamps or ponds
- Overcrowding
- Poor sanitation

🛡 **Prevention and Control**

- ✓ Provide clean drinking water
- ✓ Clean water troughs regularly
- ✓ Avoid grazing in waterlogged areas

- ✓ Control rodents
- ✓ Regular deworming and health monitoring
- ✓ Improve farm sanitatio

Field Significance

Water-borne diseases can cause:

- Diarrhea outbreaks in lambs
- Reduced growth and productivity
- Abortion in ewes
- Economic losses to farmers

Safe water supply is therefore a critical component of sheep health management.

✅ **Key Message**

"Clean water is essential for healthy sheep. Preventing water contamination helps control many infectious diseases in sheep flocks."


Air-Borne Diseases of Sheep
Causes, Symptoms, and Prevention
What are Air-Borne Diseases?
FieldvetDrMSS
Air-borne diseases are infections transmitted through:
Aerosols (coughing, sneezing)
Poor ventilation
Close confinement
Dust laden air in sheds
They mainly affect the RESPIRATORY SYSTEM and cause significant economic losses.
Major Air-Borne Diseases of Sheep
1 Pneumonic Pasteurellosis
Causative agent: Pasteurella multiocida, Mannheimia haemolytica
Predisposing Factors:
Sudden weather changes
Overcrowding
Transport stress
Symptoms
High fever
Nasal discharge
Coughing
Respiratory distress
Sudden death (acute cases)
2 Sheep Pox
Causative agent: Sheep pox Virus
Symptoms
Fever
Papules & scabs on skin
Nasal discharge
Pneumonia in severe cases
Control
Annual vaccination
Strict quarantine
Strict quarantine
Disinfection
3 Contagious Caprine Pleuropneumonia-like Syndrome (CCPP-like)
Causative agent:Mycoplasma sp
Symptoms:
Severe coughing
Rapid breathing
Nasal discharge
High mortality in outbreaks
Strict quarantine
Biosecurity
4 Ovine Respiratory Complex
Mixed infection involving:
Bacteria, Viruses,
Mycoplasma
Symptoms
Fever, Excess salivation
Nasal discharge
Breathing difficulty
Risk Factors for Air-Borne Diseases
Poor ventilation
Cold stress
Overcrowding
Nutritional deficiency
Regular vaccination
Clean & dry housing
Respiratory diseases spread faster in winter and rainy seasons -early detection saves flock.
FieldvetDrMSS

19. Air-Borne Diseases

🔬 What are Air-Borne Diseases?

Air-borne diseases are infections transmitted through:

- Aerosols (coughing, sneezing)
- Close confinement
- Poor ventilation
- Dust-laden air in sheds

They mainly affect the respiratory system and cause significant economic losses.

🦠 Major Air-Borne Diseases of Sheep:

1️⃣ Pneumonic Pasteurellosis

Causative agent: Pasteurella multocida, Mannheimia haemolytica

Predisposing factors:

- Sudden weather changes
- Overcrowding
- Transport stress

Symptoms:

- High fever
- Nasal discharge
- Coughing
- Respiratory distress
- Sudden death (acute cases)

Control:

- Timely vaccination
- Stress reduction
- Early antibiotic therapy

2️⃣ Sheep Pox:

Causative agent: Sheep pox virus

Transmission:

- Air-borne droplets
- Direct contact

Symptoms:

- Fever
- Papules & scabs on skin

- Nasal discharge
- Pneumonia in severe cases

Control:

- Annual vaccination
- Strict quarantine
- Disinfection

3️⃣ Contagious Caprine Pleuropneumonia-like Syndrome (CCPP-like):

Causative agent: Mycoplasma species

Symptoms:

- Severe coughing
- Rapid breathing
- Nasal discharge
- High mortality in outbreaks

Control:

- Isolation of affected animals
- Antibiotic therapy
- Biosecurity

4️⃣ Ovine Respiratory Complex:

Mixed infection involving:

- Bacteria
- Viruses
- Mycoplasma

Symptoms:

- Chronic coughing
- Reduced feed intake
- Poor growth
- Increased lamb mortality

Control:

- Improved housing ventilation
- Balanced nutrition
- Strategic deworming

5️⃣ Bluetongue (Air-borne spread in sheds):

Causative agent: Bluetongue virus

Transmission:

- Mainly vector-borne
- Aerosol spread possible in closed housing

Symptoms:

- Fever
- Excess salivation
- Nasal discharge
- Breathing difficulty

Control:

- Vaccination
- Vector control
- Housing management

⚠ **Risk Factors for Air-Borne Diseases:**

- Poor ventilation
- Overcrowding
- Dusty floors
- Cold stress
- Nutritional deficiency
- Mixed age groups

🛡 **Prevention & Control Measures:**

✔ Adequate ventilation

✔ Avoid overcrowding

✔ Regular vaccination

✔ Prompt isolation of sick sheep

✔ Clean & dry housing

✔ Stress-free handling

📌 **Field Tip:**

Respiratory diseases spread faster in winter and rainy seasons—early detection saves the flock.


Disease Conditions of Sheep
Grazing in Seashore / Coastal Areas
(Coastal grasslands, salt marshes, sandy pastures)
FieldvetDrMSS
1 Parasitic Diseases
a. Fasciolosis (Liver Fluke)
• Cause: Fasciola spp.
• Risk Factor: Marshy coastal land, snail population
• Symptoms: Anemia, bottle jaw, weight loss, poor fertility
• Diagnosis: Fecal examination, liver lesions
Control: Strategic flukicide use, avoid marsh grazing
b Gastrointestinal Nematodes
• Common Species: Haemonchus, Trichostrongylus
• Symptoms: Pale mucosa, diarrhea, poor growth
• Control: FAMACHA scoring, rotational grazing
2 Mineral & Nutritional Disorders
a. Copper Deficiency
• Cause: High molybdenum & sulphur in coastal soils
• Symptoms: Faded wool, anemia, infertility, sway Ꝛactl
Control: Sheep-specific mineral mixture
b Iodine Deficiency
• Symptoms: Goitre, weak lambs, reproductive failure
Control: Iodized salt supplementation
2 Mineral & Nutritional Disorders
a. Copper Deficiency
• Cause: High molybdenum & sulphur in coastal soils
• Symptoms: Faded wool, anemia, infertility, swayback in lambs
• Control: FAMACHA scoring, rotational grazing
Control: Sheep-specific mineral mixture
3 Salt-Related Disorders
a. Salt Toxicity / Dehydration
• Cause: Excess salt intake + limited fresh water
• Symptoms: Excessive thirst, nervous signs, diarrhea
Control: Continuous access to clean fresh water
4 Bacterial Diseases
• Foot Rot: Dichelobecter nodosus
• Risk Factor: Wet sand, muddy soil
• Symptoms: Lameness, foul smell, hoof lesions
• Control: Hoof trimming, zinc sulphate foot bath
5 Respiratory Diseases
a. Pneumonia
• Risk Factor: High humidity, strong coastal winds
• Symptoms: Coughing, nasal discharge, fever
Control: Wind shelter, avoid overcrowding
Preventive Measures for Coastal Sheep Farming
✓ Regular deworming & fluke control
✓ Balanced mineral supplementation
✓ Clean fresh drinking water
✓ Vaccination Clostridial & respiratory diseases)
✓ Avoid grazing in flooded / marshy areas
✓ Proper shelter against wind & rain
Veterinary Extension Education | Coastal Sheep Health | #Fieldvet

20. Coastal & Marsh Grazing Diseases

1. Fasciolosis

2. Haemonchosis

3. Copper deficiency

4. Iodine deficiency

5. Salt toxicity

6. Foot rot

7. Pneumonia

Preventive Measures:

- Regular deworming & fluke control
- Balanced mineral supplementation
- Clean fresh drinking water
- Vaccination (clostridial & respiratory diseases)
- Avoid grazing in flooded / marshy areas
- Proper shelter against wind & rain.


Disease Conditions of Sheep
Grazing in Hilly Regions
(Hill slopes · Highland pastures · Forest margins· Cool & humid zones)
FieldvetDrMSS
1 Parasitic Diseases
a. Gastro-intestinal Nematdiasis
• Common parasites: Haemonchus, Trichostrongylus, Ostertagia
• Risk factors:Cool climate,high rainfall,lush pastime Humid zones
• Symptoms: Anaemia, bottle jaw, weight loss, bottle jaw.
Diagnosis: Faecal egg count (FEC)
Control: Strategic deworming, rotational grazing
Fasciolosis (Liver Fluke)
• Cause: Fasciola gigantica / F.hepatica
• Risk factors:Marshy hill streams&Snail habitats
• Symptoms: Anaemia, submandibular oedema, reduced productivity
Control: Flukicides, avoid wet grazing areas.
2 Respiratory Diseases
a. Pneumonia
• Predisposing factors:Cold stress, fog, rain, wind
• Symptoms: Fever, nasal discharge, laboured breathing
Control: Shelter provision, timely antibiotic therapy
Maedi-Visna (in endemic areas)
• Symptoms: Progressive dyspnoea, weight loss
Control: Flock screening, marshy hill slopes.
Preventive Measures
✓ Provide shelter against cold, rain & wind
✓ Strategic deworming & fluke control
✓ Regular hoof trimming & foot baths
✓ Vaccination (Bluetorgue, bacterial diseases)
✓ Balanced nutrition with minerals
✓ Tick & vector control
✓ Avoid grazing in marshy hill slopes
2 Respiratory Diseases
a. Pneumonia
• Predisposing factors: Cold stress, fog, rain, wind
• Symptoms: Fever, nasal discharge, laboured breathing
Control: Shelter provision, timely antibiotic therapy
Maedi-Visna (in endemic areas)
• Symptoms: Progressive dyspnonea, disease.
Control: Flock screening, culling of positives
4 Vector-Borne Diseases
a. Bluetongue
• Vector. "Culicoides" midges breed in moist hill areas
• Symptoms: Fever, oral lesions, lameness
Control: Vaccination, vector control
Tick-Borne Diseases
• Anaplasmosis, piroplasnosis (region specific)
• Symptoms: Fever, anaemia, weakness
Control: Regular acaricide application.
6 Poisonous Hill Plants
Plant toxicities
• Examples: Bracken ism, Laritaita. Phododendiron
• Symptoms: Salvation, diarrhoa, neurological signs
Control: Controlled grazing, plant identification
Key Message:
Hilly region grazing predisposes sheep to parasitic, respiratory, foot and nutritional disorders.preventive health management is essential.
Veterinary Extension Education | Sheep Health Management | #Fieldvet

21. Hilly Region Disease Conditions

(Hill slopes • Highland pastures • Forest margins • Cool & humid zones)

1. Parasitic Diseases:

Gastro-intestinal Nematodiasis:

- Common parasites: Haemonchus, Trichostrongylus, Ostertagia
- Risk factors: Cool climate, high rainfall, lush pasture
- Symptoms: Anaemia, diarrhoea, weight loss, bottle jaw
- Diagnosis: Faecal egg count (FEC)
- Control: Strategic deworming, rotational grazing

Fasciolosis (Liver Fluke):

- Cause: Fasciola gigantica / F. hepatica
- Risk factor: Marshy hill streams & snail habitats
- Symptoms: Anaemia, submandibular oedema, reduced productivity
- Control: Flukicides, avoid wet grazing areas

Lungworm Disease:

- Parasite: Dictyocaulus filaria
- Symptoms: Chronic cough, dyspnoea, poor growth
- Control: Anthelmintics, pasture management

2. Respiratory Diseases:

Pneumonia

- Predisposing factors: Cold stress, fog, rain, wind
- Symptoms: Fever, nasal discharge, laboured breathing
- Control: Shelter provision, timely antibiotic therapy

Maedi-Visna (in endemic areas):

- Nature: Chronic viral respiratory disease
- Symptoms: Progressive dyspnoea, weight loss
- Control: Flock screening, culling of positives

3. Foot & Musculoskeletal Disorders:

Foot Rot

- Cause: Dichelobacter nodosus
- Risk factor: Wet, muddy hill terrain
- Symptoms: Severe lameness, hoof lesions
- Control: Hoof trimming, zinc sulphate foot bath

Foot Abscess & Laminitis:

- Risk factor: Rocky slopes, constant climbing
- Symptoms: Lameness, swelling, reluctance to move
- Control: Proper hoof care, mineral supplementation

4. Vector-Borne Diseases:

Bluetongue

- Vector: Culicoides midges (breed in moist hill areas)
- Symptoms: Fever, oral lesions, lameness
- Control: Vaccination, vector control

Tick-Borne Diseases:

- Diseases: Anaplasmosis, piroplasmosis (region-specific)
- Symptoms: Fever, anaemia, weakness
- Control: Regular acaricide application

5. Nutritional & Metabolic Disorders:

Mineral Deficiencies

- Common deficiencies: Copper, Cobalt, Selenium
- Symptoms: Poor wool quality, anaemia, low immunity
- Control: Area-specific mineral mixture

Pregnancy Toxaemia:

- Risk factor: Cold stress, poor nutrition
- Symptoms: Depression, recumbency, ketosis
- Control: Adequate energy feeding in late gestation

6. Toxicities:

Poisonous Hill Plants

- Examples: Bracken fern, Lantana, Rhododendron
- Symptoms: Salivation, diarrhoea, neurological signs
- Control: Controlled grazing, plant identification

Preventive Measures:

✓ Provide shelter against cold, rain & wind
✓ Strategic deworming & fluke control
✓ Regular hoof trimming & foot baths
✓ Vaccination (Bluetongue, bacterial diseases)
✓ Balanced nutrition with minerals
✓ Tick & vector control
✓ Avoid grazing in marshy hill slopes

Key Message:

Hilly-region grazing predisposes sheep to parasitic, respiratory, foot and nutritional disorders—preventive health management is essential.

SHEEP POX + PASTEURELLOSIS
Mixed Infection in Sheep Farms
(Field Recognition • Treatment • Prevention)
FieldvetDrMSS
CAUSATIVE AGENTS
Sheep Pox
• Caused by Sheeppox virus (Family; Poxviridae)
• Highly contagious viral disease
• Causes systemic illness & immunosuppression
Pasteurellosis
(Secondary Bacterial Pneumonia)
• Pasteurella multiocida
• Mannheimia haemolytica
• Opportunistic respiratory pathogens
• Triggered by stress or viral infection
RISK FACTORS
• Overcrowding
• Poor ventilation
• Transport stress
• Nutritional deficiency
• Sudden weather changes
• Introduction of new animals without quarantine
• Lack of vaccination
MIXED INFECTION ALERT
• Skin pox lesions + severe pneumonia
• Rapid spread within flock
• Mortality >15~20%
• Poor response if antibiotics started late
RISK FACTORS
Sheep Pox Signs
• High fever (104, 106~F)
• Papules → pustules → thick scabs
• Lesions on lips, nostrils, eyelids, udder, perineum
• Enlarged lymph nodes
• Nasal discharge
Pasteurellosis Signs
• Sudden respiratory distress
• Dyspnea & open-mouth breathing
• Mucopurulent nasal discharge
• Coughing.
• Frothy mouth (acute cases)
• Sudden death in peracute form
CLINICAL SIGNS
• Skin pox lesions + severe pneumonia
• Rapid spread within flock
• Mortality >15-20%
• Poor response if antibiotics started late
OUTBREAK ACTION PLAN
1. Immediate isolation
2. Start antibiotic therapy early
3. Provide supportive therapy to whole flock
4. Disinfect housing daily
PREVENTION (FIELD PROTOCOL)
• For Pasteurellosis Component
• Long-acting Oxytetracycline
• Enrofloxacin
• Ceftifour
• NSAIDs (Flunixin / Meloxicam)
OUTRREAK ACTION PLAN
1. Immediate isolation
2. Start antibiotic therapy early
3. Provide supportive therapy to whole flock
4. Disinfect housing daily
FieldvetDrMSS
KEY MESSAGE
Sheep Pox weakens immunity.
Pasteurellosis causes fatal pneumonia.
Early treatment • vaccination saves flocks.

Section 6: Infectious & Major Diseases

22. Sheep Pox + Pasteurellosis Mixed Infection

1. Diseases Involved

🐑 **Sheep Pox**

Caused by Sheeppox virus

Family:Poxviridae
Highly contagious viral disease of sheep.

Pasteurellosis

- Caused mainly by Pasteurella multocida and Mannheimia haemolytica
- Bacterial respiratory disease leading to pneumonia and septicemia.

2. Why Mixed Infection Occurs

Sheep pox weakens the animal's immune system and damages skin and respiratory tissues, allowing bacteria to invade.

Predisposing factors

- Stress due to transport or overcrowding
- Poor nutrition
- Sudden climate change
- Long-distance grazing migration
- Poor ventilation in sheep sheds
- Delay in vaccination

3. Clinical Signs in Mixed Infection

Skin Lesions (Sheep Pox)

- Fever (40–41°C)

Nodules and scabs on:

- Lips
- Nose
- Eyelids
- Ears
- Udder and tail region
- Thick skin lesions that later become crusts.

Respiratory Signs (Pasteurellosis)

- Severe pneumonia
- Nasal discharge

- Coughing
- Rapid breathing
- Depression and weakness

Systemic Signs

- Loss of appetite
- High mortality in lambs
- Sudden deaths in severe cases

4. Post-Mortem Findings

- Sheep Pox
- Nodular lesions in skin and lungs
- Enlarged lymph nodes
- Pox lesions in respiratory mucosa
- Pasteurellosis
- Fibrinous pneumonia
- Consolidated lung lobes
- Pleural fluid accumulation

5. Diagnosis

Clinical signs and flock history

Laboratory confirmation:

- PCR for Sheeppox virus
- Bacterial culture for Pasteurella multocida

Differential diagnosis includes:

- Goat pox
- Contagious ecthyma
- Bluetongue

6. Treatment (Supportive + Secondary Infection Control)

For bacterial pneumonia

- Broad spectrum antibiotics
- Oxytetracycline
- Enrofloxacin
- Ceftiofur

Supportive therapy

- Anti-inflammatory drugs
- Antihistamines
- Vitamin B complex

- Electrolytes and fluids

Wound care

- Antiseptic dressing for skin lesions.

(Note: No specific antiviral treatment for sheep pox.)

7. Prevention and Control

Vaccination

- Sheep pox vaccination in endemic areas.

Biosecurity

- Isolate affected animals
- Restrict animal movement
- Disinfect sheds and equipment

Management

- Avoid overcrowding
- Provide balanced nutrition
- Ensure adequate ventilation

Flock health monitoring

- Early detection and treatment of respiratory infections.

8. Economic Impact

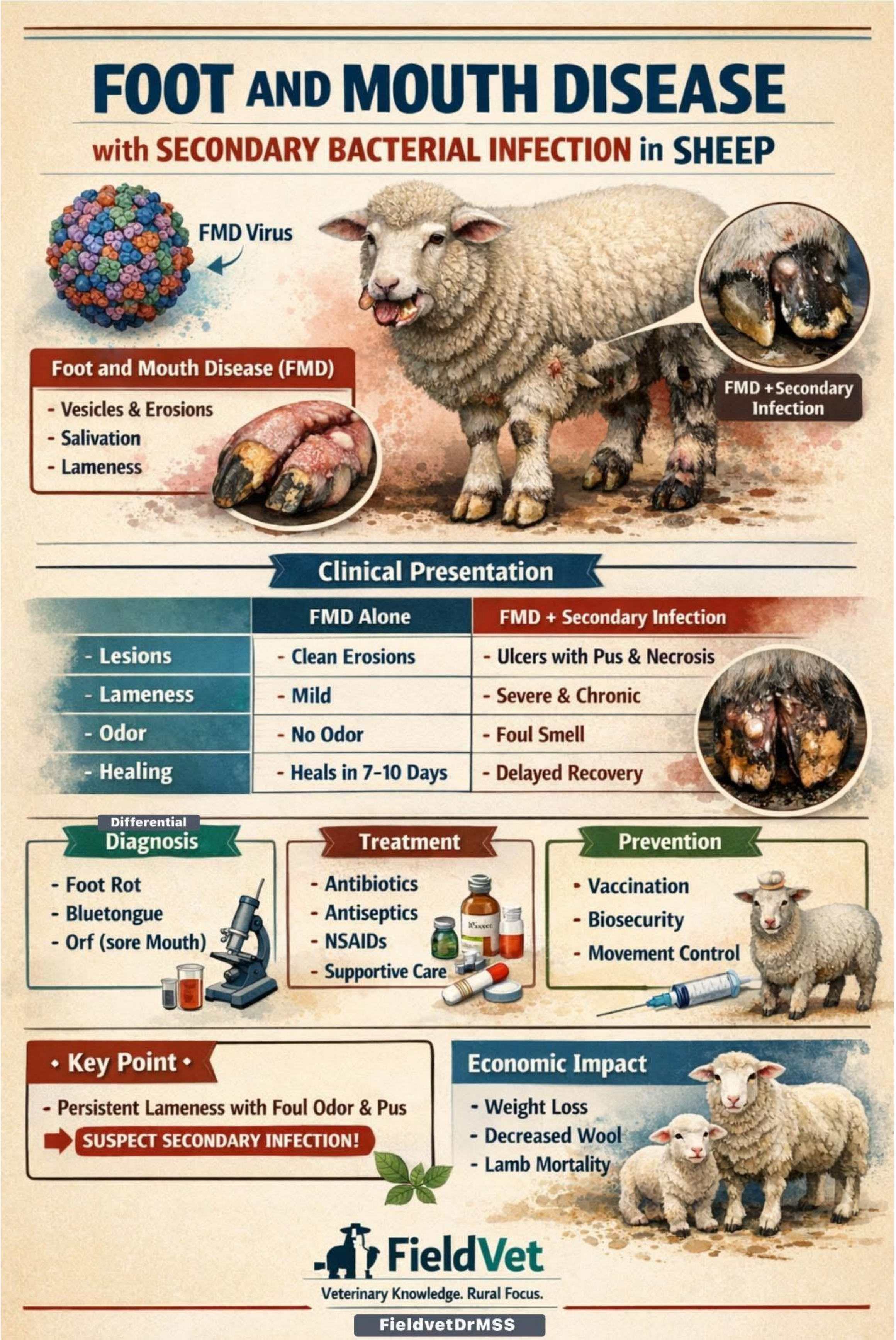
Mixed infections cause:

- High mortality in lambs
- Reduced weight gain
- Wool damage
- Increased treatment costs
- Trade restrictions during outbreak

Key Field Message

“Sheep pox weakens the animal – Pasteurellosis kills the flock.”

Early vaccination, proper management, and rapid veterinary intervention are essential to prevent severe outbreaks.


FOOT AND MOUTH DISEASE
with SECONDARY BACTERIAL INFECTION in SHEEP
FMD Virus
Foot and Mouth Disease (FMD)
- Vesicles & Erosions
- Salivation
- Lameness
FMD +Secondary Infection
Clinical Presentation
FMD Alone
FMD + Secondary Infection
- Lesions
- Clean Erosions
- Ulcers with Pus & Necrosis
- Lameness
- Mild
- Severe & Chronic
- Odor
- No Odor
- Foul Smell
- Healing
- Heals in 7-10 Days
- Delayed Recovery
Differential Diagnosis
- Foot Rot
- Bluetongue
- Orf (sore Mouth)
Treatment
- Antibiotics
- Antiseptics
- NSAIDs
- Supportive Care
Prevention
• Vaccination
- Biosecurity
- Movement Control
• Key Point •
- Persistent Lameness with Foul Odor & Pus
SUSPECT SECONDARY INFECTION!
Economic Impact
- Weight Loss
- Decreased Wool
- Lamb Mortality
FieldVet
Veterinary Knowledge. Rural Focus.
FieldvetDrMSS

23. Foot and Mouth Disease (FMD)

1️⃣ Primary Disease: Foot and Mouth Disease (FMD)

Causative Agent: Foot-and-mouth disease virus

Family: Picornaviridae

Affects: Sheep, goats, cattle, pigs, and other cloven-hoofed animals

🔬 Pathogenesis in Sheep:

Sheep often show mild or subclinical signs, making them important carriers in endemic areas (relevant in Indian small ruminant systems).

- Virus enters via inhalation or ingestion
- Replicates in pharyngeal epithelium
- Viremia → vesicle formation

Clinical Signs in Sheep

- Mild fever (often unnoticed)
- Lameness (most common sign)
- Vesicles/erosions in:
- Interdigital space
- Coronary band
- Dental pad (rare compared to cattle)
- Salivation (less obvious than cattle)
- Sudden drop in body condition
- In lambs: myocarditis ("tiger heart") → sudden death

2️⃣ Secondary Bacterial Infection:

Why Secondary Infection Occurs?

FMD causes:

- Ruptured vesicles
- Raw erosions in feet and mouth
- Compromised epithelial barrier

This predisposes to invasion by bacteria such as:

- Fusobacterium necrophorum
- Dichelobacter nodosus
- Staphylococcus spp.
- Trueperella pyogenes

Differential Diagnosis:

- Foot rot

- Bluetongue
- Contagious ecthyma (Orf)
- Sheep pox
- Vesicular stomatitis

💊 **Treatment Protocol:**

⚠ **Important: There is no specific antiviral treatment for FMD. Management is supportive.**

A. Supportive Care

- Soft bedding
- Isolation
- Electrolytes
- NSAIDs (e.g., Meloxicam)

B. Control of Secondary Bacterial Infection

• Broad-spectrum antibiotics:

- Oxytetracycline (long-acting)
- Penicillin + Streptomycin

• Topical antiseptics:

- 1% Potassium permanganate wash
- Povidone iodine
- 5% Copper sulfate footbath

C. Pain & Inflammation Control

- NSAIDs
- Avoid excessive movement

🛡 **Prevention & Control:**

Vaccination

- Regular vaccination with polyvalent FMD vaccine
- Booster as per endemic schedule (biannual in India)

Biosecurity

- Quarantine new animals (minimum 2–3 weeks)
- Disinfection of sheds
- Restrict movement during outbreaks

Herd-Level Strategy:

In endemic areas like India:

- Mass vaccination

- Movement control
- Early reporting to veterinary authorities

⚠ **Economic Importance:**

- Reduced weight gain
- Decreased wool production
- Lamb mortality
- Trade restrictions

🔬 **Key Veterinary Insight (Field Perspective):**

In sheep, lameness may be the only visible sign of FMD. If lameness persists beyond 7–10 days with foul smell and pus, suspect secondary bacterial involvement.


GOAT POX in SHEEP
FieldvetDrMSS
Capripoxvirus
• Causative Agent: Capripoxvirus (Family: Poxviridae)
• Highly Contagious • Major Loss in Small Ruminants
EPIDEMIOLOGY
• Contagious Viral Disease of Sheep & Goats
• High Fever, Skin Lesions, Mortality in Lambs
• Airborne + Contact Transmission
• Economic Loss – Wool, Weight, Trade
TRANSMISSION
Direct Contact (Lesions, Scabs)
Aerosol (Inhalation)
Contaminated Objects (Fomites)
Blood, Body Fluids
Insects (Flies, Ticks)
Incubation Period: 4 – 14 Days
CLINICAL SIGNS IN SHEEP
Macule → Papule → Vesicle → Pustule → Scab
Systemic Phase
High Fever (40–41.5°C)
Depression
Anorexia
Nasal & Eye Discharge
Lymphadenopathy
Skin Lesions - Common Sites
Face
Ears
Nose
Udder
Tail Base
Perineum
Inner Thighs
Udder
Scrotum
Severe Cases
Pneumonia
Diarrhea
Abortion
Lamb Mortality
POSTMORTEM FINDINGS
• Nodules in Skin
• Pock Lesions in Lungs
• Enlarged Lymph Nodes
• Edema
DIFFERENTIAL DIAGNOSIS
• Orf (Contagious Ecthyma)
• Bluetongue
• Dermatophilosis
• Sheep Pox
TREATMENT (No Specific Antiviral)
• Supportive Care
• Antibiotics (Oxytetracycline)
• NSAIDs (Meloxicam)
• Antipyretics, Fluids
• Wound Care (Povidone Iodine)
Isolation of Sick Animals
PREVENTION & CONTROL
• Vaccination
Live Attenuated Vaccine (Annual)
• Biosecurity
Quarantine (21–30 days)
Disinfection (Sodium Hypochlorite)
Dispose Scabs/ Carcasses
Movement Restriction
ECONOMIC IMPACT
Weight Gain
Wool Quality
Skin Damage
Mortality in Lambs
Trade Restrictions
FIELD VETERINARY INSIGHT
Sudden Fever + Generalized Nodular Skin Lesions
Enlarged Lymph Nodes → Strongly Suggests
Capripox Infection
FieldVet
Veterinary Knowledge. Rural Focus.

24. Goat Pox in Sheep

🦠 Causative Agent

Capripoxvirus

Family: Poxviridae

- Sheeppox and goatpox are caused by closely related Capripox viruses.
- Host specificity is common, but cross-infection can occur in mixed flocks.

📍 Epidemiological Importance (India & Endemic Regions):

- Highly contagious viral disease
- Significant morbidity and mortality (especially lambs)
- Major economic loss in small ruminant production
- Important transboundary disease

Transmission:

- Direct contact (skin lesions, scabs)
- Aerosol droplets
- Contaminated equipment
- Insects (mechanical vectors)

Incubation period: 4–14 days

Clinical Signs in Sheep:

1 Systemic Phase

- High fever (40–41.5°C)
- Depression
- Nasal & ocular discharge
- Enlarged superficial lymph nodes

2 Skin Lesion Progression

Macule → Papule → Vesicle → Pustule → Scab

Common Sites:

- Face
- Ears
- Perineum
- Tail region
- Inner thighs
- Udder & scrotum

Severe Cases

- Pneumonia (nodular lung lesions)
- Diarrhoea
- Abortion
- High lamb mortality

🔎 **Postmortem Findings:**

- Nodules in skin
- Pock lesions in lungs
- Enlarged lymph nodes
- Edematous skin

🧪 **Differential Diagnosis:**

- Orf (Contagious ecthyma)
- Bluetongue
- Dermatophilosis
- Sheep pox (if goat-origin virus suspected)

💊 **Treatment:**

⚠ **No specific antiviral therapy.**

Supportive Care

- Broad-spectrum antibiotics (to prevent secondary bacterial infection)
- NSAIDs for fever
- Antihistamines if needed
- Fluid therapy in severe cases
- Isolation of affected animals

🛡 **Prevention & Control:**

Vaccination

- Live attenuated sheeppox/goatpox vaccine
- Annual vaccination in endemic areas

Biosecurity

- Strict quarantine (21–30 days)
- Disinfection of sheds
- Proper disposal of scabs & carcasses
- Movement restriction during outbreak

Economic Impact:

- Reduced weight gain

- Wool damage
- Skin quality deterioration
- Lamb mortality
- Trade restrictions

Field Veterinary Insight:

In endemic rural flocks, sudden fever + generalized nodular skin lesions + enlarged lymph nodes strongly suggest Capripox infection.


ORF IN LAMBS
ZOONOTIC WARNING
Orf can infect humans causing painful skin lesions.
✓ Wear gloves while handling infected animals
✓ Wash hands thoroughly after handling
FieldvetDrMSS
Meta AI

25. ORF (Contagious Ecthyma)

Causes • Symptoms • Diagnosis • Treatment • Control

🔬 **What is Orf?**

Orf is a highly contagious viral disease of sheep and goats, commonly affecting young lambs (1–6 weeks of age). It is a zoonotic disease (can infect humans).

🦠 **Cause:**

Causative agent:

➡ Orf virus (Parapoxvirus, Poxviridae family)

Source of infection:

- Infected sheep & goats
- Scabs in the environment (virus survives for months)

Mode of transmission:

- Direct contact with lesions
- Through abrasions on lips, gums, teats
- Contaminated feeders, fencing, bedding

⚠ **Clinical Symptoms (Little Lambs):**

Early Signs

- Fever (mild)
- Reduced suckling
- Depression

Typical Lesions

• Papules → vesicles → pustules → thick scabs

Lesions mainly on:

- Lips & mouth angles
- Gums & tongue
- Nose & nostrils
- Occasionally eyelids

Advanced / Complicated Cases

- Painful mouth → starvation
- Weight loss & poor growth
- Secondary bacterial infection
- Pneumonia or death (rare but possible)

🔬 **Diagnosis:**

Field Diagnosis

- Characteristic scabby lesions around mouth
- History of outbreak in flock
- Affects young lambs primarily

Laboratory Diagnosis (Confirmatory)

- PCR
- Virus isolation
- Electron microscopy

📌 **Usually field diagnosis is sufficient**

💊 **Treatment:**

⚠ **No specific antiviral treatment available**

Supportive & Symptomatic Treatment

- Topical antiseptics (povidone iodine, potassium permanganate)
- Antibiotic ointment to prevent secondary infection
- Broad-spectrum antibiotics (if severe secondary infection)
- NSAIDs for pain & fever
- Soft, palatable feed
- Assist feeding if lambs cannot suckle

🛡 **Control & Prevention:**

🧼 **Biosecurity Measures**

✔ Isolate affected lambs

✔ Disinfect feeders & water troughs

✔ Burn or safely dispose scabs

✔ Avoid handling lesions with bare hands

💉 **Vaccination:**

✔ Live Orf vaccine (where endemic)

✔ Vaccinate only healthy animals

✔ Avoid vaccinating disease-free flocks

🐑 **Flock Management:**

✔ Avoid overcrowding

- ✔ Prevent injuries around mouth
- ✔ Maintain clean housing
- ✔ Ensure good nutrition & colostrum intake

⚠ **Zoonotic Warning:**

Orf can infect humans causing painful skin lesions.

- ✔ Wear gloves while handling infected animals
- ✔ Wash hands thoroughly after handling

📌 **Field Tip:**

Orf is self-limiting but causes heavy production loss in lambs—early isolation prevents flock spread.


Tetanus in Sheep
Tetanus is preventable but often fatal — vaccination and wound hygiene are the only reliable safeguards.
Meta AI
FieldvetDrMSS

26.

26. Tetanus In Sheep

Cause:

- Caused by Clostridium tetani, an anaerobic, spore-forming bacterium
- Spores present in soil, manure, dust

Infection enters through:

- Wounds and punctures
- Castration, docking, shearing wounds
- Lambing injuries
- Contaminated injections or surgical procedures
- Produces a potent neurotoxin (tetanospasmin)

Predisposing Factors:

- Poor wound hygiene
- Anaerobic (deep, closed) wounds
- Lack of vaccination
- Unsanitary husbandry practices

Symptoms:

- Incubation period: 3–21 days
- **Early signs:**
    - Stiff gait, reluctance to move
    - Muscle tremors
    - Difficulty opening mouth (lockjaw)
- **Progressive signs:**
    - Rigid extension of legs
    - Erect ears, wrinkled forehead
    - Tail raised stiffly
    - Hypersensitivity to sound and touch
- **Advanced stage:**
    - Opisthotonus (arched back)
    - Convulsions
    - Respiratory failure
    - High mortality rate (often >70%)

Diagnosis:

- **Primarily clinical diagnosis based on:**
  - History of recent wound or surgery
  - Characteristic muscle rigidity
  - Laboratory confirmation is rarely practical
- **Differential diagnosis:**
  - Hypocalcaemia
  - Strychnine poisoning
  - Polioencephalomalacia

Treatment:

(Treatment is often unrewarding once signs appear)

1. Neutralize toxin

- Tetanus antitoxin (if available) – early stage only

2. Eliminate bacteria

- Penicillin or broad-spectrum antibiotics

3. Wound management

- Locate and thoroughly clean wound
- Debridement and antiseptic dressing

4. Control muscle spasms

- Sedatives (diazepam, chloral hydrate)

5. Supportive care

- Quiet, dark environment
- Soft bedding
- Assisted feeding and hydration

Control & Prevention:

Most Important: Vaccination

Tetanus toxoid:

- First dose: 6–8 weeks of age
- Booster after 4 weeks
- Annual booster thereafter
- Vaccinate before castration, docking, or shearing

Management Practices:

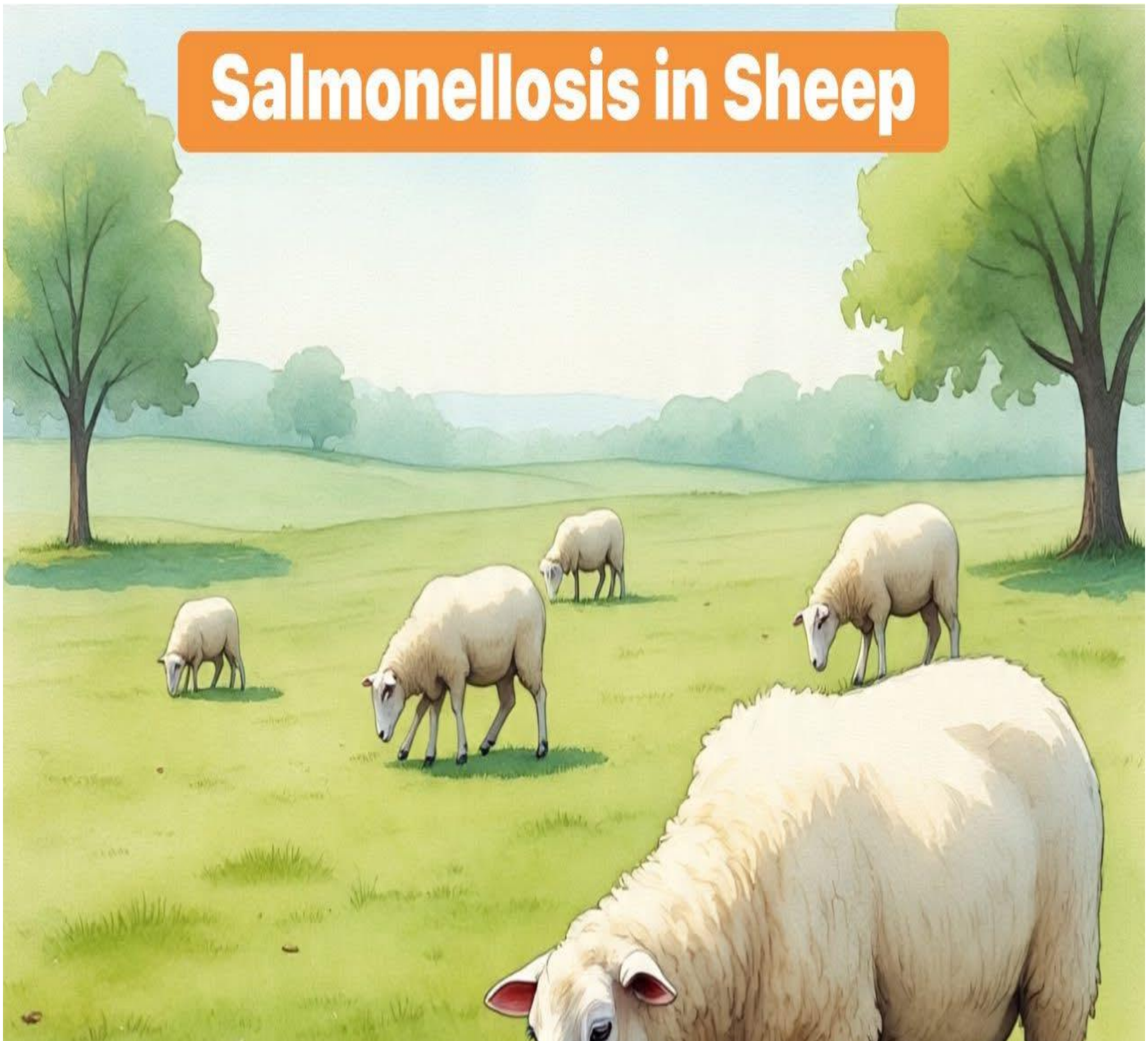
- Maintain clean housing
- Use sterile instruments
- Proper wound disinfection
- Avoid unnecessary invasive procedures
- Immediate treatment of wounds

Public Health Importance:

- Zoonotic organism present in environment
- Humans at risk through contaminated wounds
- Proper hygiene and vaccination are essential.

Key Message:

Tetanus is preventable but often fatal — vaccination and wound hygiene are the only reliable safeguards


Salmonellosis in Sheep
Salmonellosis is a highly contagious, stress-associated, water- and feed-borne disease causing diarrhoea, abortions, and sudden deaths in sheep.
Meta AI
FieldvetDrMSS

27. Salmonellosis In Sheep

Cause:

- Salmonella spp.
- Common serotypes: S. Typhimurium, S. Dublin, S. Enteritidis
- Gram-negative, facultative intracellular bacteria

Source & Mode of Infection:

- Contaminated feed and drinking water
- Fecal contamination of sheds and water troughs
- Carrier animals shedding organisms
- **Stress factors:** transport, overcrowding, poor nutrition

Symptoms / Clinical Signs:

Adult Sheep

- High fever (40–41°C)
- Profuse watery or bloody diarrhoea
- Anorexia, depression
- Dehydration
- Abortions (especially late pregnancy)
- Septicemia → sudden death (acute cases)

Lambs

- Severe diarrhoea
- Rapid dehydration
- Weakness
- High mortality

Diagnosis:

- Fecal culture and isolation of Salmonella
- Blood culture in septicemic cases
- PCR for rapid confirmation
- **Post-mortem lesions:**
  - Enteritis
  - Enlarged spleen
  - Congested liver
- **Differential diagnosis:**
  - Colibacillosis
  - Clostridial enterotoxemia
  - Coccidiosis

Treatment:

⚠ **Treat early; prognosis is poor in advanced septicemia**

Antibiotics (based on sensitivity testing):

- Enrofloxacin
- Ciprofloxacin
- Ceftriaxone
- Amoxicillin–clavulanate

Supportive Therapy:

- IV or oral fluid therapy
- Electrolytes
- NSAIDs (Flunixin meglumine, Meloxicam)
- Probiotics after antibiotic therapy

Control & Prevention:

- Provide clean, potable drinking water
- Avoid fecal contamination of feed and water
- Regular cleaning and disinfection of sheds and troughs
- Isolate sick animals immediately
- Reduce stress (transport, overcrowding)
- Proper carcass disposal
- Rodent control on the farm

Zoonotic Importance:

- Zoonotic disease ⚠

Humans may get infected via:

- Direct animal contact
- Contaminated water or meat
- Use gloves and maintain personal hygiene

Key Field Message:

Salmonellosis is a highly contagious, stress-associated, water- and feed-borne disease causing diarrhea, abortions, and sudden deaths in sheep.


sheep Leptospirosis.
Meta AI

28. Leptospirosis in Sheep

🦠 **Cause**

- Leptospira spp. (mainly Leptospira interrogans)
- Important serovars in sheep:
- Hardjo
- Pomona
- Icterohaemorrhagiae

Source of infection:

- Urine of infected animals (sheep, cattle, rodents, dogs)
- Contaminated water, pasture, mud

🔁 **Mode of Transmission:**

Entry through:

- Skin abrasions
- Mucous membranes
- Drinking contaminated water
- Contact with urine-soaked soil or bedding
- Zoonotic transmission to humans ⚠

🤒 **Clinical Signs / Symptoms:**

🔷 **In Sheep**

- Often subclinical
- Fever
- Reduced feed intake
- Lethargy
- Abortion (last trimester)
- Stillbirths / weak lambs
- Reduced fertility
- Occasional jaundice & hemoglobinuria
- Mastitis (rare)

🔷 **In Lambs**

- Fever
- Septicemia
- Sudden death (rare)

🧪 **Diagnosis:**

🔹 **Field Diagnosis**

History of:

- Abortion storms
- Wet grazing areas
- Rodent infestation

🔬 **Laboratory Diagnosis**

- MAT (Microscopic Agglutination Test) – Gold standard
- ELISA
- PCR (blood, urine)
- Culture (special media – time consuming)

📌 **Samples:**

- Blood (acute stage)
- Urine (carrier animals)
- Aborted fetus / placenta

💉 **Treatment:**

🔹 **Antibiotics (Early treatment is critical)**

- Oxytetracycline
- 10–20 mg/kg IM/IV
- Streptomycin
- 25 mg/kg IM (effective against carriers)
- Penicillin (supportive)

🔹 **Supportive Therapy**

- NSAIDs
- Fluids (if dehydrated)
- Vitamins

⚠ **Treat entire flock if outbreak suspected.**

🛡 **Control & Prevention:**

🔹 **Farm-Level Control:**

✓ Isolate affected animals
✓ Improve drainage & hygiene
✓ Rodent control (very important)
✓ Avoid stagnant water sources

Vaccination:

- Polyvalent Leptospira vaccine
- Annual booster in endemic areas

Biosecurity:

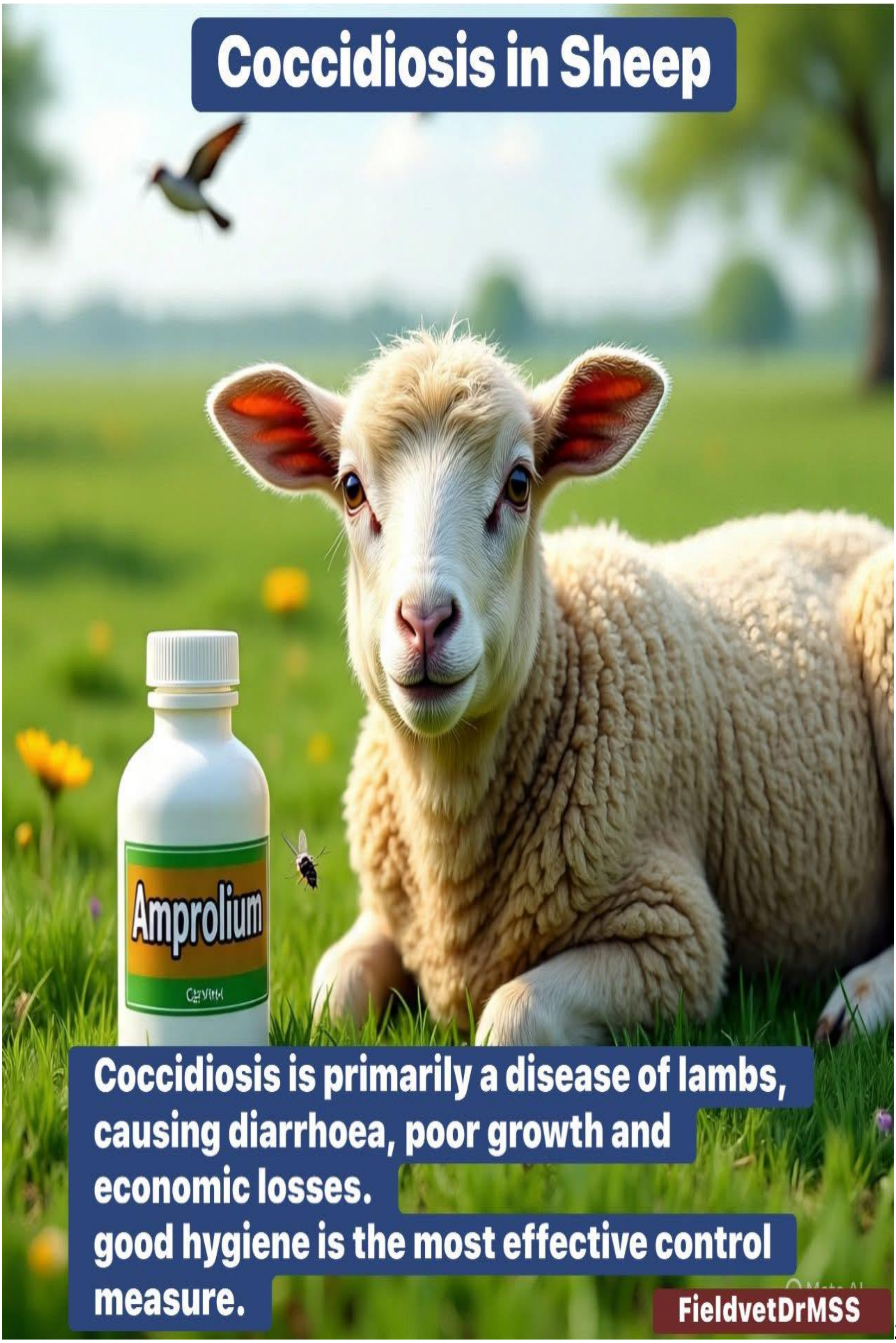
- ✓ Disinfect urine-contaminated areas
- ✓ Use gloves while handling animals
- ✓ Safe disposal of aborted materials

Public Health Importance:

- Zoonotic disease
- Causes fever, jaundice, kidney failure in humans
- High risk to:
  - Shepherds
  - Veterinarians
  - Farm workers

One-Line Summary:

"Leptospirosis in sheep is a zoonotic bacterial disease causing abortion, infertility, and fever, transmitted through urine-contaminated water and controlled by antibiotics, vaccination, and strict hygiene."


Coccidiosis in Sheep
Amprolium
Coccidiosis is primarily a disease of lambs,
causing diarrhoea, poor growth and
economic losses.
good hygiene is the most effective control
measure.
FieldvetDrMSS

29.

29. Coccidiosis In Sheep

Cause:

- Eimeria species (protozoan parasites)
- **Important species:** E. ovinoidalis, E. crandallis
- Host-specific intestinal parasites

Source & Mode of Infection:

- Ingestion of sporulated oocysts from:
  - Contaminated drinking water
  - Feed troughs
  - Wet, dirty floors and bedding
- Favoured by:
  - Overcrowding
  - Poor hygiene
  - Stress (weaning, transport)
  - Wet and humid conditions

Symptoms / Clinical Signs:

Lambs (most affected)

- Profuse watery or bloody diarrhoea
- Tenesmus (straining)
- Dehydration
- Weight loss, stunted growth
- Rough hair coat
- Weakness, recumbency
- Death in severe cases

Adult Sheep

- Usually subclinical
- Act as carriers, shedding oocysts

Diagnosis:

- **Fecal examination:**
  - Oocyst detection by flotation technique
  - Oocyst count (OPG)

- **History of:**
    - Diarrhoea in lambs
    - Poor growth
    - Overcrowded housing
- **Post-mortem lesions:**
    - Thickened intestinal mucosa
    - Hemorrhagic enteritis
- **Differential diagnosis:**
    - Colibacillosis
    - Salmonellosis
    - Enterotoxemia

Treatment:

◈ Early treatment gives best results

Anticoccidial Drugs

- Amprolium
- Sulfa drugs (Sulfadimidine, Sulfaquinoxaline)
- Toltrazuril
- Diclazuril

Supportive Therapy

- Oral/IV fluids
- Electrolytes
- Multivitamins
- Probiotics

Control & Prevention:

- Avoid overcrowding
- Maintain dry, clean housing
- Regular cleaning and disinfection of pens
- Raise feed and water troughs
- Strategic prophylactic anticoccidial treatment in lambs
- Proper drainage to avoid wet floors

• Quarantine newly introduced animals

Zoonotic Importance:

• ❌ Not zoonotic.

Key Field Message:

Coccidiosis is primarily a disease of lambs, causing diarrhoea, poor growth and economic losses—good hygiene is the most effective control measure.


Worm Infestation-Induced
Bacterial Diseases of Sheep
(Secondary Bacterial Infections Following Parasitism)
FieldvetDrMSS
Heavy gastrointestinal or external parasitic infestations weaken immunity, damage tissues, and create entry points for opportunistic bacteria. This leads to secondary bacterial diseases, increasing mortality and production losses.
1 Enterotoxemia (Pulpy Kidney Disease)
Primary Trigger: Heavy gastrointestinal worm burden causing gut damage
Bacteria: Clostridium perfringens
Mechanism: Worm-induced intestinal injury
Sudden dietary changes • parasite Stress
Rapid bacterial multiplication and toxin release
Signs: Sudden death
Abdominal pain
Neurological signs
2 Clostridial Enteritis
Associated With: Severe parasitism (especially Harnorchus spp).
Bacteria: Clostridium septicum
Pathogenesis: • Mucosal damage from worms
Bacterial invasion through damaged gut lining.
Clinical Signs: • Diarrhea
Toxemia
Sudden death in lambs
3 Pasteurellosis (Secondary Pneumonia)
Predisposing Factor: Stress • parasitic anemia
Bacteria: Mannhelmia haemolytica
Mechanism: Worm-induced immune suppression
Reduced resistance to respiratory pathogens.
Signs: Fever
Nasal discharge
Respiratory distress
4 Foot Rot Complications
Predisposing Factor: Weak immunity due
parasitic load
Management:
Regular hoof trimming
Foot baths
Strategic parasite control
6 Foot Rot Complications
Predisposing Factor: Weak immunity due parasitic load
Bacteria: Dichebocecter nodosus.
5 Liver Abscess (Following Liver Fluke Damage)
Primary Parasite: Fasciolia hepatica
Secondary Bacteria: mysperctic aurelis
Mechanism: • Pustules
Crust formation
Wool loss
Secondary wound infections
Pathophysiological Link
Worm Infestation leads to:
Nutritional
Anemia
Protem loss
Inmune suppression
• Nutritional deficiency
• Anemia
• Protem loss
• Immune suppression
• Tissue damage
6 Skin Pyoderma (Following External Parasites)
Primepsing Parasites: use:
• Mftes
• Ticks
Common Bacteria:
Pustules Crust formation
Wool loss • Secondary wound infections
Veterinary Prevention Strategy
Strategic deworming (FAMACHA based) Pasture rotation Balanced mineral supplementation
Vaccination against clostridial diseases Biosecurity & hygiene Early diagnosis and treatment
#FieldVet Worm Infestation Induced Bacterial Diseases of Sheep

Section 7: Parasitic Infestations

30. Worm Infestation–Induced Diseases

(Secondary Bacterial Infections Following Parasitism)

Heavy gastrointestinal or external parasitic infestations weaken immunity, damage tissues, and create entry points for opportunistic bacteria. This leads to secondary bacterial diseases, increasing mortality and production losses.

1 Enterotoxemia (Pulpy Kidney Disease)

Primary Trigger: Heavy gastrointestinal worm burden causing gut damage

Bacteria: Clostridium perfringens

Mechanism:

- Worm-induced intestinal injury
- Sudden dietary changes + parasite stress
- Rapid bacterial multiplication and toxin release

Signs:

- Sudden death
- Abdominal pain
- Neurological signs

Prevention:

- Strategic deworming
- Regular vaccination
- Proper nutrition management

2 Clostridial Enteritis

Associated With: Severe parasitism (especially Haemonchus spp.)

Bacteria: Clostridium septicum

Pathogenesis:

- Mucosal damage from worms
- Bacterial invasion through damaged gut lining

Clinical Signs:

- Diarrhea
- Toxemia
- Sudden death in lambs

3️⃣ Pasteurellosis (Secondary Pneumonia):

Predisposing Factor: Stress + parasitic anemia

Bacteria: Mannheimia haemolytica

Mechanism:

- Worm-induced immune suppression
- Reduced resistance to respiratory pathogens

Signs:

- Fever
- Nasal discharge
- Respiratory distress

Control:

- Deworming program
- Reduce overcrowding
- Vaccination (where available)

4️⃣ Foot Rot Complications:

Predisposing Factor: Weak immunity due to parasitic load

Bacteria: Dichelobacter nodosus

Signs:

- Lameness
- Foul-smelling hoof lesions
- Reduced grazing

Management:

- Regular hoof trimming
- Foot baths
- Strategic parasite control

5️⃣ Liver Abscess (Following Liver Fluke Damage):

Primary Parasite: Fasciola hepatica

Secondary Bacteria: Trueperella pyogenes

Mechanism:

- Fluke migration damages liver tissue
- Bacterial invasion of necrotic areas

Signs:

- Weight loss
- Reduced productivity
- Condemnation at slaughter

6 Skin Pyoderma (Following External Parasites):

Predisposing Parasites:

- Lice
- Mites
- Ticks

Common Bacteria: Staphylococcus aureus

Signs:

- Pustules
- Crust formation
- Wool loss
- Secondary wound infections

⚠ **Pathophysiological Link:**

Worm infestation leads to:

- Nutritional deficiency
- Anemia
- Protein loss
- Immune suppression
- Tissue damage

➡ Creates ideal conditions for bacterial invasion.

Veterinary Prevention Strategy:

✓ Strategic deworming (FAMACHA-based)
✓ Pasture rotation
✓ Balanced mineral supplementation
✓ Vaccination against clostridial diseases
✓ Biosecurity & hygiene
✓ Early diagnosis and treatment


Lungworm Infestation in Sheep
(Verminous Pneumonia)
FieldvetDrMSS
CAUSES
Causative Parasites
Dictyocaulus filaria (most common)
Protostrongylus rufescens
Muellerius capillaris
Mode of Infection
Ingestion of infective larvae (L3) from pasture
High risk in wet, cool climates
Overgrazing & poor pasture rotation
PREDISPOSING FACTORS
Rainy season & high humidity
Young sheep and lambs
Overstocking
Poor nutrition
Continuous grazing on same pasture
Severe Cases
Pneumonia
Secondary bacterial infection
Death (rare but possible)
SYMPTOMS
Early Signs
Frequent coughing (especially after exercise)
Nasal discharge
Mild fever
Reduced appetite
Advanced Signs
Persistent deep cough
Laboured breathing (dyspnoea)
Open-mouth breathing
Weight loss and poor growth
Exercise intolerance
DIAGNOSIS
History of grazing in wet pastures
Characteristic coughing
Poor response to antibiotics alone
Laboratory Diagnosis
Faecal examination (Baermann technique) – gold standard
Detection of lungworm larvae
TREATMENT
Anthelmintics (Highly Effective)
Albendazole
Fenbendazole
Ivermectin
Levamisole
Moxidectin
(As per body sveight and veterinary advice)
CONTROL & PREVENTION
Pasture Management
Rotational grazing
Avoid wet & marshy grazing areas
Mixed grazing with cattle
Strategic Deworming
Based on faecal egg count
Before rainy season
Mid-rainy season
Lungworm infestation is a pasture associated disease - early diagnosis, and strategie deworming prevent ecenomic losses and respiratory disaess in Sheep.
Veterinary Parasitology | Sheep Respiratory Diseases | Preventive Health | #Fieldvet

31. Lungworm Infestation

(Verminous Pneumonia)

Causes:

Causative Parasites

- Dictyocaulus filaria (most common)
- Protostrongylus rufescens
- Muellerius capillaris

Mode of Infection

- Ingestion of infective larvae (L3) from pasture
- High risk in wet, cool climates
- Overgrazing & poor pasture rotation

Predisposing Factors:

- Rainy season & high humidity
- Young sheep and lambs
- Overstocking
- Poor nutrition
- Continuous grazing on same pasture

Symptoms:

Early Signs

- Frequent coughing (especially after exercise)
- Nasal discharge
- Mild fever
- Reduced appetite

Advanced Signs

- Persistent deep cough
- Laboured breathing (dyspnoea)
- Open-mouth breathing
- Weight loss and poor growth
- Exercise intolerance

Severe Cases

- Pneumonia
- Secondary bacterial infection
- Death (rare but possible)

Diagnosis:

Field Diagnosis

- History of grazing in wet pastures
- Characteristic coughing
- Poor response to antibiotics alone

Laboratory Diagnosis

- Faecal examination (Baermann technique) – gold standard
- Detection of lungworm larvae

Post-mortem Findings

- Adult worms in trachea & bronchi
- Consolidated lungs
- Verminous pneumonia lesions

Treatment:

Anthelmintics (Highly Effective)

- Albendazole
- Fenbendazole
- Ivermectin
- Levamisole
- Moxidectin

(As per body weight and veterinary advice)

Supportive Therapy:

- Broad-spectrum antibiotics (for secondary infection)
- NSAIDs (to reduce inflammation)
- Good nutrition & rest

Control & Prevention:

Pasture Management

✓ Rotational grazing
✓ Avoid wet & marshy grazing areas
✓ Mixed grazing with cattle

Strategic Deworming

✓ Before rainy season
✓ Mid-rainy season
✓ Based on faecal egg count

Management Practices

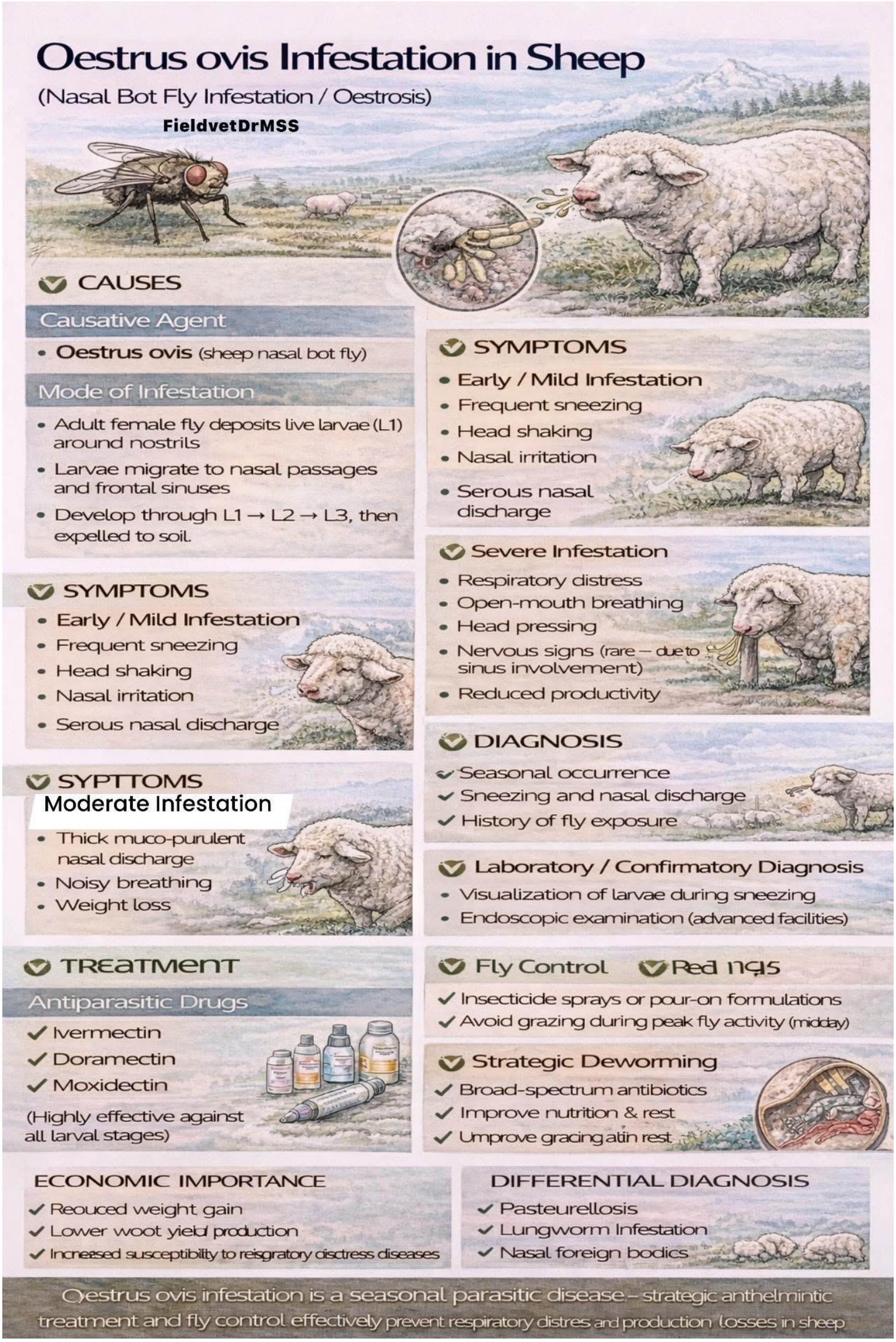
- ✓ Avoid overstocking
- ✓ Improve nutrition
- ✓ Quarantine & deworm new animals

Differential Diagnosis:

- Pasteurellosis
- PPR
- Mycoplasma pneumonia
- Maedi-Visna (chronic cases).

Key Message:

Lungworm infestation is a pasture-associated disease—early diagnosis and strategic deworming prevent economic losses and respiratory distress in sheep.



Oestrus ovis Infestation in Sheep
(Nasal Bot Fly Infestation / Oestrosis)
FieldvetDrMSS
CAUSES
Causative Agent
• Oestrus ovis (sheep nasal bot fly)
Mode of Infestation
• Adult female fly deposits live larvae (L1) around nostrils
• Larvae migrate to nasal passages and frontal sinuses
• Develop through L1 → L2 → L3, then expelled to soil.
SYMPTOMS
• Early / Mild Infestation
• Frequent sneezing
• Head shaking
• Nasal irritation
• Serous nasal discharge
SYPTTOMS
Moderate Infestation
• Thick muco-purulent nasal discharge
• Noisy breathing
• Weight loss
TREaTMenT
Antiparasitic Drugs
✓ Ivermectin
✓ Doramectin
✓ Moxidectin
(Highly effective against all larval stages)
SYMPTOMS
• Early / Mild Infestation
• Frequent sneezing
• Head shaking
• Nasal irritation
• Serous nasal discharge
Severe Infestation
• Respiratory distress
• Open-mouth breathing
• Head pressing
• Nervous signs (rare – due to sinus involvement)
• Reduced productivity
DIAGNOSIS
✓ Seasonal occurrence
✓ Sneezing and nasal discharge
✓ History of fly exposure
Laboratory / Confirmatory Diagnosis
• Visualization of larvae during sneezing
• Endoscopic examination (advanced facilities)
Fly Control
Red
✓ Insecticide sprays or pour-on formulations
✓ Avoid grazing during peak fly activity (midday)
Strategic Deworming
✓ Broad-spectrum antibiotics
✓ Improve nutrition & rest
✓ Umprove gracing atin rest
ECONOMIC IMPORTANCE
✓ Reouced weight gain
✓ Lower woot yield production
✓ Increesed susceptibility to reisgratory disctress diseases
DIFFERENTIAL DIAGNOSIS
✓ Pasteurellosis
✓ Lungworm Infestation
✓ Nasal foreign bodics
Oestrus ovis infestation is a seasonal parasitic disease – strategic anthelmintic treatment and fly control effectively prevent respiratory distres and production losses in sheep

32. Oestrus ovis Infestation

(Nasal Bot Fly Infestation / Oestrosis)

Causes

Causative Agent

- Oestrus ovis (sheep nasal bot fly)

Mode of Infestation

- Adult female fly deposits live larvae (L1) around nostrils
- Larvae migrate to nasal passages and frontal sinuses
- Develop through L1 → L2 → L3, then expelled to soil

Risk Factors

- Warm, dry climate
- Summer and post-monsoon season
- Extensive grazing systems
- Poor fly control

Symptoms

Early / Mild Infestation

- Frequent sneezing
- Head shaking
- Nasal irritation
- Serous nasal discharge

Moderate Infestation

- Thick muco-purulent nasal discharge
- Noisy breathing (snoring sound)
- Reduced feed intake
- Weight loss

Severe Infestation

- Respiratory distress
- Open-mouth breathing
- Head pressing
- Nervous signs (rare – due to sinus involvement)
- Reduced productivity

Diagnosis

Clinical Diagnosis

- Seasonal occurrence

- Sneezing and nasal discharge
- History of fly exposure

Laboratory / Confirmatory Diagnosis

- Visualization of larvae during sneezing
- Endoscopic examination (advanced facilities)

Post-Mortem Findings

- Larvae in nasal cavity & frontal sinuses
- Mucosal inflammation
- Secondary bacterial infection

Treatment

Antiparasitic Drugs

- ✓ Ivermectin
- ✓ Doramectin
- ✓ Moxidectin

(Highly effective against all larval stages)

Supportive Treatment

- Broad-spectrum antibiotics (if secondary infection)
- NSAIDs (reduce inflammation)
- Improve nutrition & rest

Control & Prevention

Fly Control

- ✓ Insecticide sprays or pour-on formulations
- ✓ Avoid grazing during peak fly activity (midday)

Strategic Deworming

- ✓ End of fly season
- ✓ Before breeding season
- ✓ Whole-flock treatment

Management Practices

- ✓ Maintain flock hygiene
- ✓ Avoid overcrowding
- ✓ Quarantine & treat new animals

Economic Importance

- Reduced weight gain

- Lower wool yield
- Reduced milk production
- Increased susceptibility to respiratory diseases

Differential Diagnosis

- Pasteurellosis
- Lungworm infestation
- Nasal foreign bodies
- Sinusitis

Key Message

Oestrus ovis infestation is a seasonal parasitic disease—strategic anthelmintic treatment and fly control effectively prevent respiratory distress and production losses in sheep.


IMPORTANT TICK INFESTATION IN SHEEP
Major Species, Diseases & Control Methods
FieldvetDrMSS
COMMON TICK SPECIES AFFECTING SHEEP
1 Rhipicephalus (Boophilus) spp.
One-host tick
Heavy infestation → anemia, poor growth
Vector of Babesiosis, Anaplasmosis
2 Hyalomma spp.
Long-legged, aggressive tick
Causes severe irritation and wounds
3 Haemaphysalis spp.
Three-host tick
Common in hilly and forest areas
Vector of Theileria, Babesia
4 Ixodes spp.
Found in cooler climates
Vector of Theileria, Babesia
TICK-BORNE DISEASES IN SHEEP
Babesiosis – Fever, anemia, hemoglobinuria
Theileriosis – Lymph node enlargement, emaciation
Anaplasmosis – Progressive anemia, weakness
Tick paralysis – Ascending paralysis, sudden death
Secondary infections – Myiasis, dermatitis
Rotate acaricides to prevent resistance.
CONTROL METHODS OF TICKS IN SHEEP
1 CHEMICAL CONTROL (Acaricides)
Commonly used drugs
Synthetic pyrethroids: Cypermethrin, Deltamethrin
Organophosphates: Chlorpyrifos
Macrocyclic lactones: Ivermectin, Doramectin
2 MANAGEMENTAL CONTROL
Regular inspection of flock
Quarantine new animals (2–3 weeks)
Avoid overstocking
Maintain clean sheds & dry bedding
Rotate Acaricides to prevent resistance
PASTURE MANAGEMENT
Rotational grazing
Pasture spelling (tick life cycle break)
Bush and weed control
Avoid grazing in heavily infested areas during peak season
BIOLOGICAL & ECOLOGICAL METHODS
Encourage natural predators (birds)
Use of tick resistant sheep breeds
Reduce chemical overuse to protect ecosystem
KEY FIELD POINTS
Effective tick control in sheep requires an integrated approach combining acaricides, good management, and pasture hygiene to reduce disease, improve producilvity, and minimize economic losses.

33. Tick Infestation in Sheep

Major Species, Diseases & Control Methods

🕷 Common Tick Species Affecting Sheep

1. Rhipicephalus (Boophilus) spp.

- One-host tick
- Heavy infestation → anemia, poor growth
- Vector of Babesiosis, Anaplasmosis

2. Hyalomma spp.

- Long-legged, aggressive tick
- Causes severe irritation and wounds
- Vector of Theileriosis

3. Haemaphysalis spp.

- Three-host tick
- Common in hilly and forest areas
- Vector of Theileria, Babesia

4. Ixodes spp.

- Found in cooler climates
- Vector of Tick paralysis, louping ill

🧬 Tick-Borne Diseases In Sheep

- Babesiosis – Fever, anemia, hemoglobinuria
- Theileriosis – Lymph node enlargement, emaciation
- Anaplasmosis – Progressive anemia, weakness
- Tick paralysis – Ascending paralysis, sudden death
- Secondary infections – Myiasis, dermatitis

⚠ Effects of Tick Infestation

- Blood loss → Anemia
- Reduced wool quality and weight gain
- Skin damage → leather defects
- Stress and reduced immunity
- Economic losses to farmers

🛡 Control Methods of Ticks in Sheep

1️⃣ Chemical Control (Acaricides)

Commonly used drugs

- Synthetic pyrethroids: Cypermethrin, Deltamethrin
- Organophosphates: Chlorpyrifos
- Macrocyclic lactones: Ivermectin, Doramectin

Application methods

- Dipping
- Spraying
- Pour-on
- Injectable (systemic)

⚠ **Rotate acaricides to prevent resistance**

2️⃣ Managemental Control

- Regular inspection of flock
- Quarantine new animals (2–3 weeks)
- Avoid overstocking
- Maintain clean sheds and dry bedding
- Strategic shearing to expose ticks

3️⃣ Pasture Management

- Rotational grazing
- Pasture spelling (tick life cycle break)
- Bush and weed control
- Avoid grazing in heavily infested areas during peak season

4️⃣ Biological & Ecological Methods

- Encourage natural predators (birds)
- Use of tick-resistant sheep breeds
- Reduce chemical overuse to protect ecosystem

5️⃣ Integrated Tick Management (Best Practice)

✓ Chemical + Management + Pasture control
✓ Strategic treatment during peak seasons
✓ Farmer education & surveillance

📌 **Key Field Points**

- Treat entire flock, not individual animals
- Follow correct dose & interval
- Observe withdrawal period
- Monitor for acaricide resistance

🔍 **Conclusion**

Effective tick control in sheep requires an integrated approach combining acaricides, good management, and pasture hygiene to reduce disease, improve productivity, and minimize economic losses.


Mite Infestation (Mange) IN SHEEP
FieldvetDrMSS
Common Causative Mites
Psoroptes ovis (Sheep Scab Mite)
Sarcoptes scabiei
Chorioptes bovis
Demodex ovis
Clinical Signs
Psoroptic Mange (Sheep Scab)
• Intense Itching & Wool Loss
• Crusty Lesions
• Weight Loss
Sarcoptes Mange
• Thickened Skin
• Hair Loss on Face & Ears
• Severe Itching
Chorioptic Mange
• Lesions on Feet & Tail
• Crusting & Scaling
Demodectic Mange
• Skin Nodules
Diagnosis
Clinical Exam
▸ Itching & Lesions
▸ Flock History
Treatment & Control
Systemic Treatment
▸ Ivermectin, Doramectin, Moxidectin
▸ Repeat Every 7–14 Days
Topical Treatment
▸ Lime Sulfur Dip
▸ Amitraz or Pyrethroid Sprays
▸ Treat Entire Flock
Wool Damage
Weight Loss
Secondary Infections
Reduced Production
Quarantine New Animals
Regular Flock Checks
Disinfect Housing
Disinfect equipment
FieldvetDrMSS

34. Mite Infestation (Mange)

🔬 Common Causative Mites

1. Psoroptes ovis – Causes psoroptic mange (Sheep scab)
2. Sarcoptes scabiei – Causes sarcoptic mange
3. Chorioptes bovis – Causes chorioptic mange (foot and tail mange)
4. Demodex ovis – Causes demodectic mange (rare in sheep)

Clinical Signs

1 Psoroptic Mange (Sheep Scab)

- Intense pruritus (severe itching)
- Sheep rubbing against objects
- Wool loss (patchy alopecia)
- Thick crust formation
- Moist exudative lesions
- Weight loss, anemia in severe cases

2 Sarcoptic Mange

- Severe itching
- Thickened, wrinkled skin
- Hair loss on face, ears, axilla, groin
- Reduced feed intake

3 Chorioptic Mange

- Mild itching
- Lesions on feet, tail, scrotum
- Crusting and scaling

4 Demodectic Mange

- Nodules in skin
- Usually non-pruritic
- Rare and often localized

🔎 Diagnosis

Clinical Diagnosis

- History of intense itching and flock spread
- Typical lesion distribution
- Seasonality (common in winter & overcrowding)

Laboratory Diagnosis

- Deep skin scraping (especially for Sarcoptes & Demodex)
- Superficial skin scraping (Psoroptes, Chorioptes)
- Microscopic examination
- Crust examination

Response to treatment (therapeutic diagnosis)

💊 **Treatment**

Systemic Treatment (Most Effective)

- Ivermectin – 0.2 mg/kg SC (repeat after 7–14 days)
- Doramectin – 0.2 mg/kg SC
- Moxidectin – As per label dose

(Repeat dose is essential to kill newly hatched mites)

Topical Treatment

- Lime sulfur dips
- Amitraz spray (0.025–0.05%)
- Synthetic pyrethroids (cypermethrin spray)

⚠ **Treat entire flock, not just affected animals.**

🛡 **Control & Prevention**

- Quarantine new animals (minimum 2–3 weeks)
- Regular flock inspection
- Avoid overcrowding
- Proper nutrition to improve immunity
- Shearing management
- Disinfection of housing and equipment
- Strategic deworming & ectoparasite control schedule

📉 **Economic Importance**

- Wool damage
- Reduced body weight gain
- Secondary bacterial infections
- Decreased reproductive performance

Trade restrictions in severe outbreaks (especially psoroptic mange)

BEST TREATMENT FOR FOOT ROT IN SHEEP

FieldvetDrMSS

1. Causes

- *Dichelobacter nodosus* (primary bacterium)
- *Fusobacterium necrophorum* (secondary infection)
- Favouring conditions: wet; muddy, unhygienic flooring

2. Clinical Signs

- Lameness, reluctance to Walk
- Foul smell from feet
- Underrunning of hoof wall
- Swollen, painful interdigital space
- Loss of body condition

3. Best Treatment Protocol

A. Foot Bathing (Essential Step)

Use any one of the following solutions:

- 10% Zinc Sulfate (most effective)
- 5% Formalin (effective but irritant; use with caution)
- Copper Sulfate 5–10%

Routine:

- Stand sheep in solution for 10–15 minutes
- Repeat every 5–7 days for 3–4 sessions

B. Systemic Antibiotics

Oxytetracyccline (Long-acting)

- Penicillin–Streptomycin combination
- Amoxicillin

C. Topical Antibiotics

Apply after cleaning and trying:

- Oxytetracycline spray
- Iodine-based antiseptics
- Chlorhexidine solution (0.5%)

D. Hoof Trimming

Remove loose, underrun horn

- Expose infected pockets for aeration

Avoid excessive trimming to prevent bleeding

E. Pain Management

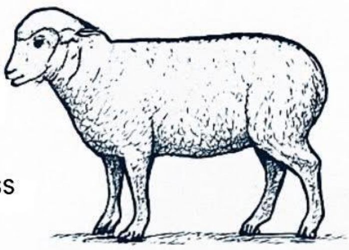
NSAIDs: Meloxicam / Flunixin

Reduces pain and improve mobility

4. Control & Prevention

- Regular hoof trimming (every 3–4 months)
- Keep flooring dry, avold muddy areas
- Quarantine and inspect new animals
- Routine foot baths in rainy season

FieldvetDrMSS



Section 8: Clinical Conditions & Disorders

35. Foot Rot Treatment Protocol

1. Causes

- Dichelobacter nodosus (primary bacterium)
- Fusobacterium necrophorum (secondary infection)
- Favouring conditions: wet, muddy, unhygienic flooring

2. Clinical Signs:

- Lameness, reluctance to walk
- Foul smell from feet
- Underrunning of hoof wall
- Swollen, painful interdigital space
- Loss of body condition

3. Best Treatment Protocol:

A. Foot Bathing (Essential Step)

Use any one of the following solutions:

- 10% Zinc Sulfate (most effective)
- 5% Formalin (effective but irritant; use with caution)
- Copper Sulfate 5–10%

Routine:

- Stand sheep in solution for 10–15 minutes
- Repeat every 5–7 days for 3–4 sessions

B. Systemic Antibiotics (Severe Cases):

- Oxytetracycline (Long-acting)
- Penicillin–Streptomycin combination
- Amoxicillin

(Administer under veterinary supervision)

C. Topical Antibiotics:

Apply after cleaning and trimming:

- Oxytetracycline spray
- Iodine-based antiseptics
- Chlorhexidine solution (0.5%)

D. Hoof Trimming:

- Remove loose, underrun horn

- Expose infected pockets for aeration
- Avoid excessive trimming to prevent bleeding

E. Pain Management:

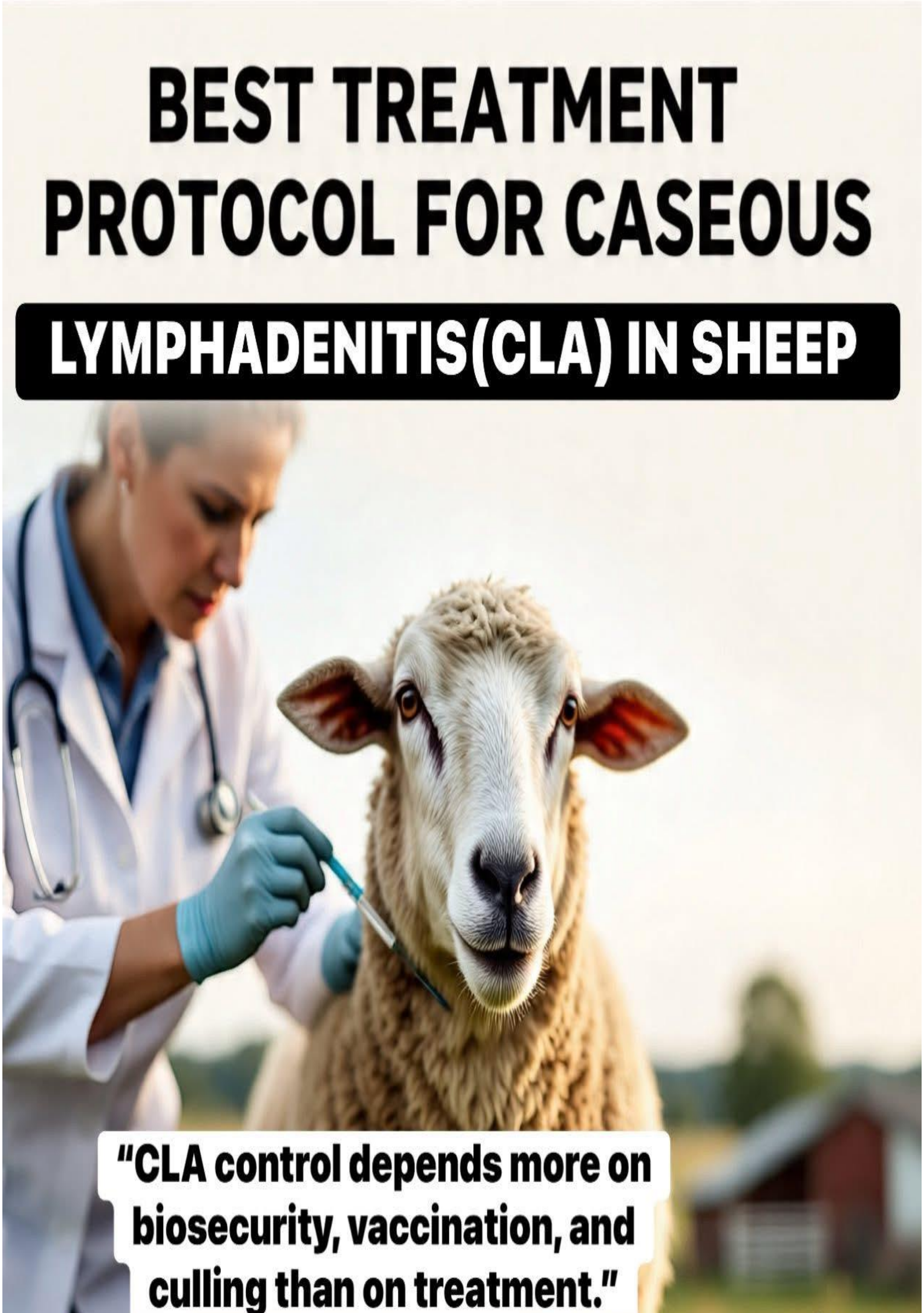
- NSAIDs: Meloxicam / Flunixin
- Reduces pain and improves mobility

4. Control & Prevention:

- Regular hoof trimming (every 3–4 months)
- Keep flooring dry; avoid muddy areas
- Quarantine and inspect new animals
- Routine foot baths in rainy season
- Culling chronically infected animals

5. Prognosis:

- Good with early treatment
- Chronic cases may need repeated therapy or culling.


BEST TREATMENT
PROTOCOL FOR CASEOUS
LYMPHADENITIS(CLA) IN SHEEP
"CLA control depends more on
biosecurity, vaccination, and
culling than on treatment."
Meta AI
FieldvetDrMSS

36. Caseous Lymphadenitis (CLA)

Causes • Symptoms • Diagnosis • Treatment • Control

1. Cause:

- Corynebacterium pseudotuberculosis → chronic, contagious disease affecting lymph nodes and internal organs.

2. Symptoms:

- Firm, enlarged lymph nodes (parotid, prescapular, prefemoral)
- Abscesses with thick "cheesy" green pus
- Weight loss, reduced wool quality
- Coughing / respiratory distress (internal CLA)
- Reduced milk and meat productivity

3. Diagnosis:

- Physical examination of lymph nodes
- Needle aspiration of abscess → characteristic pus
- Bacterial culture & PCR
- ELISA serology for flock screening
- Ultrasound for internal abscesses

4. Best Treatment Protocol:

A. Abscess Management (External CLA)

1. Isolate the affected animal
2. Clip wool & disinfect area (iodine/chlorhexidine)
3. When abscess is mature:
    - Lance at lowest point
    - Drain pus completely
    - Flush cavity with iodine/chlorhexidine
4. Collect and burn the pus & contaminated material
5. Keep animal in isolation until wound heals

B. Antibiotic Therapy:

CLA is difficult to eradicate because organism survives inside cells.

Useful options (under veterinary supervision):

- Procaine Penicillin + Streptomycin
- Tulathromycin
- Long-acting Oxytetracycline

- Rifampicin (rarely used)
- Ceftiofur (limited benefit)

Antibiotics reduce progression but do not completely cure chronic CLA.

C. Internal CLA:

- Long-term antibiotics may reduce severity
- Supportive care (fluids, NSAIDs)
- Cull chronic, severely affected animals

5. Control & Prevention (Most Important):

A. Biosecurity

- Quarantine all new animals for 30 days
- Check for abscesses before introduction
- Purchase from CLA-free flocks

B. Shearing Management

- Sterilize shearing blades between sheep
- Reduce shearing cuts → common source of infection
- Clean shearing area regularly

C. Sanitation

- Disinfect equipment (ear-taggers, castration tools)
- Burn or deep bury abscess material
- Avoid overcrowding

D. Vaccination

- CLA vaccines (where available) reduce severity and new infections
- Annual vaccination recommended for endemic flocks

E. Culling

- Remove repeat offenders or sheep with multiple abscesses
- Improves long-term flock resistance and reduces spread

Key Message

"CLA control depends more on biosecurity, vaccination, and culling than on treatment."

OBSTETRICAL DISORDERS OF SHEEP

Causes – Symptoms – Diagnosis – Treatment – Control

FieldvetDrMSS

1. Dystocia (Difficult Lambing)

Causes

- Oversized fetus
- Malpresentations (breech, head back, llmb flexion)
- Incomplete cervical dilation
- Uterine inertia (primary or secondary)

Symptoms

- Prolonged labour (>1 hour without progress)
- Straiinng without delivering
- Visible abnormal presentation

Diagnosis

Vaginal examination

Assess fetal position, size and dilation

Treatment:Correct malpresentation

Gentle traction with lubrication,Oxytocin for uterine inertia,Caesarean section if needed

3. Vaginal / Uterine Prolapse

Causes

- Late gestation abdominal pressure
- Hypocalcemia
- Over-conditioning
- Very short tall docking

Symptoms

- Tissue protruding from vulva

Diagnosis

Visual examination to identify type

Treatment:Clean and replace prolapsed tissue.Retention sutures (Bühner stitch)

Correct mineral imbalance

4. Retained Placenta (Retained Fetal Membrames

Causes

- Difficult Lambing
- Selenium / vitamin E deficiency
- Infections

Symptoms

Placenta not expelled within 6–8 hours.

Foul smell, fever in severe cases

2 Ring Womb (Failure of Cervical Dilation):

Causes:Hormonal imbalance

Older ewes,Mineral deficiencies

Symptoms:Active labour with closed cervix.No progress in delivery

No progress in lambing

Diagnosis:Vaginal exam showing undilated cervix

Treatment:Gentle manual stretching

Prostaglandins or oxytocin inj

Caesarean section if unresponsive

Control:Culling repeat cases

Balanced late-pregnancy nutrition,prevent any stress

5. Pregnancy Toxemia (Lambing Paralysis) – Obstetrical Impact

Causes

- Twin/triplet pregnancies
- Negative energy balance

Symptoms

Weakness, recumbency,Unable to lamb normally

Treatment:Propylene glycol

IV glucose,Induce lambing / C-section if ewe worsens as per vet guidance

Control

- High energy feed in late pregnancy
- Monitor high-risk ewes

7. Abortions (Obstetrical Failure

Causes

- Infectious agents (Brucelia, Chlamydia, Toxoplasma, Camplyole)
- Nutritional deficiencies
- Stress

37. Obstetrical Disorders

(Causes – Symptoms – Diagnosis – Treatment – Control)

1. Dystocia (Difficult Lambing)

Causes

- Oversized fetus
- Malpresentations (breech, head back, limb flexion)
- Incomplete cervical dilation
- Uterine inertia (primary or secondary)

Symptoms

- Prolonged labour (>1 hour without progress)
- Straining without delivering
- Visible abnormal presentation

Diagnosis

- Vaginal examination
- Assess fetal position, size and dilation

Treatment

- Correct malpresentation
- Gentle traction with lubrication
- Oxytocin for uterine inertia
- Caesarean section if needed

Control

- Use rams with moderate birth weights
- Good late-pregnancy nutrition
- Avoid breeding very young or very old ewes

2. Vaginal / Uterine Prolapse:

Causes

- Late gestation abdominal pressure
- Hypocalcemia
- Over-conditioning
- Very short tail docking

Symptoms

- Tissue protruding from vulva
- Straining, discomfort

Diagnosis: Visual examination to identify type

Treatment

- Clean and replace prolapsed tissue
- Retention sutures (Bühner stitch)
- Correct mineral imbalance

Control

- Avoid overfeeding
- Adequate minerals (Ca, P)
- Proper tail docking length

3. Ring Womb (Failure of Cervical Dilation):

Causes

- Hormonal imbalance
- Older ewes
- Mineral deficiencies

Symptoms

- Active labour with closed cervix
- No progress in lambing

Diagnosis

- Vaginal exam showing undilated cervix

Treatment

- Gentle manual stretching
- Prostaglandins or oxytocin
- Caesarean section if unresponsive

Control

- Culling repeat cases
- Balanced late-pregnancy nutrition

4. Retained Placenta (Retained Fetal Membranes):

Causes

- Difficult lambing
- Selenium / Vitamin E deficiency
- Infections

Symptoms

- Placenta not expelled within 6–8 hours
- Foul smell, fever in severe cases

Diagnosis

- Clinical signs + vaginal exam

Treatment

- Do NOT manually pull placenta
- Oxytocin or prostaglandin
- Antibiotics if metritis occurs

Control

- Adequate Se + Vitamin E nutrition
- Hygienic lambing practices

5. Pregnancy Toxemia (Lambing Paralysis) – Obstetrical Impact:

Causes

- Twin/triplet pregnancies
- Negative energy balance

Symptoms

- Weakness, recumbency
- Unable to lamb normally

Diagnosis

- Clinical signs + ketone testing

Treatment

- Propylene glycol
- IV glucose
- Induce lambing / C-section if ewe worsens

Control

- High-energy feed in late pregnancy
- Monitor high-risk ewes

6. Uterine Torsion (Rare in Sheep):

Causes

- Rolling in late pregnancy
- Large single fetus

Symptoms

- Labour without cervical dilation
- Restlessness, abdominal pain

Diagnosis

- Vaginal exam showing twist
- Manual palpation

Treatment

- Rolling ewe using plank method
- C-section if correction fails

Control

- Avoid slippery flooring
- Close monitoring in late gestation

7. Abortions (Obstetrical Failure):

Causes

- Infectious agents (Brucella, Chlamydia, Toxoplasma, Campylobacter)
- Nutritional deficiencies
- Stress

Symptoms

- Premature expulsion of fetus
- Vaginal discharge
- Retained placenta

Diagnosis

- History + clinical exam
- Laboratory testing (PCR, serology)

Treatment

- Supportive
- Antibiotics depending on pathogen

Control

- Vaccination (where available)
- Biosecurity and hygiene
- Rodent control

Key Message

"Early detection protects both ewe and lamb."

Urolithiasis in Ram

(Urinary Calculi / Obstructive Uropathy)

FieldvetDrMSS

Causes

- **Urinary stone formation** (phosphate, calcium carbonate, silica)
- **High-concentrate, low-roughage** feeding
- **Imbalanced Calcium:** Phosphorus ratio (Ca:P <2:1)
- **High phosphorus diets** (grain-based rations)
- **High magnesium** intake
- **Alkaline urine**
- **Inadequate water intake**
- **Early castration** → narrow urethra
- **Pelleted feeds** with high mineral content
- **Sudden dietary changes**
- **Cold weather** → reduced water consumption.

Symptoms

- **Straining to urinate** (dysuria)
- **Repeated** unsuccessful attempts to pass urine
- **Scanty** or absent urine flow
- **Dribbling** of blood-tinged urine
- **Stretching posture** (like urination stance)
- **Kicking** or biting abdomen
- **Swollen** prepuce or penis
- **Restlessness** & depression
- **Loss of** appetite
- **Grinding** of teeth pain)
- **Abdominal distension** (bladder rupture)
- **Sudden death** due to uroperitoneum)

Diagnosis

- **Clinical Diagnosis**
- History of concentrate feeding
- **Observation** of painful urination atempts
- **Palpation of urethra** (urethral proces, sigmoid flexure)
- **Rectal palpation** (distended bladder)

Imaging

- **Ultrasonography**
  - Distended bladder
  - Free fluid
- **Radiography** (limited use)
  - Mineralized calculi

Laboratory Diagnosis

- **Serum biochemistry:**
  - ↑ Blood urea nitrogen
  - ↑ Creatinine
  - ↑ Potassium

- **Urine examination** (if avaliable)
  - Crystals
  - Alkaline pH

Key Message

Urolithiasis is a nutritional and management disease – prevention is more effective than treatment.

#Fieldvet

Treatment

Emergency Management

- **Pain relief:** NSAIDs (Meloxicam / Flunixin)
- **IV fluids** to correct dehydration & electrolytes
- **Antispasmodics** (Butylscopolamine)

- **Specific Treatment**
  - **Urethral Process Amputation**
    - First-line treatment in rams
    - **Catheterization** / Hydropulsion ratedące)
    - **Surgical Options** (best for breeding ram.)
    - Perineal urethrostomy (salvage procedure)
    - **Bladder repair** if rupture detected early
- **Medical Management**
- **Urine acidifiers** (Ammonium chloride)
- Dietary correction

38. Urolithiasis in Ram

(Urinary Calculi / Obstructive Uropathy)

Causes

- Urinary stone formation (phosphate, calcium carbonate, silica).
- High-concentrate, low-roughage feeding.
- Imbalanced Calcium: Phosphorus ratio (Ca:P < 2:1).
- High phosphorus diets (grain-based rations).
- High magnesium intake.
- Alkaline urine.
- Inadequate water intake.
- Early castration → narrow urethra.
- Pelleted feeds with high mineral content.
- Sudden dietary changes.
- Cold weather → reduced water consumption.

Symptoms:

- Straining to urinate (dysuria).
- Repeated unsuccessful attempts to pass urine.
- Scanty or absent urine flow.
- Dribbling of blood-tinged urine.
- Stretching posture (like urination stance).
- Kicking or biting abdomen.
- Swollen prepuce or penis.
- Restlessness and depression.
- Loss of appetite.
- Grinding of teeth due to pain.
- Abdominal distension (bladder rupture).
- Sudden death due to uroperitoneum.

Diagnosis:

Clinical Diagnosis

- History of concentrate feeding.
- Observation of painful urination attempts.
- Palpation of urethra (urethral process, sigmoid flexure).
- Rectal palpation (distended bladder).

Laboratory Diagnosis

- Serum biochemistry:
    - ↑ Blood urea nitrogen (BUN)

- ↑ Creatinine
- ↑ Potassium
- Urine examination (if available):
  - Crystals
  - Alkaline pH

Imaging

- Ultrasonography:
  - Distended bladder
  - Free abdominal fluid (rupture)
- Radiography (limited use):
  - Mineralized calculi

Treatment:

Emergency Management

- Pain relief: NSAIDs (Meloxicam / Flunixin).
- IV fluids to correct dehydration & electrolytes.
- Antispasmodics (Butylscopolamine).

Specific Treatment

1. Urethral Process Amputation

- First-line treatment in rams.
- Relieves obstruction at distal urethra.

2. Catheterization / Hydropulsion

- Flushing of small calculi.
- Requires sedation.

3. Surgical Treatment

- Tube cystostomy – best for breeding rams.
- Perineal urethrostomy – salvage procedure.
- Bladder repair if rupture detected early.

4. Medical Management

- Urine acidifiers: Ammonium chloride.
- Dietary correction.

5. Antibiotics

- To prevent secondary urinary tract infection.

Control & Prevention:

- Maintain Ca:P ratio = 2:1.
- Increase roughage; reduce concentrates.
- Avoid excess phosphorus & magnesium.
- Provide clean, abundant drinking water.
- Add salt (1–3%) in ration to increase water intake.
- Use ammonium chloride (0.5–1%) as urine acidifier.
- Avoid early castration (delay >3 months).
- Avoid sudden feed changes.
- Encourage grazing rather than stall feeding.
- Monitor mineral content of feed and water.

Key Message:

□ Urolithiasis is a nutritional and management disease—prevention is more effective than treatment.


JOINT ILL (SEPTIC ARTHRITIS) IN SHEEP
Causes • Symptoms • Diagnosis • Treatment • Control
1. DEFINITION
FieldvetDrMSS
Joint ill is infectious inflammation of joints seen mainly in young lambs, caused by bacteria entering the bloodstream—most commonly through the navel (umbilical infection) — and localizing in joints.
2. CAUSES
A. Predisposing Factors
• Poor navel hygiene after birth
• Dirty, wet lambing pens
• Delayed or absent navel dipping
• Weak immunity due poor colostrum intake
• Overcrowding and poor ventilation
2. CAUSES
A. Predisposing Factors
• Poor navel hygiene after birth
• Dirty, wet lambing pens
• Delayed or absent navel dipping
• Weak immunity due to poor colostrum intake
• Overcrowding and poor ventilation
4. SYMPTOMS (CLINICAL SIGNS)
Local Signs
• Hot, swollen, painful joints
Commonly affected joints:
• Knee (carpal)
• Hock (tarsal), Stifle
• Stifle
• Reluctance to stand or walk
• Lameness or recumbency.
4. DIAGNOSIS
• Field Diagnosis
• History of poor lambing hygiene
• Swollen joints in young lambs (1–4 weeks)
• Clinical & Laboratory Diagnosis
• Joint palpation (pain, heat, fluctuation)
• Joint fluid aspiration: Turbid or purulent fluid.
• Bacterial culture & antibiotic sensitivity
• Blood tests (leukocytosis)
5. DIAGNOSIS
A. Medical Treatment (Early Cases)
• Systemic antibiotics (5-7 days or longer.)
• Peniclllin-Streptomycin
• Amoxielllin
• Oxytetracycline
• NSAIDS for pain & inflammation.
• Supportive care
• Fluids, Good nutrition
• Warm, dry bedding
6. TREATMENT
A. Medical Treatment (Early Cases)
• Joint lavege with sterile saline
• Intra-articular antibiotics (veternary supervision only)
c. Chronic Cases
✓ Poor prognosis
✓ Often equire culling due to-permanent joint damage
KEY MESSAGE
Joint ill is preventable—clean lambing practices and early treatment are the most effective control measures.
#SheepHealth · #NavelDipping · #LambCare · #VeterinaryPublicHealth · #Fieldvet
FieldvetDrMSS

39. Joint ILL in Sheep (Navel ILL/ Infectious Arthritis)

Joint ill is a common bacterial infection of joints in newborn lambs, usually occurring during the first few weeks of life. It is mainly associated with poor navel hygiene and contaminated lambing environments

Causes

Joint ill occurs when bacteria enter the body through the umbilical cord (navel) soon after birth.

Common bacteria involved:

- Streptococcus spp.
- Staphylococcus spp.
- Trueperella (Arcanobacterium) pyogenes
- E. coli

Mode of Infection

- Lamb is born in a dirty lambing area.
- Bacteria enter through the open umbilical cord.
- Bacteria enter the bloodstream (septicemia).
- Infection settles in joints, causing inflammation and swelling.

Clinical Signs

- Swollen, hot and painful joints
- Lameness or difficulty walking
- Lamb reluctant to stand
- Fever and dullness
- Reduced suckling
- Multiple joints may be affected (knees, hocks)

In severe cases:

- Joint deformity
- Septicemia and death

Diagnosis

- Clinical examination
- History of poor lambing hygiene
- Joint swelling and lameness in young lambs
- Joint fluid examination (if required)

Treatment

- Early treatment is very important.

- Antibiotics
    - Long acting penicillin
    - Oxytetracycline
    - Other antibiotics as advised by veterinarian
- Supportive care
- Anti-inflammatory drugs
- Proper nursing care
- Isolate affected lambs

Chronic cases may have permanent joint damage.

🛡 **Prevention (Most Important)**

- Maintain clean lambing pens
- Disinfect the navel immediately after birth using iodine solution
- Provide adequate colostrum within first 1–2 hours
- Keep bedding dry and hygienic
- Avoid overcrowding
- Ensure good flock hygiene

🩺 **Field Veterinary Significance**

- Causes high lamb mortality and poor growth
- Leads to economic losses in sheep farms
- Preventable with simple hygiene measures

✅ **Key Message:**

Joint ill is largely preventable through proper lambing hygiene, navel disinfection, and early colostrum feeding.


ORTHOPAEDIC ISSUES OF SHEEP
Causes • Clinical Signs • Management • Prevention
WHY ORTHOPAEDIC HEALTH MATTERS
FieldvetDrMSS
Lameness is a major cause of production loss
Leads to poor grówth, reduced fertility, welfare issues
Early diagnosis = better recovery & lower economic loss
COMMON ORTHOPAEDIC DISORDERS IN SHEEP
1. FOOT ROT
CAUSE: Dichelobacter, nodesus + Fusobacterium neccophotium
SIGNS: Severe lameness, underunning of hoof horn
MANAGEMENT: Hoof trimming, footbath (2eSO), / $CuSO_3$), antibiotics
PREVENTION: Dry flooring, regular foot care, biosecurity
2. FOOT SCALD (INTERDIGITAL DERMATITIS)
CAUSE: Moist conditions, bacterial infection
SIGNS: Miid-moderate lameness, reddened interalgital skin
MANAGEMENT: Topicl antiseptics, footbath
4. FRACTURES
CAUSE: Poor hoof wear, soft soil, neglect
SIGNS: Abnornal gait, uneven weight bearing
MANAGEMENT: Corrective hoof trimming & trimming
5. JOINT ILL (SEPTIC ARTHRITIS)
Cause: Bacterial infection via navel (lambs)
SIGNS: Abnormal gait, uneven weight bearing
Management: Antibiotics, joint lavage
6. OSTEOMYELITIS
CAUSE: Extension of infection from wounds or joints
SIGNS: joint swelling, stiffness
MANAGEMENT: Anti-inflammatories, antibiotics if infectious
RISK FACTORS
Wet & muddy housing
Poor nutrition (Ca, Pimbalance)
Rough handling & transport
Poor hygiene in lambing pens
Lack of routine hoof care
ROLE OF VETERINARIANS
Early lameness diagnosis
Correct hoof trimming guidance
Rational antibiotic use (AMR prevention
Farmer (education & flock health planning
PREVENTION STRATEGIES
Regular hoof trimming
Clean & dry housing
Balanced mineral nutrition
Footbath protocols
Prompt treatment of lame animals
Biosecurity for new stock
ECONOMIC & WELFARE IMPACT
Reduced treatment costs
Improved mobility & productivity
Better animal welfare
Increased flock profitability
KEY MESSAGE
Healthy feet and joints are essential for productive and welfare-friendly sheep farming,
#SheepHealth • #Orthopaedics • #Lameness • #VeterinaryEducation • #Fieldvet
FieldvetDrMSS

40.

40. Orthopaedic Issues in Sheep

Orthopaedic disorders in sheep affect bones, joints, tendons, and hooves, leading to lameness, poor mobility, reduced grazing ability, and economic losses in sheep farming. Early detection and proper management are essential for flock health.

Common Orthopaedic Problems in Sheep

1️⃣ Foot Rot

Cause: Bacterial infection (mainly Dichelobacter nodosus and Fusobacterium necrophorum)

Signs

- Severe lameness
- Foul-smelling hoof lesions
- Separation of hoof horn

Significance

- Reduced grazing
- Weight loss
- Decreased productivity

2️⃣ Foot Abscess

Cause: Bacterial infection entering through hoof injuries.

Signs

- Swollen hoof
- Sudden severe lameness
- Heat and pain in the hoof

Significance

- Difficulty walking
- Reduced feed intake

3️⃣ Joint Ill (Septic Arthritis in Lambs)

Cause: Bacterial infection entering through the navel in newborn lambs.

Signs

- Swollen joints
- Lameness
- Fever and weakness

Significance

- High lamb mortality
- Permanent joint deformities

4 Fractures

Cause

- Trauma
- Accidents during handling
- Dog attacks
- Falls

Common sites

- Leg bones
- Pelvis

Significance

- Severe pain
- Reduced mobility
- May require euthanasia in severe cases

5 Laminitis

Cause

- Sudden diet changes
- Grain overload
- Metabolic disturbances

Signs

- Reluctance to walk
- Stiff gait
- Painful hooves

Significance

- Chronic lameness
- Poor growth

6 Tendon and Ligament Injuries

Cause

- Overexertion
- Slippery floors
- Trauma

Signs

- Limping
- Swelling around joints

- Abnormal gait

Significance

- Reduced grazing ability

⚠ Risk Factors

- Wet and muddy grazing lands
- Poor hoof care
- Rough terrain
- Nutritional imbalance
- Overcrowding
- Poor lambing hygiene

Prevention Strategies

✓ Regular hoof trimming
✓ Maintain dry bedding and clean pens
✓ Provide balanced nutrition with minerals
✓ Footbath using zinc sulphate or copper sulphate
✓ Good lambing hygiene
✓ Early treatment of lameness

Veterinary & Farm Significance

Orthopaedic problems can cause:

- Reduced grazing efficiency
- Weight loss and poor growth
- Lower meat and wool production
- Increased treatment costs

Early detection of lameness is a key indicator of flock health.

✅ Key Message

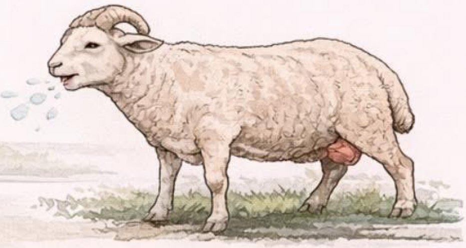
Healthy feet and joints are essential for productive sheep farming.

Regular hoof care, good hygiene, and early veterinary intervention prevent major losses.

Maedi in Sheep

(Ovine Progressive Pneumonia – OPP)

FieldvetDrMSS



Cause

- Maedi–Visna virus (MVV)
- A **lentivirus** belonging to the family *Retroviridae*

1 Epidemiology

- **Affects adult sheep** (usually > 2 years)
- **Chronic**, slowly progressive disease
- More common in:
  - Intensively managed flocks
  - Closed housing
  - Poor ventilation
- Worldwide distribution

2 Pathogenesis

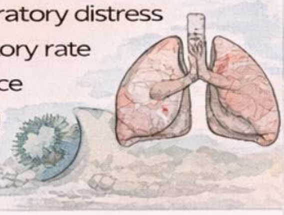
Vertical transmission
- Via colostrum and milk

Horizontal transmission
- *Aerosol* (close contact,–coughing)
- Introduction of **carrier animals**
- Long incubation period (months to years)

Clinical Signs

Maedi Form – (Maedi Form - Respiratory)

- Progressive respiratory distress
- Increased respiratory rate
- Exercise intolerance
- Chronic coughing
- Weight loss (emaciation)



Post-mortem Lesions

- Heavy, firm, non-collapsing lungs
- Greyish lung surface
- Interstitial pneumonia
- Enlarged mediastinal lymph nodes

Key Message

Maedi is a slow, incurable viral disease early testing, and strict biosecurity are the only effective control measures.Differential Diagnosis is very important.

Clinical Signs (Maedi Form – Respiratory)

- Progressive respiratory distress
- Increased respiratory rate
- Exercise intolerance
- Chronic coughing
- Weight loss (emaciation)
- **Normal or subnormal** temperature
- **Joints**

Post-mortem Lesions

- Heavy, firm, non-collapsing lungs
- Greyish lung surface
- Interstitial pneumonia
- Enlarged mediastinal lymph nodes

Diagnosis

Field Diagnosis
- Chronic respiratory signs
- Progressive weight loss
- Poor response to antibiotics

Laboratory
- **Serology** (ELISA, AGID) – most common
- PCR for MVV
- **Histopathology** (interstitial pneumonia)

Differential Diagnosis

- **Pasteurellosis**
- **CCPP-like** disease
- Lungworm infestation
- PPR
- Chronic pneumonia

Economic Importance

- Progressive production losses
- Reduced body weight and wool yield
- Reduced milk production
- Premature culling in advanced stages

Veterinary Extension Education | #Fieldvet

Section 9: Advanced & Miscellaneous Conditions

41. Maedi in Sheep

Maedi-Visna (MVV) is a globally widespread, persistent viral disease in sheep, causing chronic pneumonia (Maedi) and neurological issues (Visna), leading to significant economic losses, with major exceptions being Australia and New Zealand.

It affects most sheep-keeping regions, including Europe, Asia, Africa, and the Americas, though prevalence varies, with higher rates in developed nations linked to intensive farming practices like pooled colostrum feeding, favoring transmission via milk, aerosols, and fluids.

Global Presence:

Widespread: Occurs in most sheep-farming countries, with reported cases across continents.

Exceptions:

Australia and New Zealand are considered largely free of MVV, notes ScienceDirect.com, and Dove Medical Press.

High Prevalence:

Asia (especially China) and Europe show high flock prevalence, with significant presence in North America, South America, and Africa.

Key Characteristics:

Causative Agent: Small Ruminant Lentivirus (SRLV).

Clinical Signs: Progressive lung disease (Maedi), arthritis, mastitis, and encephalitis (Visna).

Economic Impact: Causes substantial losses due to reduced production, involuntary culling, and trade restrictions.

Transmission & Control

Main Routes: Colostrum/milk (milk-borne) and contact with infected fluids (aerosols, respiratory exudates).

Control Challenges: No effective treatment or vaccine; long infection-to-seroconversion period; difficulty in early diagnosis.

Management: Emphasizes surveillance (serology), biosecurity, hygiene, and culling seropositive animals.

Other Names:

- Ovine Progressive Pneumonia (OPP) in the U.S.
- La bouhite (France)
- Graaff-Reinet disease (South Africa)
- Zwoegerziekte (Netherlands)


PAPILLOMATOSIS IN SHEEP
CAUSE
FieldvetDrMSS
Ovine Papillomavirus (OPV)
A DNA virus causing warts.
EPIDEMIOLOGY
Young sheep (< 2 years)
Poor hygiene, overcrowding
Spread via contact & fomites.
PATHOGENESIS
1 Virus enters through skin abrasions.
2 Viral infection of epithelial cells.
3 Formation of papillomas.
CLINICAL SIGNS
Lips
Eyelid
Teats
Cauliflower-like warts
Usually self-limiting.
DIAGNOSIS
Clinical Appearance
PCR & Histopathology
DIFFERENTIAL DIAGNOSIS
Contagious Ecthyma (Orf)
Squamous Cell Carcinoma
Fibroma & Abscess
TREATMENT
Most Regress in 2–6 Months
Improve Hygiene.
Isolate Severe Cases.
Surgical Removal (if needed).
PREVENTION
Avoid Skin Trauma.
Disinfect Equipment.
Isolate Affected Sheep.
Reduce Overcrowding.
ECONOMIC IMPACT
Milk Production
Secondary Infections
Carcass Losses
NOT ZOONOTIC!
Does NOT infect humans.
FieldvetDrMSS
42.

42. Papillomatosis in Sheep

1. Definition

Papillomatosis is a viral disease characterized by benign epithelial tumors (warts or papillomas) affecting the skin and mucous membranes of sheep. It is caused by Ovine Papillomavirus (OPV).

2. Etiology:

- **Causative agent:** Ovine Papillomavirus
- **Family:** Papillomaviridae
- **Type:** Non-enveloped, double-stranded DNA virus
- **Resistance:**
    - Resistant to environmental conditions
    - Survives for months in dry environments
    - Sensitive to disinfectants like formalin and sodium hypochlorite

3. Epidemiology:

- Occurs worldwide
- More common in young sheep (< 2 years)
- Higher incidence in:
- Poor hygiene conditions
- Overcrowded flocks
- Animals with skin abrasions

Transmission

- Direct contact between animals
- Indirect transmission via:
- Fomites (shearing equipment, tagging tools)
- Contaminated housing
- Virus enters through skin wounds or abrasions

4. Pathogenesis:

1. Virus enters through micro-abrasions
2. Infects basal epithelial cells
3. Viral DNA integrates and causes:
    - Hyperplasia
    - Excessive keratinocyte proliferation
4. Formation of papillomas(warts)

⚠ **Usually benign and self-limiting**

⚠️ **Rarely may undergo malignant transformation (very uncommon in sheep)**

5. Clinical Signs:

Incubation Period

- 1–3 months

Lesion Characteristics

- Cauliflower-like growths
- Pedunculated or sessile masses
- Rough, dry surface

Common Sites

- Lips
- Muzzle
- Eyelids
- Ears
- Teats and udder
- Occasionally genital region

General Condition

- Usually normal
- No systemic illness
- May interfere with:
- Grazing (if oral lesions severe)
- Milking (if teat lesions present)

6. Differential Diagnosis:

- Contagious ecthyma (Orf)
- Squamous cell carcinoma
- Fibroma
- Cutaneous abscess
- Dermatophytosis

🔎 **Key differentiation:**

Orf lesions are proliferative but accompanied by scab formation and inflammation, whereas papillomas are dry, firm, and non-inflammatory.

7. Diagnosis:

Clinical Diagnosis

- Based on appearance and age group
- Typical wart-like lesions

Laboratory Diagnosis (if required)

- Histopathology
- Hyperkeratosis
- Acanthosis
- Koilocytosis
- PCR for viral DNA (research/confirmation cases)

8. Treatment:

⚠ **Most cases are self-limiting and regress within 2–6 months.**

Field Management

- No specific antiviral therapy
- Improve hygiene
- Isolate severely affected animals

Surgical Removal

- Indicated when:
- Large obstructive growths
- Interference with feeding or milking
- Use aseptic technique

Autogenous Vaccine

- Prepared from wart tissue
- Occasionally used in persistent flock outbreaks

9. Control & Prevention:

- Avoid skin trauma during shearing
- Disinfect instruments
- Isolate affected animals
- Maintain proper nutrition
- Reduce overcrowding

No widely available commercial vaccine for sheep papillomatosis.

10. Economic Importance:

- Usually minor economic impact
- Losses occur when:
- Lesions interfere with suckling or milking
- Secondary bacterial infection develops
- Carcass condemnation (rare)

11. Zoonotic Significance:

- ✗ Not zoonotic

(Ovine papillomavirus does not infect humans)

12. Prognosis

- ✓ Excellent
- ✓ Most cases regress spontaneously
- ✓ Immunity develops after recovery


DIFFERENT TYPES OF TUMORS IN SHEEP
(Symptoms, Diagnosis, Differential Dx, Treatment & Control)
1 Squamous Cell Carcinoma (SCC)
Common Sites:
Eyelids
Vulva
Perineum
Skin (especially non-pigmented areas)
Symptoms:
Clinical appearance
Ulcerative mass:
Biopsy (keratinp pearls).
Non-healing wound
Progressive enlargement.
Weight loss in advanced cases.
Diagnosis:
Clinical appearance.
Biopsy (keratin pearls)
Early surgical excision
Cryotherapy (small lesions)
Culling (advanced cases)
Most tumors are uncommon
2 Ovine Pulmonary Adenocarcinoma (Jaagsiekte)
Cause:
Jaagsiekte Sheep Retrovirus (JSRV)
Symptoms:
Progressive respiratory distress.
Chronic weight loss.
Nasal discharge (frothy fuid when lifting hindquarters).
Exercise intolerance.
Diagnosis:
Clinical signs;
Thoracic ultrussond,
Postmortem; Firm gray lung nodules.
PCR (omfirmat).
Differential Diagnosis:
Major effective disease in some regions.
Maedi-Visña.
Chroniç pneumonia
Lung Abscess.
Test and cull:Separate young stock
,Maintain closed flock
Strict biosecurity
3 Lymphoma (Enzootic/Sporadic)
Symptoms:
Enlarged lymph nodes.
Weight loss.
Organ enlargement.
Diagnosis:
Clinical examination,
Blood smear,
Lymph node biopsy.
Differential Diagnosis:
Caseous lymphadenitis.
Caseous lymphadenitis.
Abscess.
Abscess.
Tuberculosis.
4 Papilloma (Warts)
Ovine Papillomavirus
Symptoms:
Firm, slow-growing skin
Usually-solitary.
Diagnosis:
Clinical appearance
Histopathology if required
Differential Diagnosis:
Orf,
Fibroma.
Orf,
SCC.
Fibroma.
5 Fibroma / Fibrosarcoma
Symptoms:
Firm, slow-growing skin mass.
Usually solitary.
Diagnosis:
Biopsy
Surgical Excision
7 Ocular Tumors
Includes: SCC.
Melanoma.
Symptoms:
Eye mass:
Cinçlil exam.
Tearing.
Biopsy,
Blindness:
Surgical removal if obstructive.
GENERAL DIAGNOSTIC & TREATMENT GUIDELINES
1. History & Age
2. Growth rate
3. Location
4. Palpation (inm ucit), Biopsy (goid ctandard)
6. Histopatnology
TEST, BIOPSY & EARLY REMOVAL - KEYS TO SUCCESS
ECONOMIC IMPORTANCE
Cood Fisck management:
Avoid chronic irritation.
Biosecurity (for vinal tumors like OPA).
Genetic selection.
Fieldvet

43. Tumours in Sheep

(Symptoms, Diagnosis, Differential Diagnosis, Treatment & Control)

Tumors in sheep are generally uncommon compared to dogs and cattle, but certain neoplasms have economic and clinical importance.

1 Squamous Cell Carcinoma (SCC):

Common Sites

- Eyelids
- Vulva
- Perineum
- Skin (especially non-pigmented areas)

Symptoms

- Ulcerative, proliferative mass
- Non-healing wound
- Bleeding lesion
- Progressive enlargement
- Weight loss in advanced cases

Diagnosis

- Clinical appearance
- Biopsy and histopathology (keratin pearls)

Differential Diagnosis

- Papilloma
- Orf (Contagious ecthyma)
- Chronic abscess
- Fibroma

Treatment

- Early surgical excision
- Cryotherapy (small lesions)
- Culling (advanced cases)

Control

- Avoid excessive UV exposure
- Select pigmented breeds
- Early detection and removal

2️⃣ Ovine Pulmonary Adenocarcinoma:(Jaagsiekte)

Cause

- Jaagsiekte Sheep Retrovirus (JSRV)

Symptoms

- Progressive respiratory distress
- Chronic weight loss
- Nasal discharge (frothy fluid on lifting hindquarters)
- Exercise intolerance

Diagnosis

- Clinical signs
- Thoracic ultrasound
- **Postmortem:** Firm gray lung nodules
- PCR (confirmation)

Differential Diagnosis

- Maedi-Visna
- Chronic pneumonia
- Lung abscess

Treatment

❌ No effective treatment

Control

- Test and cull
- Separate young stock
- Maintain closed flock
- Strict biosecurity

⚠ Major economic disease in some regions.

3️⃣ Lymphoma (Enzootic/ Sporadic):

Symptoms

- Enlarged lymph nodes
- Weight loss
- Weakness
- Organ enlargement

Diagnosis

- Clinical examination

- Blood smear
- Lymph node biopsy

Differential Diagnosis

- Caseous lymphadenitis
- Abscess
- Tuberculosis

Treatment

❌ Usually not economical, Culling recommended

Control

- Remove affected animals
- Improve herd health management

4 Papilloma (Warts):

Cause

- Ovine Papillomavirus

Symptoms

- Cauliflower-like skin growths
- Common in young sheep
- Usually non-painful

Diagnosis

- Clinical appearance
- Histopathology if required

Differential Diagnosis

- Orf
- Fibroma
- SCC

Treatment

- Usually self-limiting
- Surgical removal if obstructive

Control

- Improve hygiene
- Disinfect equipment
- Avoid skin trauma

5️⃣ Fibroma / Fibrosarcoma:

Symptoms

- Firm, slow-growing skin mass
- Usually solitary

Diagnosis

- Biopsy

Differential Diagnosis

- Papilloma
- Abscess
- SCC

Treatment

- Surgical excision

Control

- Early removal

6️⃣ Melanoma:

More common in pigmented breeds

Symptoms

- Dark pigmented mass
- May ulcerate

Diagnosis

- Biopsy

Differential Diagnosis

- Pigmented SCC
- Hematoma

Treatment

- Surgical excision

Control

- Genetic selection

7️⃣ Ocular Tumours:

Includes: SCC, Melanoma

Symptoms

- Eye mass
- Tearing
- Blindness

Diagnosis

- Clinical exam
- Biopsy

Treatment

- Surgical removal
- Enucleation (advanced cases)

General Diagnostic Approach to Tumors in Sheep:

1. History & Age
2. Growth rate
3. Location
4. Palpation (firm vs soft)
5. Biopsy (gold standard)
6. Histopathology

General Treatment Principles:

- Early detection is critical
- Surgical excision is main treatment
- Advanced tumors → culling often economical
- No routine chemotherapy in sheep

Control & Prevention:

- Good flock management
- Avoid chronic irritation
- Biosecurity (for viral tumors like OPA)
- Genetic selection
- Remove affected animals early

Economic Importance:

- Reduced productivity
- Carcass condemnation
- Mortality (OPA)
- Treatment cost


CAPTURE MYOPATHY IN SHEEP
A Stress-Induced, Potentially Fatal Condition
What is Capture Myopathy?
Capture Myopathy is a metabolic muscle disorder caused by severe stress, fear, struggle, and exhaustion during handling, restraint, transport, chasing or shearing.
Results in muscle damage, metabolic acidosis, kidney failure & sudden death.
Risk Factors in Sheep
✓ Rough handling & prolonged restraint
✓ Chasing during catching or transport
✓ Heat stress & dehydration
✓ Long-distance transport
✓ Overcrowding
✓ Poor handling facilities
✓ Temperamental or wild sheep ✓ Pregnant or debilitated animals
Pathophysiology (Simple Flow)
Stress → Catecholamine surge → Muscle ischemia → Lactic acid buildup → Myonecrosis
→ Myoglobin release → Renal failure → Death →
Clinical Signs
✓ Rapid breathing (tachypnoea)
✓ Muscle tremors
✓ Hyperthermia
✓ Anxiety, panic
✓ Stiff or reluctant movement
Clinical Signs
✓ Severe muscle rigidity
✓ Recumbency
✓ Dark / coffee-colored urine (myoglobinuria)
✓ Collapse
✓ Sudden death
Laboratory Findings
✓ ↑Creatine Kinase (CK)
✓ ↑AST
✓ Metabolic acidosis ✓ Myoglobinuria
✓ Electrolite imbalance ✓ Urinalysis
Diagnosis
✓ History of recent stress or exertion
✓ Clinical signs
✓ Blood biochemistry
✓ Urinalysis
Emergency Management (Field Level)
✓ Stop all handling immediately
✓ Early intervention improves outcome
✓ Survivors may have permanent muscle damage:
ANIMAL WELFARE MESSAGE
Capture Myopathy is PREVENTABLE.
Stress kills sheep silently.
FieldvetDrMSS
Stress + Struggle + Heat = Death Risk

44. Capture Myopathy in Sheep

Capture myopathy is a stress-induced muscular disorder that occurs when sheep experience extreme physical exertion, fear, or prolonged handling. It is commonly seen during capture, transport, chasing, or handling of wild or semi-managed animals.

🔬 What is Capture Myopathy?

Capture myopathy is a metabolic muscle disease caused by excessive stress leading to:

- Muscle damage
- Metabolic acidosis
- Circulatory failure

It can occur in both domestic sheep and wild sheep species.

⚠ Causes

Common triggering factors include:

- Prolonged chasing during herding
- Rough handling or restraint
- Long-distance transportation
- Extreme fear or panic
- Overcrowding in holding pens
- High environmental temperatures

These stresses cause excessive muscle activity and oxygen depletion, leading to muscle degeneration.

🧬 Pathophysiology

During severe stress:

- Adrenaline and stress hormones increase
- Muscles consume large amounts of energy
- Oxygen deficiency develops
- Lactic acid accumulates in muscles
- Muscle fibers undergo necrosis

This leads to systemic metabolic disturbances.

▯ Clinical Signs

Signs may appear immediately or within several hours to days.

- Weakness and reluctance to move
- Muscle stiffness
- Rapid breathing
- Elevated body temperature

- Dark-colored urine (myoglobinuria)
- Recumbency
- Sudden death in severe cases

🔍 Diagnosis

Diagnosis is based on:

- History of recent stress or capture
- Clinical signs
- Elevated muscle enzymes (CK, AST)
- Post-mortem findings of muscle degeneration

📄 Post-Mortem Findings

- Pale or darkened skeletal muscles
- Muscle hemorrhage
- Kidney damage due to myoglobin
- Evidence of metabolic acidosis

🛡 Prevention

Prevention is more effective than treatment.

Key preventive measures:

- ✓ Gentle handling practices
- ✓ Avoid prolonged chasing of sheep
- ✓ Provide rest during transportation
- ✓ Maintain adequate space in holding pens
- ✓ Avoid handling during extreme heat
- ✓ Gradual movement of flocks

💊 Treatment (Supportive)

Treatment success is often limited but may include:

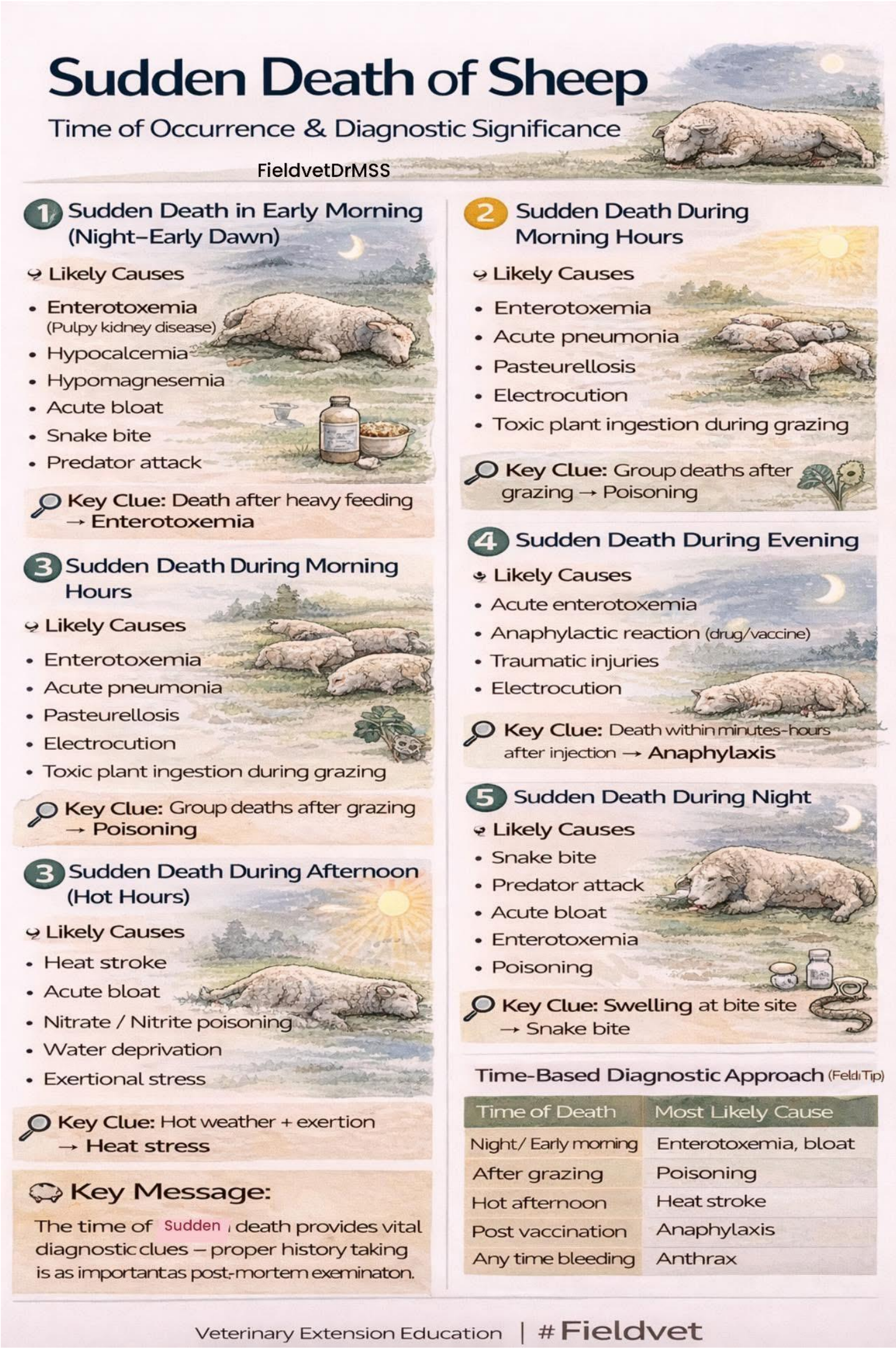
- Fluid therapy
- Anti-inflammatory drugs
- Sedation to reduce stress
- Vitamin E and Selenium supplementation
- Cooling and rest

Early veterinary intervention improves survival chances.

🌾 Key Message for Shepherds

Minimize stress during handling and transport to prevent capture myopathy.

Stress management is essential for animal welfare and flock productivity.


Sudden Death of Sheep
Time of Occurrence & Diagnostic Significance
FieldvetDrMSS
1 Sudden Death in Early Morning (Night–Early Dawn)
Likely Causes
• Enterotoxemia (Pulpy kidney disease)
• Hypocalcemia
• Hypomagnesemia
• Acute bloat
• Snake bite
• Predator attack
Key Clue: Death after heavy feeding → Enterotoxemia
2 Sudden Death During Morning Hours
Likely Causes
• Enterotoxemia
• Acute pneumonia
• Pasteurellosis
• Electrocution
• Toxic plant ingestion during grazing
Key Clue: Group deaths after grazing → Poisoning
3 Sudden Death During Morning Hours
Likely Causes
• Enterotoxemia
• Acute pneumonia
• Pasteurellosis
• Electrocution
• Toxic plant ingestion during grazing
Key Clue: Group deaths after grazing → Poisoning
4 Sudden Death During Evening
Likely Causes
• Acute enterotoxemia
• Anaphylactic reaction (drug/vaccine)
• Traumatic injuries
• Electrocution
Key Clue: Death within minutes-hours after injection → Anaphylaxis
3 Sudden Death During Afternoon (Hot Hours)
Likely Causes
• Heat stroke
• Acute bloat
• Nitrate / Nitrite poisoning
• Water deprivation
• Exertional stress
Key Clue: Hot weather + exertion → Heat stress
5 Sudden Death During Night
Likely Causes
• Snake bite
• Predator attack
• Acute bloat
• Enterotoxemia
• Poisoning
Key Clue: Swelling at bite site → Snake bite
Key Message:
The time of Sudden death provides vital diagnostic clues – proper history taking is as importantas post-mortem exeminaton.
Time-Based Diagnostic Approach (Feldi Tip)
Time of Death
Most Likely Cause
Night/ Early morning
Enterotoxemia, bloat
After grazing
Poisoning
Hot afternoon
Heat stroke
Post vaccination
Anaphylaxis
Any time bleeding
Anthrax
Veterinary Extension Education | #Fieldvet

45. Sudden Death of Sheep

(Time of Occurrence & Diagnostic Significance)

1. Sudden Death in Early Morning (Night–Early Dawn):

Likely Causes

- Enterotoxemia (Pulpy kidney disease)
- Hypocalcemia
- Hypomagnesemia
- Acute bloat
- Snake bite
- Predator attack

Diagnostic Significance

- Animals appear normal the previous evening
- Often found dead without struggle
- Full rumen, rapid post-mortem changes
- Strong association with high concentrate feeding

🔍 **Key Clue:** Death after heavy feeding → Enterotoxemia

2. Sudden Death During Morning Hours:

Likely Causes

- Enterotoxemia
- Acute pneumonia
- Pasteurellosis
- Electrocution
- Toxic plant ingestion during grazing

Diagnostic Significance

- Death soon after grazing or movement
- Froth at nostrils in respiratory causes
- Multiple animals affected → Infectious or toxic cause

🔍 **Key Clue:** Group deaths after grazing → Poisoning

3. Sudden Death During Afternoon (Hot Hours):

Likely Causes

- Heat stroke
- Acute bloat
- Nitrate / Nitrite poisoning
- Water deprivation

- Exertional stress

Diagnostic Significance

- Occurs during high ambient temperature
- Panting, open-mouth breathing before death
- Dark or chocolate-colored blood (nitrate poisoning)

🔍 **Key Clue:** Hot weather + exertion → Heat stress

4. Sudden Death During Evening:

Likely Causes

- Acute enterotoxemia
- Anaphylactic reaction (drug/vaccine)
- Traumatic injuries
- Electrocution

Diagnostic Significance

- Often follows vaccination, injection, or handling
- History of recent treatment is crucial

🔍 **Key Clue:** Death within minutes–hours after injection → Anaphylaxis

5. Sudden Death During Night:

Likely Causes

- Snake bite
- Predator attack
- Acute bloat
- Enterotoxemia
- Poisoning

Diagnostic Significance

- Bite marks, bleeding, swelling
- Signs of struggle or enclosure damage
- Often single-animal death

🔍 **Key Clue:** Swelling at bite site → Snake bite

6. Sudden Death at Any Time of Day:

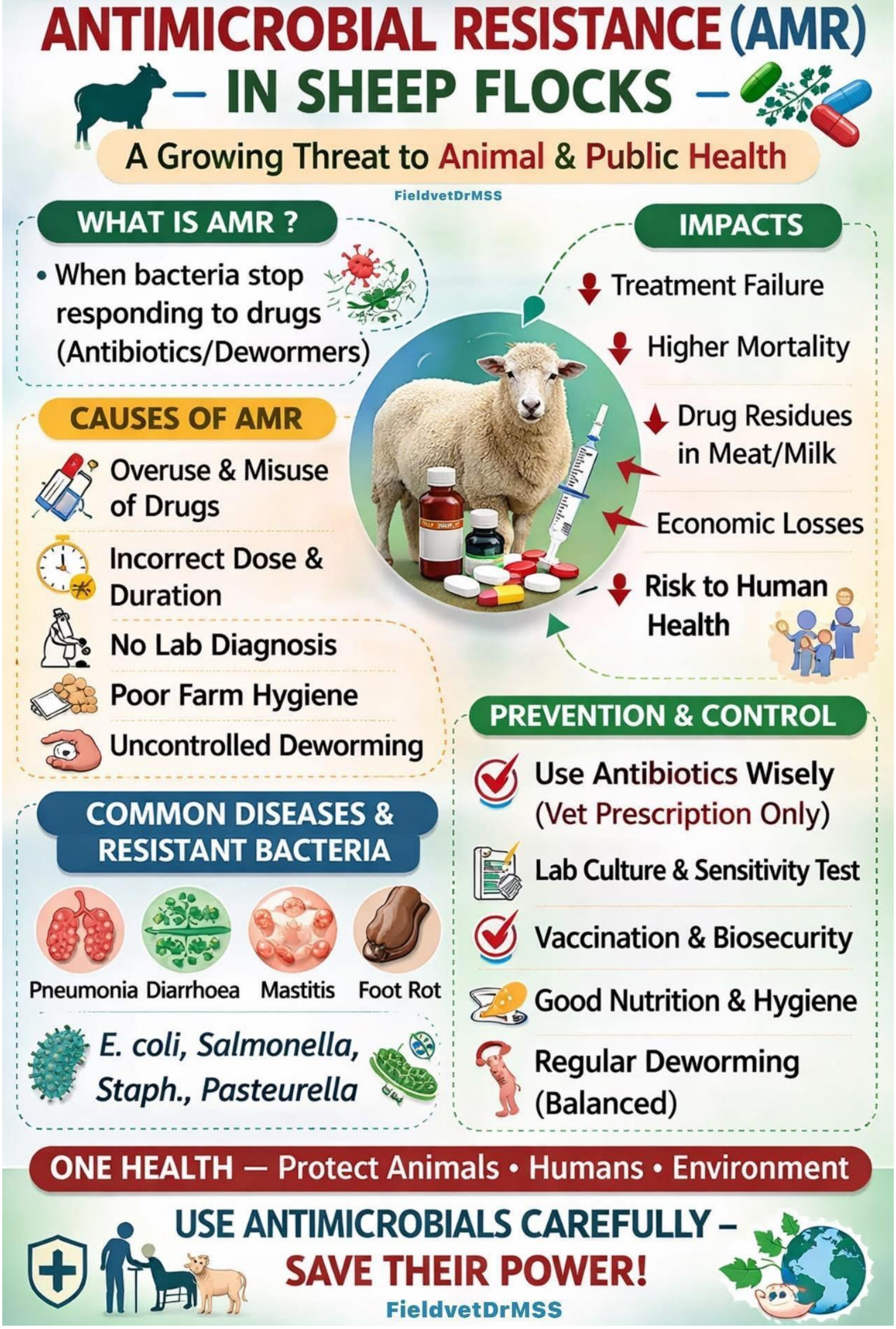
Possible Causes

- Anthrax
- Acute septicemia
- Clostridial diseases
- Lightning strike

- Severe toxicosis

⚠ **Important Warning:**

- Do NOT open carcass if anthrax suspected
- Signs: bleeding from natural orifices, no rigor mortis


ANTIMICROBIAL RESISTANCE (AMR)
– IN SHEEP FLOCKS –
A Growing Threat to Animal & Public Health
FieldvetDrMSS
WHAT IS AMR ?
• When bacteria stop responding to drugs (Antibiotics/Dewormers)
IMPACTS
Treatment Failure
Higher Mortality
Drug Residues in Meat/Milk
Economic Losses
Risk to Human Health
CAUSES OF AMR
Overuse & Misuse of Drugs
Incorrect Dose & Duration
No Lab Diagnosis
Poor Farm Hygiene
Uncontrolled Deworming
PREVENTION & CONTROL
Use Antibiotics Wisely (Vet Prescription Only)
Lab Culture & Sensitivity Test
Vaccination & Biosecurity
Good Nutrition & Hygiene
Regular Deworming (Balanced)
COMMON DISEASES & RESISTANT BACTERIA
Pneumonia
Diarrhoea
Mastitis
Foot Rot
E. coli, Salmonella, Staph., Pasteurella
ONE HEALTH — Protect Animals • Humans • Environment
USE ANTIMICROBIALS CAREFULLY – SAVE THEIR POWER!
FieldvetDrMSS

46. Antimicrobial Resistance (AMR)

🦠 What is Antimicrobial Resistance (AMR)?

Antimicrobial Resistance occurs when bacteria, parasites, or other microbes evolve to survive drugs that were once effective. In sheep farming, this mainly results from misuse or overuse of antibiotics and antiparasitic drugs. AMR threatens animal health, farm productivity, and public health under the One Health framework.

🐑 Why AMR is Important in Sheep Farming

- Reduced effectiveness of antibiotics and dewormers
- Increased treatment failures
- Higher mortality in lambs
- Increased treatment costs
- Risk of drug residues in meat and milk
- Spread of resistant microbes to humans and environment

⚠ Common Causes of AMR in Sheep Flocks

- Overuse of antibiotics
- Routine use without diagnosis
- Incorrect dosage
- Under-dosing promotes resistant bacteria
- Incomplete treatment courses
- Frequent prophylactic use
- Antibiotics used for prevention instead of treatment
- Use of the same drug repeatedly
- Uncontrolled use of dewormers
- Leading to anthelmintic resistance

🔬 Diseases in Sheep Where AMR is Emerging

- Pneumonia complex
- Enteritis / diarrhoea in lambs
- Mastitis
- Foot rot
- Caseous lymphadenitis

Common bacteria involved:

- Escherichia coli
- Pasteurella multocida
- Staphylococcus aureus
- Salmonella spp.

🧪 Signs That AMR May Be Present

- Antibiotic treatment fails repeatedly
- Disease reappears soon after treatment
- Higher number of animals affected
- Slow recovery of treated animals

🛑 How to Prevent AMR in Sheep Farms

1️⃣ Use Antibiotics Responsibly Only under veterinary prescription
Use correct dose and duration

2️⃣ Laboratory Diagnosis

Perform culture and sensitivity tests whenever possible.

3️⃣ Improve Farm Biosecurity

- Isolate sick animals
- Quarantine new animals

4️⃣ Vaccination Programs

Vaccinate against diseases such as:

- Sheep pox
- Enterotoxaemia
- Pasteurellosis

5️⃣ Good Nutrition

Balanced feeding improves immune resistance

6️⃣ Hygienic Management

- Clean sheds
- Dry bedding
- Proper ventilation

🌿 Reduce Dependence on Antibiotics

Alternative strategies include:

- Vaccination
- Probiotics
- Mineral supplementation
- Improved grazing management
- Integrated parasite management

🌍 AMR and One Health

Antimicrobial resistance spreads between:

- Animals
- Humans
- Environment

Responsible antimicrobial use protects:

- ✓ Animal health
- ✓ Human health
- ✓ Ecosystem health

📊 Key Message

"Use Antimicrobials Carefully Today to Protect Their Power Tomorrow."

▯ Role of Veterinarians

Veterinarians help by:

- Diagnosing diseases accurately
- Advising correct drug use
- Monitoring drug resistance
- Educating farmers on judicious antimicrobial use

Conclusion:

Toward a Resilient Future in Sheep Husbandry

The management of disease in sheep is a continuous journey that transitions from the microscopic level of the laboratory to the expansive reality of the grazing field. It is a discipline that requires equal parts vigilance, empathy, and scientific rigor. As we conclude this exploration, it is clear that the "art" of management lies in the ability to harmonize technical protocols with the unique rhythms of the flock.

The Pillars of Sustained Health

- **Proactive over Reactive:** The shift from treating sick individuals to managing the health of the entire population is the hallmark of a successful enterprise. Through strategic vaccination, rigorous biosecurity, and integrated parasite management, we reduce the burden of disease before it can take hold.
- **Welfare as a Diagnostic Tool:** Adhering to the Five Freedoms is not a luxury but a clinical necessity. High welfare standards directly correlate with lower cortisol levels and stronger immune responses, making animal rights a cornerstone of animal productivity.
- **The Power of Indigenous Resilience:** Our native breeds carry centuries of adapted immunity within their DNA. Protecting and promoting these indigenous genetic resources is our most sustainable defense against the changing landscapes of climate and disease.

A Vision for the "One Health" Practitioner

As we look forward, the role of the veterinarian and the farm manager must continue to evolve. By embracing Precision Livestock Farming and data-driven diagnostics, we can ensure that every intervention is as targeted and minimal as possible, safeguarding our environment from chemical runoff and our communities from antimicrobial resistance.

Ultimately, the goal of disease management is to create a system where sheep can thrive in their natural state, supported by a framework of "One Health" that protects the animal, the owner, and the ecosystem alike. Let this work serve as a reminder that in the service of these animals, our greatest tools are observation, education, and an unwavering commitment to the ethics of care.

Dedicated to the Advancement of Veterinary Knowledge.